

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399326
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Waldern; Jonathan David et al.

Grating structures for color waveguides

Abstract

Disclosed herein is a waveguide-based display comprising: a waveguide, a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection (TIR) path within the waveguide; and a grating structure for providing beam expansion of the TIR light in at least one direction and extracting the TIR light from the waveguide. The grating structure may include a plurality of grating elements having at least two different grating prescriptions and at least one surface relief grating formed by a phase separation process.

Inventors: Waldern; Jonathan David (Diablo, CA), Grant; Alastair John (San Jose, CA), Popovich; Milan Momcilo (Leicester, GB)

Applicant: DigiLens Inc. (Sunnyvale, CA)

Family ID: 1000008777920

Assignee: DigiLens Inc. (Sunnyvale, CA)

Appl. No.: 17/647408

Filed: January 07, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220214503 A1	Jul. 07, 2022

Related U.S. Application Data

us-provisional-application US 63134898 20210107

Publication Classification

Int. Cl.: G02B6/34 (20060101); F21V8/00 (20060101); G02B6/35 (20060101)

U.S. Cl.:

CPC G02B6/34 (20130101); G02B6/0011 (20130101); G02B6/3522 (20130101);

Field of Classification Search

CPC: G02B (6/0011)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1043938	12/1911	Huttenlocher	N/A	N/A
2141884	12/1937	Sonnefeld	N/A	N/A
3482498	12/1968	Becker	N/A	N/A
3620601	12/1970	Leonard et al.	N/A	N/A
3741716	12/1972	Johne et al.	N/A	N/A
3804496	12/1973	Crane et al.	N/A	N/A
3843231	12/1973	Borel et al.	N/A	N/A
3851303	12/1973	Muller	N/A	N/A
3885095	12/1974	Wolfson et al.	N/A	N/A
3940204	12/1975	Withrington	N/A	N/A
3965029	12/1975	Arora	N/A	N/A
3975711	12/1975	McMahon	N/A	N/A
4028725	12/1976	Lewis	N/A	N/A
4035068	12/1976	Rawson	N/A	N/A
4038110	12/1976	Feng	N/A	N/A
4066334	12/1977	Fray et al.	N/A	N/A
4082432	12/1977	Kirschner	N/A	N/A
4099841	12/1977	Ellis	N/A	N/A
4133152	12/1978	Penrose	N/A	N/A
4178074	12/1978	Heller	N/A	N/A
4218111	12/1979	Withrington et al.	N/A	N/A
4232943	12/1979	Rogers	N/A	N/A
4248093	12/1980	Andersson et al.	N/A	N/A
4251137	12/1980	Knop et al.	N/A	N/A
4309070	12/1981	St. Leger Searle	N/A	N/A
4322163	12/1981	Schiller	N/A	N/A
4386361	12/1982	Simmonds	N/A	N/A
4389612	12/1982	Simmonds et al.	N/A	N/A
4403189	12/1982	Simmonds	N/A	N/A
4418993	12/1982	Lipton	N/A	N/A
4472037	12/1983	Lipton	N/A	N/A
4523226	12/1984	Lipton et al.	N/A	N/A
4544267	12/1984	Schiller	N/A	N/A
4562463	12/1984	Lipton	N/A	N/A
4566758	12/1985	Bos et al.	N/A	N/A

4583117	12/1985	Lipton et al.	N/A	N/A
4643515	12/1986	Upatnieks	N/A	N/A
4647967	12/1986	Kirschner et al.	N/A	N/A
4688900	12/1986	Doane et al.	N/A	N/A
4711512	12/1986	Upatnieks	N/A	N/A
4714320	12/1986	Banbury	N/A	N/A
4728547	12/1987	Vaz et al.	N/A	N/A
4729640	12/1987	Sakata et al.	N/A	N/A
4741926	12/1987	White et al.	N/A	N/A
4743083	12/1987	Schimpe	N/A	N/A
4749256	12/1987	Bell et al.	N/A	N/A
4765703	12/1987	Suzuki et al.	N/A	N/A
4775218	12/1987	Wood et al.	N/A	N/A
4791788	12/1987	Simmonds et al.	N/A	N/A
4792850	12/1987	Liptoh et al.	N/A	N/A
4794021	12/1987	Potter	N/A	N/A
4799765	12/1988	Ferrer	N/A	N/A
4811414	12/1988	Fishbine et al.	N/A	N/A
4848093	12/1988	Simmonds et al.	N/A	N/A
4852988	12/1988	Velez et al.	N/A	N/A
4854688	12/1988	Hayford et al.	N/A	N/A
4860294	12/1988	Winzer et al.	N/A	N/A
4884876	12/1988	Lipton et al.	N/A	N/A
4890902	12/1989	Doane et al.	N/A	N/A
4928301	12/1989	Smoot	N/A	N/A
4933976	12/1989	Fishbine et al.	N/A	N/A
4938568	12/1989	Margerum et al.	N/A	N/A
4946245	12/1989	Chamberlin et al.	N/A	N/A
4960311	12/1989	Moss et al.	N/A	N/A
4964701	12/1989	Dorschner et al.	N/A	N/A
4967268	12/1989	Lipton et al.	N/A	N/A
4970129	12/1989	Ingwall et al.	N/A	N/A
4971719	12/1989	Vaz et al.	N/A	N/A
4994204	12/1990	Doane et al.	N/A	N/A
5004323	12/1990	West	N/A	N/A
5007711	12/1990	Wood et al.	N/A	N/A
5009483	12/1990	Rockwell et al.	N/A	N/A
5011624	12/1990	Yamagishi et al.	N/A	N/A
5016953	12/1990	Moss et al.	N/A	N/A
5033814	12/1990	Brown et al.	N/A	N/A
5035734	12/1990	Honkanen et al.	N/A	N/A
5053834	12/1990	Simmonds	N/A	N/A
5063441	12/1990	Lipton et al.	N/A	N/A
5076664	12/1990	Migozzi	N/A	N/A
5079416	12/1991	Filipovich	N/A	N/A
5096282	12/1991	Margerum et al.	N/A	N/A
5099343	12/1991	Margerum et al.	N/A	N/A
5109465	12/1991	Klopotek	N/A	N/A
5110034	12/1991	Simmonds et al.	N/A	N/A
5117285	12/1991	Nelson et al.	N/A	N/A

5117302	12/1991	Lipton	N/A	N/A
5119454	12/1991	McMahon et al.	N/A	N/A
5124821	12/1991	Antier et al.	N/A	N/A
5138687	12/1991	Horie et al.	N/A	N/A
5139192	12/1991	Simmonds et al.	N/A	N/A
5142357	12/1991	Lipton et al.	N/A	N/A
5142644	12/1991	Vansteenkiste et al.	N/A	N/A
5148302	12/1991	Nagano et al.	N/A	N/A
5150234	12/1991	Takahashi et al.	N/A	N/A
5151958	12/1991	Honkanen	N/A	N/A
5153751	12/1991	Ishikawa et al.	N/A	N/A
5159445	12/1991	Gitlin et al.	N/A	N/A
5160523	12/1991	Honkanen et al.	N/A	N/A
5181133	12/1992	Lipton	N/A	N/A
5183545	12/1992	Branca et al.	N/A	N/A
5187597	12/1992	Kato et al.	N/A	N/A
5193000	12/1992	Lipton et al.	N/A	N/A
5198912	12/1992	Ingwall et al.	N/A	N/A
5198914	12/1992	Arns	N/A	N/A
5200861	12/1992	Moskovich et al.	N/A	N/A
5210624	12/1992	Matsumoto et al.	N/A	N/A
5210801	12/1992	Fournier et al.	N/A	N/A
5218360	12/1992	Goetz et al.	N/A	N/A
5218480	12/1992	Moskovich et al.	N/A	N/A
5224198	12/1992	Jachimowicz et al.	N/A	N/A
5225918	12/1992	Taniguchi et al.	N/A	N/A
5239372	12/1992	Lipton	N/A	N/A
5240636	12/1992	Doane et al.	N/A	N/A
5241337	12/1992	Betensky et al.	N/A	N/A
5242476	12/1992	Bartel et al.	N/A	N/A
5243413	12/1992	Gitlin et al.	N/A	N/A
5251048	12/1992	Doane et al.	N/A	N/A
5264950	12/1992	West et al.	N/A	N/A
5268792	12/1992	Kreitzer et al.	N/A	N/A
5284499	12/1993	Harvey et al.	N/A	N/A
5289315	12/1993	Makita et al.	N/A	N/A
5295208	12/1993	Caulfield et al.	N/A	N/A
5296967	12/1993	Moskovich et al.	N/A	N/A
5299289	12/1993	Omae et al.	N/A	N/A
5303085	12/1993	Rallison	N/A	N/A
5306923	12/1993	Kazmierski et al.	N/A	N/A
5309283	12/1993	Kreitzer et al.	N/A	N/A
5313330	12/1993	Betensky	N/A	N/A
5315324	12/1993	Kubelik et al.	N/A	N/A
5315419	12/1993	Saupe et al.	N/A	N/A
5315440	12/1993	Betensky et al.	N/A	N/A
5317405	12/1993	Kuriki et al.	N/A	N/A
5327269	12/1993	Tilton et al.	N/A	N/A
5329363	12/1993	Moskovich et al.	N/A	N/A
5341230	12/1993	Smith	N/A	N/A

5343147	12/1993	Sager et al.	N/A	N/A
5351151	12/1993	Levy	N/A	N/A
5359362	12/1993	Lewis et al.	N/A	N/A
5363220	12/1993	Kuwayama et al.	N/A	N/A
5368770	12/1993	Saupe et al.	N/A	N/A
5369511	12/1993	Amos	N/A	N/A
5371626	12/1993	Betensky	N/A	N/A
5400069	12/1994	Braun et al.	N/A	N/A
5408346	12/1994	Trissel et al.	N/A	N/A
5410370	12/1994	Janssen	N/A	N/A
5410376	12/1994	Cornsweet et al.	N/A	N/A
5416510	12/1994	Lipton et al.	N/A	N/A
5416514	12/1994	Janssen et al.	N/A	N/A
5418584	12/1994	Larson	N/A	N/A
5418871	12/1994	Revelli et al.	N/A	N/A
5428480	12/1994	Betensky et al.	N/A	N/A
5437811	12/1994	Doane et al.	N/A	N/A
5438357	12/1994	McNelley	N/A	N/A
5452385	12/1994	Izumi et al.	N/A	N/A
5453863	12/1994	West et al.	N/A	N/A
5455693	12/1994	Wreede et al.	N/A	N/A
5455713	12/1994	Kreitzer et al.	N/A	N/A
5462700	12/1994	Beeson et al.	N/A	N/A
5463428	12/1994	Lipton et al.	N/A	N/A
5465311	12/1994	Caulfield et al.	N/A	N/A
5471326	12/1994	Hall et al.	N/A	N/A
5473222	12/1994	Thoeny et al.	N/A	N/A
5476611	12/1994	Nolan et al.	N/A	N/A
5481321	12/1995	Lipton	N/A	N/A
5481385	12/1995	Zimmerman et al.	N/A	N/A
5485313	12/1995	Betensky	N/A	N/A
5493430	12/1995	Lu et al.	N/A	N/A
5493448	12/1995	Betensky et al.	N/A	N/A
5496621	12/1995	Makita et al.	N/A	N/A
5499140	12/1995	Betensky	N/A	N/A
5500671	12/1995	Andersson et al.	N/A	N/A
5500769	12/1995	Betensky	N/A	N/A
5510913	12/1995	Hashimoto et al.	N/A	N/A
5515184	12/1995	Caulfield et al.	N/A	N/A
5516455	12/1995	Jacobine et al.	N/A	N/A
5524272	12/1995	Podowski et al.	N/A	N/A
5528720	12/1995	Winston et al.	N/A	N/A
5530566	12/1995	Kumar	N/A	N/A
5532736	12/1995	Kuriki et al.	N/A	N/A
5532875	12/1995	Betensky	N/A	N/A
5537232	12/1995	Biles	N/A	N/A
RE35310	12/1995	Moskovich	N/A	N/A
5543950	12/1995	Lavrentovich et al.	N/A	N/A
5544268	12/1995	Bischel et al.	N/A	N/A
5559637	12/1995	Moskovich et al.	N/A	N/A

5572248	12/1995	Allen et al.	N/A	N/A
5572250	12/1995	Lipton et al.	N/A	N/A
5576888	12/1995	Betensky	N/A	N/A
5579026	12/1995	Tabata	N/A	N/A
5583795	12/1995	Smyth	N/A	N/A
5585035	12/1995	Nerad et al.	N/A	N/A
5593615	12/1996	Nerad et al.	N/A	N/A
5604611	12/1996	Saburi et al.	N/A	N/A
5606433	12/1996	Yin et al.	N/A	N/A
5612733	12/1996	Flohr	N/A	N/A
5612734	12/1996	Nelson et al.	N/A	N/A
5619254	12/1996	McNelley	N/A	N/A
5619586	12/1996	Sibbald et al.	N/A	N/A
5621529	12/1996	Gordon et al.	N/A	N/A
5621552	12/1996	Coates et al.	N/A	N/A
5625495	12/1996	Moskovich et al.	N/A	N/A
5629259	12/1996	Akada et al.	N/A	N/A
5631107	12/1996	Tarumi et al.	N/A	N/A
5633100	12/1996	Mickish et al.	N/A	N/A
5646785	12/1996	Gilboa et al.	N/A	N/A
5648857	12/1996	Ando et al.	N/A	N/A
5661577	12/1996	Jenkins et al.	N/A	N/A
5661603	12/1996	Hanano et al.	N/A	N/A
5665494	12/1996	Kawabata et al.	N/A	N/A
5668614	12/1996	Chien et al.	N/A	N/A
5668907	12/1996	Veligdan	N/A	N/A
5677797	12/1996	Betensky et al.	N/A	N/A
5680231	12/1996	Grinberg et al.	N/A	N/A
5680411	12/1996	Ramdane et al.	N/A	N/A
5682255	12/1996	Friesem et al.	N/A	N/A
5686931	12/1996	Fuenfschilling et al.	N/A	N/A
5686975	12/1996	Lipton	N/A	N/A
5691795	12/1996	Doane et al.	N/A	N/A
5694230	12/1996	Welch	N/A	N/A
5695682	12/1996	Doane et al.	N/A	N/A
5701132	12/1996	Kollin et al.	N/A	N/A
5706108	12/1997	Ando et al.	N/A	N/A
5706136	12/1997	Okuyama et al.	N/A	N/A
5707925	12/1997	Akada et al.	N/A	N/A
5710645	12/1997	Phillips et al.	N/A	N/A
5724189	12/1997	Ferrante	N/A	N/A
5724463	12/1997	Deacon et al.	N/A	N/A
5726782	12/1997	Kato et al.	N/A	N/A
5727098	12/1997	Jacobson	N/A	N/A
5729242	12/1997	Margerum et al.	N/A	N/A
5731060	12/1997	Hirukawa et al.	N/A	N/A
5731853	12/1997	Taketomi et al.	N/A	N/A
5736424	12/1997	Prybyla et al.	N/A	N/A
5742262	12/1997	Tabata et al.	N/A	N/A
5745266	12/1997	Smith et al.	N/A	N/A

5745301	12/1997	Betensky et al.	N/A	N/A
5748272	12/1997	Tanaka et al.	N/A	N/A
5748277	12/1997	Huang et al.	N/A	N/A
5751452	12/1997	Tanaka et al.	N/A	N/A
5757546	12/1997	Lipton et al.	N/A	N/A
5760931	12/1997	Saburi et al.	N/A	N/A
5760960	12/1997	Lin et al.	N/A	N/A
5764414	12/1997	King et al.	N/A	N/A
5771320	12/1997	Stone	N/A	N/A
5790288	12/1997	Jager et al.	N/A	N/A
5790314	12/1997	Duck et al.	N/A	N/A
5798641	12/1997	Spagna et al.	N/A	N/A
5804609	12/1997	Ohnishi et al.	N/A	N/A
5808804	12/1997	Moskovich	N/A	N/A
5812608	12/1997	Valimaki et al.	N/A	N/A
5822089	12/1997	Phillips et al.	N/A	N/A
5822127	12/1997	Chen et al.	N/A	N/A
5825448	12/1997	Bos et al.	N/A	N/A
5831700	12/1997	Li et al.	N/A	N/A
5835661	12/1997	Tai et al.	N/A	N/A
5841507	12/1997	Barnes	N/A	N/A
5841587	12/1997	Moskovich et al.	N/A	N/A
5847787	12/1997	Fredley et al.	N/A	N/A
5856842	12/1998	Tedesco	N/A	N/A
5857043	12/1998	Cook et al.	N/A	N/A
5867238	12/1998	Miller et al.	N/A	N/A
5867618	12/1998	Ito et al.	N/A	N/A
5868951	12/1998	Schuck, III et al.	N/A	N/A
5870228	12/1998	Kreitzer et al.	N/A	N/A
5875012	12/1998	Crawford et al.	N/A	N/A
5877826	12/1998	Yang et al.	N/A	N/A
5886822	12/1998	Spitzer	N/A	N/A
5892598	12/1998	Asakawa et al.	N/A	N/A
5892599	12/1998	Bahuguna	N/A	N/A
5898511	12/1998	Mizutani et al.	N/A	N/A
5900987	12/1998	Kreitzer et al.	N/A	N/A
5900989	12/1998	Kreitzer	N/A	N/A
5903395	12/1998	Rallison et al.	N/A	N/A
5903396	12/1998	Rallison	N/A	N/A
5907416	12/1998	Hegg et al.	N/A	N/A
5907436	12/1998	Perry et al.	N/A	N/A
5911018	12/1998	Bischel et al.	N/A	N/A
5917459	12/1998	Son et al.	N/A	N/A
5926147	12/1998	Sehm et al.	N/A	N/A
5929946	12/1998	Sharp et al.	N/A	N/A
5929960	12/1998	West et al.	N/A	N/A
5930433	12/1998	Williamson et al.	N/A	N/A
5936776	12/1998	Kreitzer	N/A	N/A
5937115	12/1998	Domash	N/A	N/A
5942157	12/1998	Sutherland et al.	N/A	N/A

5945893	12/1998	Plessky et al.	N/A	N/A
5949302	12/1998	Sarkka	N/A	N/A
5949508	12/1998	Kumar et al.	N/A	N/A
5956113	12/1998	Crawford	N/A	N/A
5962147	12/1998	Shalhub et al.	N/A	N/A
5963375	12/1998	Kreitzer	N/A	N/A
5966223	12/1998	Friesem et al.	N/A	N/A
5969874	12/1998	Moskovich	N/A	N/A
5969876	12/1998	Kreitzer et al.	N/A	N/A
5973727	12/1998	McGrew et al.	N/A	N/A
5974162	12/1998	Metz et al.	N/A	N/A
5985422	12/1998	Krauter	N/A	N/A
5986746	12/1998	Metz et al.	N/A	N/A
5991087	12/1998	Rallison	N/A	N/A
5999089	12/1998	Carlson et al.	N/A	N/A
5999282	12/1998	Suzuki et al.	N/A	N/A
5999314	12/1998	Asakura et al.	N/A	N/A
6014187	12/1999	Taketomi et al.	N/A	N/A
6023375	12/1999	Kreitzer	N/A	N/A
6042947	12/1999	Asakura et al.	N/A	N/A
6043585	12/1999	Plessky et al.	N/A	N/A
6046585	12/1999	Simmonds	N/A	N/A
6052540	12/1999	Koyama	N/A	N/A
6061107	12/1999	Yang	N/A	N/A
6061463	12/1999	Metz et al.	N/A	N/A
6069728	12/1999	Huignard et al.	N/A	N/A
6075626	12/1999	Mizutani et al.	N/A	N/A
6078427	12/1999	Fontaine et al.	N/A	N/A
6084998	12/1999	Straayer	N/A	N/A
6094311	12/1999	Moskovich	N/A	N/A
6097551	12/1999	Kreitzer	N/A	N/A
6104448	12/1999	Doane et al.	N/A	N/A
6107943	12/1999	Schroeder	N/A	N/A
6115152	12/1999	Popovich et al.	N/A	N/A
6118908	12/1999	Bischel et al.	N/A	N/A
6121899	12/1999	Theriault	N/A	N/A
6127066	12/1999	Ueda et al.	N/A	N/A
6128058	12/1999	Walton et al.	N/A	N/A
6133971	12/1999	Silverstein et al.	N/A	N/A
6133975	12/1999	Li et al.	N/A	N/A
6137630	12/1999	Tsou et al.	N/A	N/A
6141074	12/1999	Bos et al.	N/A	N/A
6141154	12/1999	Kreitzer et al.	N/A	N/A
6151142	12/1999	Phillips et al.	N/A	N/A
6154190	12/1999	Yang et al.	N/A	N/A
6156243	12/1999	Kosuga et al.	N/A	N/A
6167169	12/1999	Brinkman et al.	N/A	N/A
6169594	12/2000	Aye et al.	N/A	N/A
6169613	12/2000	Amitai et al.	N/A	N/A
6169636	12/2000	Kreitzer et al.	N/A	N/A

6172792	12/2000	Jepsen et al.	N/A	N/A
6176837	12/2000	Foxlin	N/A	N/A
6185015	12/2000	Reinhorn et al.	N/A	N/A
6185016	12/2000	Popovich	N/A	N/A
6188462	12/2000	Lavrentovich et al.	N/A	N/A
6191887	12/2000	Michaloski et al.	N/A	N/A
6195206	12/2000	Yona et al.	N/A	N/A
6195209	12/2000	Kreitzer et al.	N/A	N/A
6204835	12/2000	Yang et al.	N/A	N/A
6211976	12/2000	Popovich et al.	N/A	N/A
6218316	12/2000	Marsh	N/A	N/A
6222297	12/2000	Perdue	N/A	N/A
6222675	12/2000	Mall et al.	N/A	N/A
6222971	12/2000	Veligdan et al.	N/A	N/A
6249386	12/2000	Yona et al.	N/A	N/A
6259423	12/2000	Tokito et al.	N/A	N/A
6259559	12/2000	Kobayashi et al.	N/A	N/A
6266166	12/2000	Katsumata et al.	N/A	N/A
6268839	12/2000	Yang et al.	N/A	N/A
6269203	12/2000	Davies et al.	N/A	N/A
6275031	12/2000	Simmonds et al.	N/A	N/A
6278429	12/2000	Ruth et al.	N/A	N/A
6285813	12/2000	Schultz et al.	N/A	N/A
6297860	12/2000	Moskovich et al.	N/A	N/A
6301056	12/2000	Kreitzer et al.	N/A	N/A
6301057	12/2000	Kreitzer et al.	N/A	N/A
6317083	12/2000	Johnson et al.	N/A	N/A
6317227	12/2000	Mizutani et al.	N/A	N/A
6317228	12/2000	Popovich et al.	N/A	N/A
6317528	12/2000	Gadkaree et al.	N/A	N/A
6320563	12/2000	Yang et al.	N/A	N/A
6321069	12/2000	Piirainen	N/A	N/A
6323970	12/2000	Popovich	N/A	N/A
6323989	12/2000	Jacobson et al.	N/A	N/A
6324014	12/2000	Moskovich et al.	N/A	N/A
6327089	12/2000	Hosaki et al.	N/A	N/A
6330109	12/2000	Ishii et al.	N/A	N/A
6333819	12/2000	Svedenkrans	N/A	N/A
6339486	12/2001	Popovich	N/A	N/A
6340540	12/2001	Ueda et al.	N/A	N/A
6351273	12/2001	Lemelson et al.	N/A	N/A
6351333	12/2001	Araki et al.	N/A	N/A
6356172	12/2001	Koivisto et al.	N/A	N/A
6356674	12/2001	Davis et al.	N/A	N/A
6359730	12/2001	Tervonen	N/A	N/A
6359737	12/2001	Stringfellow	N/A	N/A
6366281	12/2001	Lipton et al.	N/A	N/A
6366369	12/2001	Ichikawa et al.	N/A	N/A
6366378	12/2001	Tervonen et al.	N/A	N/A
6377238	12/2001	McPheters	N/A	N/A

6377321	12/2001	Khan et al.	N/A	N/A
6388797	12/2001	Lipton et al.	N/A	N/A
6392812	12/2001	Howard	N/A	N/A
6407724	12/2001	Waldern et al.	N/A	N/A
6409687	12/2001	Foxlin	N/A	N/A
6411444	12/2001	Moskovich et al.	N/A	N/A
6414760	12/2001	Lopez et al.	N/A	N/A
6417971	12/2001	Moskovich et al.	N/A	N/A
6437563	12/2001	Simmonds et al.	N/A	N/A
6445512	12/2001	Moskovich et al.	N/A	N/A
6449095	12/2001	Ohtaki et al.	N/A	N/A
6456584	12/2001	Nagata et al.	N/A	N/A
6470132	12/2001	Nousiainen et al.	N/A	N/A
6473209	12/2001	Popovich	N/A	N/A
6476974	12/2001	Kreitzer et al.	N/A	N/A
6483303	12/2001	Simmonds et al.	N/A	N/A
6486997	12/2001	Bruzzzone et al.	N/A	N/A
6504518	12/2002	Kuwayama et al.	N/A	N/A
6504629	12/2002	Popovich et al.	N/A	N/A
6509937	12/2002	Moskovich et al.	N/A	N/A
6510263	12/2002	Maisenhoelder et al.	N/A	N/A
6518747	12/2002	Sager et al.	N/A	N/A
6519088	12/2002	Lipton	N/A	N/A
6522794	12/2002	Bischel et al.	N/A	N/A
6522795	12/2002	Jordan et al.	N/A	N/A
6524771	12/2002	Maeda et al.	N/A	N/A
6529336	12/2002	Kreitzer et al.	N/A	N/A
6534977	12/2002	Duncan et al.	N/A	N/A
6545778	12/2002	Ono et al.	N/A	N/A
6545808	12/2002	Ehbets et al.	N/A	N/A
6550949	12/2002	Bauer et al.	N/A	N/A
6552789	12/2002	Modro	N/A	N/A
6557413	12/2002	Nieminen et al.	N/A	N/A
6559813	12/2002	DeLuca et al.	N/A	N/A
6560019	12/2002	Nakai	N/A	N/A
6563648	12/2002	Gleckman et al.	N/A	N/A
6563650	12/2002	Moskovich et al.	N/A	N/A
6567014	12/2002	Hansen et al.	N/A	N/A
6567573	12/2002	Domash et al.	N/A	N/A
6577411	12/2002	David et al.	N/A	N/A
6577429	12/2002	Kurtz et al.	N/A	N/A
6580529	12/2002	Amitai et al.	N/A	N/A
6583838	12/2002	Hoke et al.	N/A	N/A
6583873	12/2002	Goncharov et al.	N/A	N/A
6587619	12/2002	Kinoshita	N/A	N/A
6594090	12/2002	Kruschwitz et al.	N/A	N/A
6596193	12/2002	Coates et al.	N/A	N/A
6597176	12/2002	Simmonds et al.	N/A	N/A
6597475	12/2002	Shirakura et al.	N/A	N/A
6598987	12/2002	Parikka	N/A	N/A

6600590	12/2002	Roddy et al.	N/A	N/A
6608720	12/2002	Freeman	N/A	N/A
6611253	12/2002	Cohen	N/A	N/A
6618104	12/2002	Date et al.	N/A	N/A
6624943	12/2002	Nakai et al.	N/A	N/A
6625381	12/2002	Roddy et al.	N/A	N/A
6646772	12/2002	Popovich et al.	N/A	N/A
6646810	12/2002	Harter, Jr. et al.	N/A	N/A
6661578	12/2002	Hedrick	N/A	N/A
6667134	12/2002	Sutherland et al.	N/A	N/A
6674578	12/2003	Sugiyama et al.	N/A	N/A
6677086	12/2003	Suthehrland et al.	N/A	N/A
6678093	12/2003	Scobey et al.	N/A	N/A
6686815	12/2003	Mirshekarl-Syahkal et al.	N/A	N/A
6690516	12/2003	Aritake et al.	N/A	N/A
6692666	12/2003	Sutherland et al.	N/A	N/A
6699407	12/2003	Suthehrland et al.	N/A	N/A
6706086	12/2003	Emig et al.	N/A	N/A
6706451	12/2003	Sutherland et al.	N/A	N/A
6721096	12/2003	Bruzzzone et al.	N/A	N/A
6730442	12/2003	Sutherland et al.	N/A	N/A
6731434	12/2003	Hua et al.	N/A	N/A
6738105	12/2003	Hannah et al.	N/A	N/A
6741189	12/2003	Gibbons, II et al.	N/A	N/A
6744478	12/2003	Asakura et al.	N/A	N/A
6747781	12/2003	Trisnadi et al.	N/A	N/A
6748342	12/2003	Dickhaus	N/A	N/A
6750941	12/2003	Satoh et al.	N/A	N/A
6750995	12/2003	Dickson	N/A	N/A
6750996	12/2003	Jagt et al.	N/A	N/A
6757105	12/2003	Niv et al.	N/A	N/A
6771403	12/2003	Endo et al.	N/A	N/A
6776339	12/2003	Piikivi	N/A	N/A
6781701	12/2003	Sweetser et al.	N/A	N/A
6791629	12/2003	Moskovich et al.	N/A	N/A
6791739	12/2003	Ramanujan et al.	N/A	N/A
6804066	12/2003	Ha et al.	N/A	N/A
6805490	12/2003	Levola	N/A	N/A
6821457	12/2003	Natarajan et al.	N/A	N/A
6822713	12/2003	Yaroshchuk et al.	N/A	N/A
6825987	12/2003	Repetto et al.	N/A	N/A
6829095	12/2003	Amitai	N/A	N/A
6830789	12/2003	Doane et al.	N/A	N/A
6833955	12/2003	Niv	N/A	N/A
6836369	12/2003	Fujikawa et al.	N/A	N/A
6842563	12/2004	Zhang et al.	N/A	N/A
6844212	12/2004	Bond et al.	N/A	N/A
6844980	12/2004	He et al.	N/A	N/A
6844989	12/2004	Jo et al.	N/A	N/A

6847274	12/2004	Salmela et al.	N/A	N/A
6847488	12/2004	Travis	N/A	N/A
6850210	12/2004	Lipton et al.	N/A	N/A
6853491	12/2004	Ruhle et al.	N/A	N/A
6853493	12/2004	Kreitzer et al.	N/A	N/A
6861107	12/2004	Klasen-Memmer et al.	N/A	N/A
6864861	12/2004	Schehrer et al.	N/A	N/A
6864927	12/2004	Cathey	N/A	N/A
6864931	12/2004	Kumar et al.	N/A	N/A
6867888	12/2004	Sutherland et al.	N/A	N/A
6873443	12/2004	Joubert et al.	N/A	N/A
6876791	12/2004	Murashima et al.	N/A	N/A
6878494	12/2004	Sutehrland et al.	N/A	N/A
6885483	12/2004	Takada	N/A	N/A
6903872	12/2004	Schrader	N/A	N/A
6909345	12/2004	Salmela et al.	N/A	N/A
6917375	12/2004	Akada et al.	N/A	N/A
6919003	12/2004	Ikeda et al.	N/A	N/A
6922267	12/2004	Endo et al.	N/A	N/A
6926429	12/2004	Barlow et al.	N/A	N/A
6927570	12/2004	Simmonds et al.	N/A	N/A
6927694	12/2004	Smith et al.	N/A	N/A
6940361	12/2004	Jokio et al.	N/A	N/A
6943788	12/2004	Tomono	N/A	N/A
6950173	12/2004	Sutherland et al.	N/A	N/A
6950227	12/2004	Schrader	N/A	N/A
6951393	12/2004	Koide	N/A	N/A
6952312	12/2004	Weber et al.	N/A	N/A
6952435	12/2004	Lai et al.	N/A	N/A
6958662	12/2004	Salmela et al.	N/A	N/A
6958868	12/2004	Pender	N/A	N/A
6963454	12/2004	Martins et al.	N/A	N/A
6972788	12/2004	Robertson et al.	N/A	N/A
6975345	12/2004	Lipton et al.	N/A	N/A
6980365	12/2004	Moskovich	N/A	N/A
6985296	12/2005	Lipton et al.	N/A	N/A
6987908	12/2005	Bond et al.	N/A	N/A
6999239	12/2005	Martins et al.	N/A	N/A
7002618	12/2005	Lipton et al.	N/A	N/A
7002753	12/2005	Moskovich et al.	N/A	N/A
7003075	12/2005	Miyake et al.	N/A	N/A
7003187	12/2005	Frick et al.	N/A	N/A
7006732	12/2005	Gunn, III et al.	N/A	N/A
7009773	12/2005	Chaoulov et al.	N/A	N/A
7018563	12/2005	Sutherland et al.	N/A	N/A
7018686	12/2005	Sutehrland et al.	N/A	N/A
7018744	12/2005	Otaki et al.	N/A	N/A
7019793	12/2005	Moskovich et al.	N/A	N/A
7021777	12/2005	Amitai	N/A	N/A

7026892	12/2005	Kajiya	N/A	N/A
7027671	12/2005	Huck et al.	N/A	N/A
7034748	12/2005	Kajiya	N/A	N/A
7046439	12/2005	Kaminsky et al.	N/A	N/A
7050674	12/2005	Lee et al.	N/A	N/A
7053735	12/2005	Salmela et al.	N/A	N/A
7053991	12/2005	Sandusky	N/A	N/A
7054045	12/2005	McPheters et al.	N/A	N/A
7058434	12/2005	Wang et al.	N/A	N/A
7068405	12/2005	Sutherland et al.	N/A	N/A
7068898	12/2005	Buretea et al.	N/A	N/A
7072020	12/2005	Sutherland et al.	N/A	N/A
7075273	12/2005	O'Gorman et al.	N/A	N/A
7077984	12/2005	Natarajan et al.	N/A	N/A
7081215	12/2005	Natarajan et al.	N/A	N/A
7088457	12/2005	Zou et al.	N/A	N/A
7088515	12/2005	Lipton	N/A	N/A
7095562	12/2005	Peng et al.	N/A	N/A
7099080	12/2005	Lipton et al.	N/A	N/A
7101048	12/2005	Travis	N/A	N/A
7108383	12/2005	Mitchell et al.	N/A	N/A
7110184	12/2005	Yona et al.	N/A	N/A
7119965	12/2005	Rolland et al.	N/A	N/A
7123418	12/2005	Weber et al.	N/A	N/A
7123421	12/2005	Moskovich et al.	N/A	N/A
7126418	12/2005	Hunton et al.	N/A	N/A
7126583	12/2005	Breed	N/A	N/A
7132200	12/2005	Ueda et al.	N/A	N/A
7133084	12/2005	Moskovich et al.	N/A	N/A
7139109	12/2005	Mukawa	N/A	N/A
RE39424	12/2005	Moskovich	N/A	N/A
7145729	12/2005	Kreitzer et al.	N/A	N/A
7149385	12/2005	Parikka et al.	N/A	N/A
7151246	12/2005	Fein et al.	N/A	N/A
7158095	12/2006	Jenson et al.	N/A	N/A
7167286	12/2006	Anderson et al.	N/A	N/A
7167616	12/2006	Ling et al.	N/A	N/A
7175780	12/2006	Sutherland et al.	N/A	N/A
7181105	12/2006	Teramura et al.	N/A	N/A
7181108	12/2006	Levola	N/A	N/A
7184002	12/2006	Lipton et al.	N/A	N/A
7184615	12/2006	Levola	N/A	N/A
7186567	12/2006	Sutherland et al.	N/A	N/A
7190849	12/2006	Katase	N/A	N/A
7198737	12/2006	Natarajan et al.	N/A	N/A
7199934	12/2006	Yamasaki	N/A	N/A
7205960	12/2006	David	N/A	N/A
7205964	12/2006	Yokoyama et al.	N/A	N/A
7206107	12/2006	Levola	N/A	N/A
7212175	12/2006	Magee et al.	N/A	N/A

7218817	12/2006	Magnusson et al.	N/A	N/A
7230767	12/2006	Walck et al.	N/A	N/A
7230770	12/2006	Kreitzer et al.	N/A	N/A
7242527	12/2006	Spitzer et al.	N/A	N/A
7248128	12/2006	Mattila et al.	N/A	N/A
7248765	12/2006	Lee et al.	N/A	N/A
7256915	12/2006	Sutherland et al.	N/A	N/A
7259906	12/2006	Islam	N/A	N/A
7265882	12/2006	Sutherland et al.	N/A	N/A
7265903	12/2006	Sutherland et al.	N/A	N/A
7268946	12/2006	Wang	N/A	N/A
7285903	12/2006	Cull et al.	N/A	N/A
7286272	12/2006	Mukawa	N/A	N/A
7289069	12/2006	Ranta	N/A	N/A
RE39911	12/2006	Moskovich	N/A	N/A
7299983	12/2006	Piikivi	N/A	N/A
7301601	12/2006	Lin et al.	N/A	N/A
7312906	12/2006	Sutherland et al.	N/A	N/A
7313291	12/2006	Okhotnikov et al.	N/A	N/A
D559250	12/2007	Pombo	N/A	N/A
7319573	12/2007	Nishiyama	N/A	N/A
7320534	12/2007	Sugikawa et al.	N/A	N/A
7323275	12/2007	Otaki et al.	N/A	N/A
7333685	12/2007	Stone et al.	N/A	N/A
7336271	12/2007	Ozeki et al.	N/A	N/A
7339737	12/2007	Urey et al.	N/A	N/A
7339742	12/2007	Amitai et al.	N/A	N/A
7349612	12/2007	Nishii et al.	N/A	N/A
7356218	12/2007	Kato et al.	N/A	N/A
7356224	12/2007	Levner et al.	N/A	N/A
7369911	12/2007	Volant et al.	N/A	N/A
7375870	12/2007	Schorpp	N/A	N/A
7375886	12/2007	Lipton et al.	N/A	N/A
7376068	12/2007	Khoury	N/A	N/A
7376307	12/2007	Singh et al.	N/A	N/A
7391573	12/2007	Amitai	N/A	N/A
7394865	12/2007	Borran et al.	N/A	N/A
7394961	12/2007	Kornilovich et al.	N/A	N/A
7395181	12/2007	Foxlin	N/A	N/A
7397606	12/2007	Peng et al.	N/A	N/A
7401920	12/2007	Kranz et al.	N/A	N/A
7404644	12/2007	Evans et al.	N/A	N/A
7410286	12/2007	Travis	N/A	N/A
7411637	12/2007	Weiss	N/A	N/A
7413678	12/2007	Natarajan et al.	N/A	N/A
7413679	12/2007	Sutherland et al.	N/A	N/A
7415173	12/2007	Kassamakov et al.	N/A	N/A
7416818	12/2007	Sutherland et al.	N/A	N/A
7418170	12/2007	Mukawa et al.	N/A	N/A
7420733	12/2007	Natarajan et al.	N/A	N/A

7433116	12/2007	Islam	N/A	N/A
7436568	12/2007	Kuykendall, Jr.	N/A	N/A
D581447	12/2007	Yee	N/A	N/A
7447967	12/2007	Onggosanusi et al.	N/A	N/A
7453612	12/2007	Mukawa	N/A	N/A
7454103	12/2007	Parriaux	N/A	N/A
7457040	12/2007	Amitai	N/A	N/A
7466994	12/2007	Pihlaja et al.	N/A	N/A
7477206	12/2008	Cowan et al.	N/A	N/A
7479354	12/2008	Ueda et al.	N/A	N/A
7480215	12/2008	Makela et al.	N/A	N/A
7482996	12/2008	Larson et al.	N/A	N/A
7483604	12/2008	Levola	N/A	N/A
7492512	12/2008	Niv et al.	N/A	N/A
7496293	12/2008	Shamir et al.	N/A	N/A
7499217	12/2008	Cakmakci et al.	N/A	N/A
7500104	12/2008	Goland	N/A	N/A
7511891	12/2008	Messerschmidt	N/A	N/A
7513668	12/2008	Peng et al.	N/A	N/A
7522344	12/2008	Curatu et al.	N/A	N/A
7525448	12/2008	Wilson et al.	N/A	N/A
7528385	12/2008	Volodin et al.	N/A	N/A
7542210	12/2008	Chirieleison	N/A	N/A
7545429	12/2008	Travis	N/A	N/A
7550234	12/2008	Otaki et al.	N/A	N/A
7558446	12/2008	Wimberger-Friedl et al.	N/A	N/A
7567372	12/2008	Schorpp	N/A	N/A
7570322	12/2008	Sutherland et al.	N/A	N/A
7570405	12/2008	Sutherland et al.	N/A	N/A
7570429	12/2008	Maliah et al.	N/A	N/A
7572555	12/2008	Takizawa et al.	N/A	N/A
7573640	12/2008	Nivon et al.	N/A	N/A
7576916	12/2008	Amitai	N/A	N/A
7577326	12/2008	Amitai	N/A	N/A
7579119	12/2008	Ueda et al.	N/A	N/A
7583423	12/2008	Sutherland et al.	N/A	N/A
7587110	12/2008	Singh et al.	N/A	N/A
7588863	12/2008	Takizawa et al.	N/A	N/A
7589900	12/2008	Powell	N/A	N/A
7589901	12/2008	DeJong et al.	N/A	N/A
7592988	12/2008	Katase	N/A	N/A
7593575	12/2008	Houle et al.	N/A	N/A
7597447	12/2008	Larson et al.	N/A	N/A
7599012	12/2008	Nakamura et al.	N/A	N/A
7600893	12/2008	Laino et al.	N/A	N/A
7602552	12/2008	Blumenfeld	N/A	N/A
7605719	12/2008	Wenger et al.	N/A	N/A
7605774	12/2008	Brandt et al.	N/A	N/A
7605882	12/2008	Sutherland et al.	N/A	N/A

7616270	12/2008	Hirabayashi et al.	N/A	N/A
7617022	12/2008	Wood et al.	N/A	N/A
7618750	12/2008	Ueda et al.	N/A	N/A
7619739	12/2008	Sutherland et al.	N/A	N/A
7619825	12/2008	Peng et al.	N/A	N/A
7629086	12/2008	Otaki et al.	N/A	N/A
7639208	12/2008	Ha et al.	N/A	N/A
7639911	12/2008	Lee et al.	N/A	N/A
7643214	12/2009	Amitai	N/A	N/A
7643225	12/2009	Tsai	N/A	N/A
7656585	12/2009	Powell et al.	N/A	N/A
7660047	12/2009	Travis et al.	N/A	N/A
7672024	12/2009	Kuan	N/A	N/A
7672055	12/2009	Amitai	N/A	N/A
7672549	12/2009	Ghosh et al.	N/A	N/A
7675021	12/2009	Lapstun	N/A	N/A
7675684	12/2009	Weissman et al.	N/A	N/A
7691248	12/2009	Ikeda et al.	N/A	N/A
7710622	12/2009	Takabayashi et al.	N/A	N/A
7710654	12/2009	Ashkenazi et al.	N/A	N/A
7711228	12/2009	Noda et al.	N/A	N/A
7724441	12/2009	Amitai	N/A	N/A
7724442	12/2009	Amitai	N/A	N/A
7724443	12/2009	Amitai	N/A	N/A
7733571	12/2009	Li	N/A	N/A
7733572	12/2009	Brown et al.	N/A	N/A
7740387	12/2009	Schultz et al.	N/A	N/A
7747113	12/2009	Mukawa et al.	N/A	N/A
7751122	12/2009	Amitai	N/A	N/A
7751662	12/2009	Kleemann et al.	N/A	N/A
7764413	12/2009	Levola	N/A	N/A
7777819	12/2009	Simmonds	N/A	N/A
7778305	12/2009	Parriaux et al.	N/A	N/A
7778508	12/2009	Hirayama	N/A	N/A
7843642	12/2009	Shaoulov et al.	N/A	N/A
7847235	12/2009	Krupkin et al.	N/A	N/A
7864427	12/2010	Korenaga et al.	N/A	N/A
7865080	12/2010	Hecker et al.	N/A	N/A
7866869	12/2010	Karakawa	N/A	N/A
7872707	12/2010	Sutherland et al.	N/A	N/A
7872804	12/2010	Moon et al.	N/A	N/A
7884593	12/2010	Simmonds et al.	N/A	N/A
7884985	12/2010	Amitai et al.	N/A	N/A
7887186	12/2010	Watanabe	N/A	N/A
7903921	12/2010	Ostergard	N/A	N/A
7907342	12/2010	Simmonds et al.	N/A	N/A
7920787	12/2010	Gentner et al.	N/A	N/A
7928862	12/2010	Matthews	N/A	N/A
7936513	12/2010	Wu et al.	N/A	N/A
7936519	12/2010	Mukawa et al.	N/A	N/A

7944428	12/2010	Travis	N/A	N/A
7944616	12/2010	Mukawa	N/A	N/A
7949214	12/2010	DeJong et al.	N/A	N/A
D640310	12/2010	Suzuki et al.	N/A	N/A
7961117	12/2010	Zimmerman et al.	N/A	N/A
7969644	12/2010	Tilleman et al.	N/A	N/A
7969657	12/2010	Cakmakci et al.	N/A	N/A
7970246	12/2010	Travis et al.	N/A	N/A
7976208	12/2010	Travis	N/A	N/A
7984884	12/2010	Iliev et al.	N/A	N/A
7999982	12/2010	Endo et al.	N/A	N/A
8000020	12/2010	Amitai et al.	N/A	N/A
8000491	12/2010	Brodkin et al.	N/A	N/A
8004765	12/2010	Amitai	N/A	N/A
8014050	12/2010	McGrew	N/A	N/A
8016475	12/2010	Travis	N/A	N/A
8018579	12/2010	Krah	N/A	N/A
8022942	12/2010	Bathiche et al.	N/A	N/A
8023783	12/2010	Mukawa et al.	N/A	N/A
RE42992	12/2010	David	N/A	N/A
8073296	12/2010	Mukawa et al.	N/A	N/A
8077274	12/2010	Sutherland et al.	N/A	N/A
8079713	12/2010	Ashkenazi	N/A	N/A
8082222	12/2010	Rangarajan et al.	N/A	N/A
8086030	12/2010	Gordon et al.	N/A	N/A
8089568	12/2011	Brown et al.	N/A	N/A
8093451	12/2011	Spangenberg et al.	N/A	N/A
8098439	12/2011	Amitai et al.	N/A	N/A
8107023	12/2011	Simmonds et al.	N/A	N/A
8107780	12/2011	Simmonds	N/A	N/A
8120548	12/2011	Barber	N/A	N/A
8120848	12/2011	Isano	N/A	N/A
8132948	12/2011	Owen et al.	N/A	N/A
8132976	12/2011	Odell et al.	N/A	N/A
8134434	12/2011	Diederichs et al.	N/A	N/A
8136690	12/2011	Fang et al.	N/A	N/A
8137981	12/2011	Andrew et al.	N/A	N/A
8142016	12/2011	Legerton et al.	N/A	N/A
8149086	12/2011	Klein et al.	N/A	N/A
8152315	12/2011	Travis et al.	N/A	N/A
8152353	12/2011	Yang et al.	N/A	N/A
8155489	12/2011	Saarikko et al.	N/A	N/A
8159752	12/2011	Wertheim et al.	N/A	N/A
8160409	12/2011	Large	N/A	N/A
8160411	12/2011	Levola et al.	N/A	N/A
D659137	12/2011	Matsumoto	N/A	N/A
8167173	12/2011	Simmonds et al.	N/A	N/A
8186874	12/2011	Sinbar et al.	N/A	N/A
8188925	12/2011	DeJean	N/A	N/A
8189263	12/2011	Wang et al.	N/A	N/A

8189973	12/2011	Travis et al.	N/A	N/A
D661334	12/2011	Cho et al.	N/A	N/A
D661335	12/2011	Jeon	N/A	N/A
8194325	12/2011	Levola et al.	N/A	N/A
8199803	12/2011	Hauske et al.	N/A	N/A
8213065	12/2011	Mukawa	N/A	N/A
8213755	12/2011	Mukawa et al.	N/A	N/A
8220966	12/2011	Mukawa	N/A	N/A
8224133	12/2011	Popovich et al.	N/A	N/A
8233204	12/2011	Robbins et al.	N/A	N/A
8253914	12/2011	Kajiya et al.	N/A	N/A
8254031	12/2011	Levola	N/A	N/A
8264498	12/2011	Vanderkamp et al.	N/A	N/A
8294749	12/2011	Cable	N/A	N/A
8295710	12/2011	Marcus	N/A	N/A
8301031	12/2011	Gentner et al.	N/A	N/A
8305577	12/2011	Kivioja et al.	N/A	N/A
8306423	12/2011	Gottwald et al.	N/A	N/A
8310327	12/2011	Willers et al.	N/A	N/A
8314819	12/2011	Kimmel et al.	N/A	N/A
8314993	12/2011	Levola et al.	N/A	N/A
8320032	12/2011	Levola	N/A	N/A
8321810	12/2011	Heintze	N/A	N/A
8325166	12/2011	Akutsu et al.	N/A	N/A
8329773	12/2011	Fäcke et al.	N/A	N/A
8335040	12/2011	Mukawa et al.	N/A	N/A
8335414	12/2011	Zinoviev et al.	N/A	N/A
D673996	12/2012	Kim et al.	N/A	N/A
8351744	12/2012	Travis et al.	N/A	N/A
8354640	12/2012	Hamre et al.	N/A	N/A
8354806	12/2012	Travis et al.	N/A	N/A
8355610	12/2012	Simmonds	N/A	N/A
8369019	12/2012	Baker et al.	N/A	N/A
8376548	12/2012	Schultz	N/A	N/A
8382293	12/2012	Phillips, III et al.	N/A	N/A
8384504	12/2012	Diederichs et al.	N/A	N/A
8384694	12/2012	Powell et al.	N/A	N/A
8384730	12/2012	Vanderkamp et al.	N/A	N/A
8396339	12/2012	Mukawa et al.	N/A	N/A
8396341	12/2012	Lee et al.	N/A	N/A
8398242	12/2012	Yamamoto et al.	N/A	N/A
8403490	12/2012	Sugiyama et al.	N/A	N/A
8422840	12/2012	Large	N/A	N/A
8427439	12/2012	Larsen et al.	N/A	N/A
8432363	12/2012	Saarikko et al.	N/A	N/A
8432372	12/2012	Butler et al.	N/A	N/A
8432614	12/2012	Amitai	N/A	N/A
8441731	12/2012	Sprague	N/A	N/A
8447365	12/2012	Imanuel	N/A	N/A
8466953	12/2012	Levola	N/A	N/A

8472119	12/2012	Kelly	N/A	N/A
8472120	12/2012	Border et al.	N/A	N/A
8477261	12/2012	Travis et al.	N/A	N/A
8481130	12/2012	Harding et al.	N/A	N/A
8482858	12/2012	Sprague	N/A	N/A
8488246	12/2012	Border et al.	N/A	N/A
8491121	12/2012	Tilleman et al.	N/A	N/A
8491136	12/2012	Travis et al.	N/A	N/A
8493366	12/2012	Bathiche et al.	N/A	N/A
8493662	12/2012	Noui	N/A	N/A
8494229	12/2012	Jarvenpaa et al.	N/A	N/A
8508848	12/2012	Saarikko	N/A	N/A
8520309	12/2012	Sprague	N/A	N/A
D691192	12/2012	Stanley et al.	N/A	N/A
8547638	12/2012	Levola	N/A	N/A
8548290	12/2012	Travers et al.	N/A	N/A
8565560	12/2012	Popovich et al.	N/A	N/A
D694310	12/2012	Cho et al.	N/A	N/A
D694311	12/2012	Cho et al.	N/A	N/A
8578038	12/2012	Kaikuranta et al.	N/A	N/A
8581831	12/2012	Travis	N/A	N/A
8582206	12/2012	Travis	N/A	N/A
8593734	12/2012	Laakkonen	N/A	N/A
8611014	12/2012	Valera et al.	N/A	N/A
8619062	12/2012	Powell et al.	N/A	N/A
D697130	12/2013	Lövgren	N/A	N/A
8633786	12/2013	Ermolov et al.	N/A	N/A
8634120	12/2013	Popovich et al.	N/A	N/A
8634139	12/2013	Brown et al.	N/A	N/A
8639072	12/2013	Popovich et al.	N/A	N/A
8643691	12/2013	Rosenfeld et al.	N/A	N/A
8643948	12/2013	Amitai et al.	N/A	N/A
8649099	12/2013	Schultz et al.	N/A	N/A
8654420	12/2013	Simmonds	N/A	N/A
8659826	12/2013	Brown et al.	N/A	N/A
D701206	12/2013	Luckey et al.	N/A	N/A
8670029	12/2013	McEldowney	N/A	N/A
8693087	12/2013	Nowatzyk et al.	N/A	N/A
8698705	12/2013	Burke	N/A	N/A
8731350	12/2013	Lin et al.	N/A	N/A
8736802	12/2013	Kajiya et al.	N/A	N/A
8736963	12/2013	Robbins et al.	N/A	N/A
8742952	12/2013	Bold	N/A	N/A
8746008	12/2013	Mauritsen et al.	N/A	N/A
8749886	12/2013	Gupta	N/A	N/A
8749890	12/2013	Wood et al.	N/A	N/A
8767294	12/2013	Chen et al.	N/A	N/A
8786923	12/2013	Chuang et al.	N/A	N/A
8810600	12/2013	Bohn et al.	N/A	N/A
8810913	12/2013	Simmonds et al.	N/A	N/A

8810914	12/2013	Amitai	N/A	N/A
8814691	12/2013	Haddick et al.	N/A	N/A
8816578	12/2013	Peng et al.	N/A	N/A
8817350	12/2013	Robbins et al.	N/A	N/A
8824836	12/2013	Sugiyama	N/A	N/A
8830143	12/2013	Pitchford et al.	N/A	N/A
8830584	12/2013	Saarikko et al.	N/A	N/A
8830588	12/2013	Brown et al.	N/A	N/A
8842368	12/2013	Simmonds et al.	N/A	N/A
8859412	12/2013	Jain	N/A	N/A
8872435	12/2013	Kreitzer et al.	N/A	N/A
8873149	12/2013	Bohn et al.	N/A	N/A
8873150	12/2013	Amitai	N/A	N/A
D718304	12/2013	Heinrich	N/A	N/A
D718366	12/2013	Mehin et al.	N/A	N/A
8885112	12/2013	Popovich et al.	N/A	N/A
8885997	12/2013	Nguyen et al.	N/A	N/A
8903207	12/2013	Brown et al.	N/A	N/A
8906088	12/2013	Pugh et al.	N/A	N/A
8913324	12/2013	Schrader	N/A	N/A
8913865	12/2013	Bennett	N/A	N/A
8917453	12/2013	Bohn	N/A	N/A
8917962	12/2013	Nichol et al.	N/A	N/A
8929589	12/2014	Publicover et al.	N/A	N/A
8933144	12/2014	Enomoto et al.	N/A	N/A
8934743	12/2014	Nishiwaki et al.	N/A	N/A
8937771	12/2014	Robbins et al.	N/A	N/A
8937772	12/2014	Burns et al.	N/A	N/A
8938141	12/2014	Magnusson	N/A	N/A
8950867	12/2014	Macnamara	N/A	N/A
8964298	12/2014	Haddick et al.	N/A	N/A
8965152	12/2014	Simmonds	N/A	N/A
D725102	12/2014	Lee et al.	N/A	N/A
8985803	12/2014	Bohn	N/A	N/A
8989535	12/2014	Robbins	N/A	N/A
D726180	12/2014	Roat et al.	N/A	N/A
9019595	12/2014	Jain	N/A	N/A
9025253	12/2014	Hadad et al.	N/A	N/A
9035344	12/2014	Jain	N/A	N/A
D733709	12/2014	Kawai	N/A	N/A
9075184	12/2014	Popovich et al.	N/A	N/A
9081178	12/2014	Simmonds et al.	N/A	N/A
9097890	12/2014	Miller et al.	N/A	N/A
9103978	12/2014	Nishiwaki et al.	N/A	N/A
9122015	12/2014	Shimizu	N/A	N/A
9128226	12/2014	Fattal et al.	N/A	N/A
9129295	12/2014	Border et al.	N/A	N/A
9164290	12/2014	Robbins et al.	N/A	N/A
9176324	12/2014	Scherer et al.	N/A	N/A
9188717	12/2014	Nishiwaki	N/A	N/A

9201270	12/2014	Fattal et al.	N/A	N/A
9215293	12/2014	Miller	N/A	N/A
D746896	12/2015	Markovitz et al.	N/A	N/A
9239507	12/2015	Chen et al.	N/A	N/A
9244275	12/2015	Li	N/A	N/A
9244280	12/2015	Tiana et al.	N/A	N/A
9244281	12/2015	Zimmerman et al.	N/A	N/A
D749074	12/2015	Cazalet et al.	N/A	N/A
9253359	12/2015	Takahashi	N/A	N/A
9269854	12/2015	Jain	N/A	N/A
D751551	12/2015	Ho et al.	N/A	N/A
D752129	12/2015	Lee et al.	N/A	N/A
9274338	12/2015	Robbins et al.	N/A	N/A
9274339	12/2015	Brown et al.	N/A	N/A
9274349	12/2015	Popovich et al.	N/A	N/A
D754782	12/2015	Kokinakis et al.	N/A	N/A
9310566	12/2015	Valera et al.	N/A	N/A
9316786	12/2015	Nishiwaki et al.	N/A	N/A
9329325	12/2015	Simmonds et al.	N/A	N/A
9335548	12/2015	Cakmakci et al.	N/A	N/A
9335604	12/2015	Popovich et al.	N/A	N/A
9341846	12/2015	Popovich et al.	N/A	N/A
9354366	12/2015	Jain	N/A	N/A
9366862	12/2015	Haddick et al.	N/A	N/A
9366864	12/2015	Brown et al.	N/A	N/A
9372347	12/2015	Levola et al.	N/A	N/A
9377623	12/2015	Robbins et al.	N/A	N/A
9377852	12/2015	Shapiro et al.	N/A	N/A
9389415	12/2015	Fattal et al.	N/A	N/A
9400395	12/2015	Travers et al.	N/A	N/A
9423360	12/2015	Kostamo et al.	N/A	N/A
9429692	12/2015	Saarikko et al.	N/A	N/A
9431794	12/2015	Jain	N/A	N/A
9435961	12/2015	Jiang	N/A	N/A
9456744	12/2015	Popovich et al.	N/A	N/A
9459451	12/2015	Saarikko et al.	N/A	N/A
9464779	12/2015	Popovich et al.	N/A	N/A
9465213	12/2015	Simmonds	N/A	N/A
9465227	12/2015	Popovich et al.	N/A	N/A
9484482	12/2015	Hsu et al.	N/A	N/A
9494799	12/2015	Robbins et al.	N/A	N/A
9507150	12/2015	Stratton et al.	N/A	N/A
9513480	12/2015	Saarikko et al.	N/A	N/A
9516193	12/2015	Aramaki	N/A	N/A
9519089	12/2015	Brown et al.	N/A	N/A
9523852	12/2015	Brown et al.	N/A	N/A
9535253	12/2016	Levola et al.	N/A	N/A
9541383	12/2016	Abovitz et al.	N/A	N/A
9541763	12/2016	Heberlein et al.	N/A	N/A
9547174	12/2016	Gao et al.	N/A	N/A

9551468	12/2016	Jones	N/A	N/A
9551874	12/2016	Amitai	N/A	N/A
9551880	12/2016	Amitai	N/A	N/A
9599813	12/2016	Stratton et al.	N/A	N/A
9612403	12/2016	Abovitz et al.	N/A	N/A
9632226	12/2016	Waldern et al.	N/A	N/A
9635352	12/2016	Henry et al.	N/A	N/A
9648313	12/2016	Henry et al.	N/A	N/A
9651368	12/2016	Abovitz et al.	N/A	N/A
9664824	12/2016	Simmonds et al.	N/A	N/A
9664910	12/2016	Mansharof et al.	N/A	N/A
9671612	12/2016	Kress et al.	N/A	N/A
9674413	12/2016	Tiana et al.	N/A	N/A
9678345	12/2016	Melzer et al.	N/A	N/A
9679367	12/2016	Wald	N/A	N/A
9715067	12/2016	Brown et al.	N/A	N/A
9715110	12/2016	Brown et al.	N/A	N/A
D793468	12/2016	Yu et al.	N/A	N/A
D795865	12/2016	Porter et al.	N/A	N/A
D795866	12/2016	Porter et al.	N/A	N/A
9726540	12/2016	Popovich et al.	N/A	N/A
9727772	12/2016	Popovich et al.	N/A	N/A
9733475	12/2016	Brown et al.	N/A	N/A
9739950	12/2016	Sqalli et al.	N/A	N/A
9746688	12/2016	Popovich et al.	N/A	N/A
9754507	12/2016	Wenger et al.	N/A	N/A
9762895	12/2016	Henry et al.	N/A	N/A
9766465	12/2016	Tiana et al.	N/A	N/A
9785231	12/2016	Zimmerman	N/A	N/A
9791694	12/2016	Haverkamp et al.	N/A	N/A
9791696	12/2016	Woltman et al.	N/A	N/A
9791703	12/2016	Vallius et al.	N/A	N/A
9804316	12/2016	Drolet et al.	N/A	N/A
9804389	12/2016	Popovich et al.	N/A	N/A
9823423	12/2016	Waldern et al.	N/A	N/A
9857605	12/2017	Popovich et al.	N/A	N/A
9874931	12/2017	Koenck et al.	N/A	N/A
9899800	12/2017	Ferrotti et al.	N/A	N/A
9915825	12/2017	Robbins et al.	N/A	N/A
9933684	12/2017	Brown et al.	N/A	N/A
9939577	12/2017	Inoue et al.	N/A	N/A
9939628	12/2017	Basset et al.	N/A	N/A
9977247	12/2017	Brown et al.	N/A	N/A
D827641	12/2017	Morisawa	N/A	N/A
10088686	12/2017	Robbins et al.	N/A	N/A
10089516	12/2017	Popovich et al.	N/A	N/A
10095045	12/2017	Robbins et al.	N/A	N/A
10107966	12/2017	Horibe et al.	N/A	N/A
10114220	12/2017	Grey et al.	N/A	N/A
10156681	12/2017	Waldern et al.	N/A	N/A

10162181	12/2017	Webster et al.	N/A	N/A
10185154	12/2018	Popovich et al.	N/A	N/A
D840454	12/2018	Han et al.	N/A	N/A
10197804	12/2018	Stenberg et al.	N/A	N/A
10209517	12/2018	Popovich et al.	N/A	N/A
10216061	12/2018	Popovich et al.	N/A	N/A
10234696	12/2018	Popovich et al.	N/A	N/A
10241330	12/2018	Popovich et al.	N/A	N/A
10241332	12/2018	Vallius	N/A	N/A
10281725	12/2018	Yokoyama	N/A	N/A
10330777	12/2018	Popovich et al.	N/A	N/A
10345519	12/2018	Miller et al.	N/A	N/A
10359736	12/2018	Popovich et al.	N/A	N/A
D855687	12/2018	Villalpando	N/A	N/A
D859510	12/2018	Harmon et al.	N/A	N/A
10409144	12/2018	Popovich et al.	N/A	N/A
10423222	12/2018	Popovich et al.	N/A	N/A
10423813	12/2018	Popovich et al.	N/A	N/A
10437051	12/2018	Popovich et al.	N/A	N/A
10437064	12/2018	Popovich et al.	N/A	N/A
10444510	12/2018	Lee et al.	N/A	N/A
10459145	12/2018	Popovich et al.	N/A	N/A
10459311	12/2018	Popovich et al.	N/A	N/A
D871494	12/2018	Yamada et al.	N/A	N/A
D872170	12/2019	Evans et al.	N/A	N/A
D872794	12/2019	Wilkins	N/A	N/A
10527797	12/2019	Waldern et al.	N/A	N/A
10532594	12/2019	Akahane et al.	N/A	N/A
10545346	12/2019	Waldern et al.	N/A	N/A
10569449	12/2019	Curts et al.	N/A	N/A
10578876	12/2019	Lam et al.	N/A	N/A
10591756	12/2019	Popovich et al.	N/A	N/A
10598938	12/2019	Huang et al.	N/A	N/A
D880575	12/2019	Thixton	N/A	N/A
10613268	12/2019	Colburn et al.	N/A	N/A
10642058	12/2019	Popovich et al.	N/A	N/A
10649119	12/2019	Mohanty et al.	N/A	N/A
10670876	12/2019	Popovich et al.	N/A	N/A
10678053	12/2019	Waldern et al.	N/A	N/A
10690831	12/2019	Calafiore	N/A	N/A
10690915	12/2019	Popovich et al.	N/A	N/A
10690916	12/2019	Popovich et al.	N/A	N/A
10698214	12/2019	Vallius et al.	N/A	N/A
10705281	12/2019	Fattal et al.	N/A	N/A
10712571	12/2019	Popovich et al.	N/A	N/A
10725304	12/2019	Ratnam et al.	N/A	N/A
10725312	12/2019	Popovich et al.	N/A	N/A
10732351	12/2019	Colburn et al.	N/A	N/A
10732569	12/2019	Waldern et al.	N/A	N/A
10747982	12/2019	Popovich et al.	N/A	N/A

10823887	12/2019	Calafiore et al.	N/A	N/A
10859768	12/2019	Popovich et al.	N/A	N/A
10890707	12/2020	Waldern et al.	N/A	N/A
10942430	12/2020	Waldern et al.	N/A	N/A
10983257	12/2020	Colburn et al.	N/A	N/A
11103892	12/2020	Liao et al.	N/A	N/A
11106048	12/2020	Popovich et al.	N/A	N/A
11107972	12/2020	Diest et al.	N/A	N/A
11137603	12/2020	Zhang	N/A	N/A
11175512	12/2020	Waldern et al.	N/A	N/A
11194098	12/2020	Waldern et al.	N/A	N/A
11194162	12/2020	Waldern et al.	N/A	N/A
11204540	12/2020	Popovich et al.	N/A	N/A
11243333	12/2021	Ouderkirk et al.	N/A	N/A
11256155	12/2021	Popovich et al.	N/A	N/A
11281013	12/2021	Popovich et al.	N/A	N/A
11300795	12/2021	Stanley et al.	N/A	N/A
11306193	12/2021	Lane et al.	N/A	N/A
11307357	12/2021	Mohanty	N/A	N/A
11340386	12/2021	Ouderkirk et al.	N/A	N/A
11378732	12/2021	Waldern et al.	N/A	N/A
11391950	12/2021	Calafiore	N/A	N/A
11460621	12/2021	Popovich et al.	N/A	N/A
11543594	12/2022	Grant et al.	N/A	N/A
11604314	12/2022	Popovich et al.	N/A	N/A
11703645	12/2022	Waldern et al.	N/A	N/A
11747568	12/2022	Waldern et al.	N/A	N/A
12140764	12/2023	Popovich et al.	N/A	N/A
2001/0024177	12/2000	Popovich	N/A	N/A
2001/0033400	12/2000	Sutherland et al.	N/A	N/A
2001/0036012	12/2000	Nakai et al.	N/A	N/A
2001/0043163	12/2000	Waldern et al.	N/A	N/A
2001/0046142	12/2000	Van Santen et al.	N/A	N/A
2001/0050756	12/2000	Lipton et al.	N/A	N/A
2002/0003509	12/2001	Lipton et al.	N/A	N/A
2002/0009299	12/2001	Lipton	N/A	N/A
2002/0011969	12/2001	Lipton et al.	N/A	N/A
2002/0012064	12/2001	Yamaguchi	N/A	N/A
2002/0018040	12/2001	Aye et al.	N/A	N/A
2002/0021407	12/2001	Elliott	N/A	N/A
2002/0021461	12/2001	Ono et al.	N/A	N/A
2002/0036825	12/2001	Lipton et al.	N/A	N/A
2002/0047837	12/2001	Suyama et al.	N/A	N/A
2002/0075240	12/2001	Lieberman et al.	N/A	N/A
2002/0093701	12/2001	Zhang et al.	N/A	N/A
2002/0110077	12/2001	Drobot et al.	N/A	N/A
2002/0126332	12/2001	Popovich	N/A	N/A
2002/0127497	12/2001	Brown et al.	N/A	N/A
2002/0131175	12/2001	Yagi et al.	N/A	N/A
2002/0150032	12/2001	Nishiuchi et al.	N/A	N/A

2002/0150337	12/2001	Fujimaki	N/A	N/A
2002/0167462	12/2001	Lewis et al.	N/A	N/A
2002/0196332	12/2001	Lipton et al.	N/A	N/A
2003/0007070	12/2002	Lipton et al.	N/A	N/A
2003/0025881	12/2002	Hwang	N/A	N/A
2003/0030912	12/2002	Gleckman et al.	N/A	N/A
2003/0038912	12/2002	Broer et al.	N/A	N/A
2003/0039442	12/2002	Bond et al.	N/A	N/A
2003/0063042	12/2002	Friesem et al.	N/A	N/A
2003/0063884	12/2002	Smith et al.	N/A	N/A
2003/0067685	12/2002	Niv	N/A	N/A
2003/0076590	12/2002	Kramer	N/A	N/A
2003/0086670	12/2002	Moridaira et al.	N/A	N/A
2003/0107809	12/2002	Chen et al.	N/A	N/A
2003/0129542	12/2002	Shih et al.	N/A	N/A
2003/0149346	12/2002	Arnone et al.	N/A	N/A
2003/0175004	12/2002	Garito et al.	N/A	N/A
2003/0197154	12/2002	Manabe et al.	N/A	N/A
2003/0197157	12/2002	Sutherland et al.	N/A	N/A
2003/0202247	12/2002	Niv et al.	N/A	N/A
2003/0206329	12/2002	Ikeda et al.	N/A	N/A
2003/0228019	12/2002	Eichler et al.	N/A	N/A
2004/0004767	12/2003	Song	N/A	N/A
2004/0004989	12/2003	Shigeoka	N/A	N/A
2004/0012833	12/2003	Newswanger et al.	N/A	N/A
2004/0047938	12/2003	Kosuga et al.	N/A	N/A
2004/0057138	12/2003	Tanijiri et al.	N/A	N/A
2004/0075830	12/2003	Miyake et al.	N/A	N/A
2004/0089842	12/2003	Suthehrland et al.	N/A	N/A
2004/0108971	12/2003	Waldern et al.	N/A	N/A
2004/0109234	12/2003	Levola	N/A	N/A
2004/0112862	12/2003	Willson et al.	N/A	N/A
2004/0125454	12/2003	Kawasaki et al.	N/A	N/A
2004/0130797	12/2003	Leigh	N/A	N/A
2004/0141217	12/2003	Endo et al.	N/A	N/A
2004/0156008	12/2003	Reznikov et al.	N/A	N/A
2004/0174348	12/2003	David	N/A	N/A
2004/0175627	12/2003	Sutherland et al.	N/A	N/A
2004/0179764	12/2003	Melikechi et al.	N/A	N/A
2004/0184156	12/2003	Gunn, III et al.	N/A	N/A
2004/0188617	12/2003	Devitt et al.	N/A	N/A
2004/0200368	12/2003	Ogino et al.	N/A	N/A
2004/0208446	12/2003	Bond et al.	N/A	N/A
2004/0208466	12/2003	Mossberg et al.	N/A	N/A
2004/0225025	12/2003	Sullivan et al.	N/A	N/A
2004/0263969	12/2003	Lipton et al.	N/A	N/A
2004/0263971	12/2003	Lipton et al.	N/A	N/A
2005/0018304	12/2004	Lipton et al.	N/A	N/A
2005/0047705	12/2004	Domash et al.	N/A	N/A
2005/0079663	12/2004	Masutani et al.	N/A	N/A

2005/0083564	12/2004	Mallya et al.	N/A	N/A
2005/0105909	12/2004	Stone	N/A	N/A
2005/0122395	12/2004	Lipton et al.	N/A	N/A
2005/0134404	12/2004	Kajiya et al.	N/A	N/A
2005/0135747	12/2004	Greiner et al.	N/A	N/A
2005/0136260	12/2004	Garcia	N/A	N/A
2005/0141066	12/2004	Ouchi	N/A	N/A
2005/0141811	12/2004	Yang et al.	N/A	N/A
2005/0174321	12/2004	Ikeda et al.	N/A	N/A
2005/0180687	12/2004	Amitai	N/A	N/A
2005/0195276	12/2004	Lipton et al.	N/A	N/A
2005/0218377	12/2004	Lawandy	N/A	N/A
2005/0231774	12/2004	Hayashi et al.	N/A	N/A
2005/0232530	12/2004	Kekas	N/A	N/A
2005/0254752	12/2004	Domash et al.	N/A	N/A
2005/0259217	12/2004	Lin et al.	N/A	N/A
2005/0259302	12/2004	Metz et al.	N/A	N/A
2005/0259944	12/2004	Anderson et al.	N/A	N/A
2005/0265585	12/2004	Rowe	N/A	N/A
2005/0269481	12/2004	David et al.	N/A	N/A
2005/0271258	12/2004	Rowe	N/A	N/A
2005/0286133	12/2004	Lipton	N/A	N/A
2006/0002274	12/2005	Kihara et al.	N/A	N/A
2006/0012878	12/2005	Lipton et al.	N/A	N/A
2006/0013977	12/2005	Duke et al.	N/A	N/A
2006/0043938	12/2005	O'Gorman et al.	N/A	N/A
2006/0055993	12/2005	Kobayashi et al.	N/A	N/A
2006/0093012	12/2005	Singh et al.	N/A	N/A
2006/0093793	12/2005	Miyakawa et al.	N/A	N/A
2006/0114564	12/2005	Sutherland et al.	N/A	N/A
2006/0119837	12/2005	Raguin et al.	N/A	N/A
2006/0119916	12/2005	Sutherland et al.	N/A	N/A
2006/0126179	12/2005	Levola	N/A	N/A
2006/0132914	12/2005	Weiss et al.	N/A	N/A
2006/0142455	12/2005	Agarwal et al.	N/A	N/A
2006/0146422	12/2005	Koike	N/A	N/A
2006/0159864	12/2005	Natarajan et al.	N/A	N/A
2006/0164593	12/2005	Peyghambarian et al.	N/A	N/A
2006/0171647	12/2005	Ye et al.	N/A	N/A
2006/0177180	12/2005	Tazawa et al.	N/A	N/A
2006/0181683	12/2005	Bhowmik et al.	N/A	N/A
2006/0191293	12/2005	Kuczma	N/A	N/A
2006/0215244	12/2005	Yosha et al.	N/A	N/A
2006/0215976	12/2005	Singh et al.	N/A	N/A
2006/0221063	12/2005	Ishihara	N/A	N/A
2006/0221448	12/2005	Nivon et al.	N/A	N/A
2006/0228073	12/2005	Mukawa et al.	N/A	N/A
2006/0262250	12/2005	Hobbs	N/A	N/A
2006/0268104	12/2005	Cowan et al.	N/A	N/A

2006/0268412	12/2005	Downing et al.	N/A	N/A
2006/0279662	12/2005	Kapellner et al.	N/A	N/A
2006/0284974	12/2005	Lipton et al.	N/A	N/A
2006/0285205	12/2005	Lipton et al.	N/A	N/A
2006/0291021	12/2005	Mukawa	N/A	N/A
2006/0291052	12/2005	Lipton et al.	N/A	N/A
2006/0292493	12/2005	Shinotsuka et al.	N/A	N/A
2007/0012777	12/2006	Tsikos et al.	N/A	N/A
2007/0019152	12/2006	Caputo et al.	N/A	N/A
2007/0019297	12/2006	Stewart et al.	N/A	N/A
2007/0034600	12/2006	Willson et al.	N/A	N/A
2007/0041684	12/2006	Popovich et al.	N/A	N/A
2007/0045596	12/2006	King et al.	N/A	N/A
2007/0052929	12/2006	Allman et al.	N/A	N/A
2007/0053032	12/2006	Popovich	N/A	N/A
2007/0070476	12/2006	Yamada et al.	N/A	N/A
2007/0070504	12/2006	Akutsu et al.	N/A	N/A
2007/0070859	12/2006	Hirayama	N/A	N/A
2007/0089625	12/2006	Grinberg et al.	N/A	N/A
2007/0097502	12/2006	Lipton et al.	N/A	N/A
2007/0109400	12/2006	Woodgate et al.	N/A	N/A
2007/0109401	12/2006	Lipton et al.	N/A	N/A
2007/0115553	12/2006	Chang-Hasnain et al.	N/A	N/A
2007/0116409	12/2006	Bryan et al.	N/A	N/A
2007/0127348	12/2006	Ooi et al.	N/A	N/A
2007/0133089	12/2006	Lipton et al.	N/A	N/A
2007/0133920	12/2006	Lee et al.	N/A	N/A
2007/0133983	12/2006	Traff	N/A	N/A
2007/0146624	12/2006	Duston et al.	N/A	N/A
2007/0146625	12/2006	Ooi et al.	N/A	N/A
2007/0154153	12/2006	Fomitchov et al.	N/A	N/A
2007/0160325	12/2006	Son et al.	N/A	N/A
2007/0177007	12/2006	Lipton et al.	N/A	N/A
2007/0182915	12/2006	Osawa et al.	N/A	N/A
2007/0183650	12/2006	Lipton et al.	N/A	N/A
2007/0188602	12/2006	Cowan et al.	N/A	N/A
2007/0188837	12/2006	Shimizu et al.	N/A	N/A
2007/0195409	12/2006	Yun et al.	N/A	N/A
2007/0206155	12/2006	Lipton	N/A	N/A
2007/0211164	12/2006	Olsen et al.	N/A	N/A
2007/0236560	12/2006	Lipton et al.	N/A	N/A
2007/0237456	12/2006	Blauvelt et al.	N/A	N/A
2007/0247687	12/2006	Handschy et al.	N/A	N/A
2007/0258138	12/2006	Cowan et al.	N/A	N/A
2007/0263169	12/2006	Lipton	N/A	N/A
2008/0001909	12/2007	Lim	N/A	N/A
2008/0018851	12/2007	Lipton et al.	N/A	N/A
2008/0024598	12/2007	Perlin et al.	N/A	N/A
2008/0043334	12/2007	Itzkovitch et al.	N/A	N/A

2008/0049100	12/2007	Lipton et al.	N/A	N/A
2008/0062259	12/2007	Lipton et al.	N/A	N/A
2008/0063808	12/2007	Stumpe et al.	N/A	N/A
2008/0089073	12/2007	Hikmet	N/A	N/A
2008/0106775	12/2007	Amitai et al.	N/A	N/A
2008/0106779	12/2007	Peterson et al.	N/A	N/A
2008/0117289	12/2007	Schowengerdt et al.	N/A	N/A
2008/0136916	12/2007	Wolff	N/A	N/A
2008/0136923	12/2007	Inbar et al.	N/A	N/A
2008/0138013	12/2007	Parriaux	N/A	N/A
2008/0143964	12/2007	Cowan et al.	N/A	N/A
2008/0143965	12/2007	Cowan et al.	N/A	N/A
2008/0149517	12/2007	Lipton et al.	N/A	N/A
2008/0151370	12/2007	Cook et al.	N/A	N/A
2008/0151379	12/2007	Amitai	N/A	N/A
2008/0186573	12/2007	Lipton	N/A	N/A
2008/0186574	12/2007	Robinson et al.	N/A	N/A
2008/0186604	12/2007	Amitai	N/A	N/A
2008/0193085	12/2007	Singh et al.	N/A	N/A
2008/0198471	12/2007	Amitai	N/A	N/A
2008/0225187	12/2007	Yamanaka	N/A	N/A
2008/0226281	12/2007	Lipton	N/A	N/A
2008/0239067	12/2007	Lipton	N/A	N/A
2008/0239068	12/2007	Lipton	N/A	N/A
2008/0273081	12/2007	Lipton	N/A	N/A
2008/0278812	12/2007	Amitai	N/A	N/A
2008/0285137	12/2007	Simmonds et al.	N/A	N/A
2008/0285140	12/2007	Amitai	N/A	N/A
2008/0297731	12/2007	Powell et al.	N/A	N/A
2008/0297807	12/2007	Feldman et al.	N/A	N/A
2008/0298649	12/2007	Ennis et al.	N/A	N/A
2008/0298740	12/2007	Hlousek et al.	N/A	N/A
2008/0303895	12/2007	Akka et al.	N/A	N/A
2008/0303896	12/2007	Lipton et al.	N/A	N/A
2008/0304111	12/2007	Queenan et al.	N/A	N/A
2008/0309586	12/2007	Vitale	N/A	N/A
2008/0316303	12/2007	Chiu et al.	N/A	N/A
2008/0316375	12/2007	Lipton et al.	N/A	N/A
2009/0010135	12/2008	Ushiro et al.	N/A	N/A
2009/0017424	12/2008	Yoeli et al.	N/A	N/A
2009/0019222	12/2008	Verma et al.	N/A	N/A
2009/0052017	12/2008	Sasaki	N/A	N/A
2009/0052046	12/2008	Amitai	N/A	N/A
2009/0052047	12/2008	Amitai	N/A	N/A
2009/0067774	12/2008	Magnusson	N/A	N/A
2009/0074356	12/2008	Sanchez et al.	N/A	N/A
2009/0097122	12/2008	Niv	N/A	N/A
2009/0097127	12/2008	Amitai	N/A	N/A
2009/0116790	12/2008	Mossberg et al.	N/A	N/A
2009/0121301	12/2008	Chang	N/A	N/A

2009/0122413	12/2008	Hoffman et al.	N/A	N/A
2009/0122414	12/2008	Amitai	N/A	N/A
2009/0128495	12/2008	Kong et al.	N/A	N/A
2009/0128781	12/2008	Li	N/A	N/A
2009/0128902	12/2008	Niv et al.	N/A	N/A
2009/0128911	12/2008	Itzkovitch et al.	N/A	N/A
2009/0136246	12/2008	Murakami	N/A	N/A
2009/0141324	12/2008	Mukawa	N/A	N/A
2009/0153437	12/2008	Aharoni	N/A	N/A
2009/0169152	12/2008	Oestergard	N/A	N/A
2009/0190222	12/2008	Simmonds et al.	N/A	N/A
2009/0213208	12/2008	Glatt	N/A	N/A
2009/0237804	12/2008	Amitai et al.	N/A	N/A
2009/0242021	12/2008	Petkie et al.	N/A	N/A
2009/0296218	12/2008	Ryytty	N/A	N/A
2009/0303599	12/2008	Levola	N/A	N/A
2009/0316246	12/2008	Asai et al.	N/A	N/A
2010/0014312	12/2009	Travis et al.	N/A	N/A
2010/0039796	12/2009	Mukawa	N/A	N/A
2010/0053565	12/2009	Mizushima et al.	N/A	N/A
2010/0060551	12/2009	Sugiyama et al.	N/A	N/A
2010/0060990	12/2009	Wertheim et al.	N/A	N/A
2010/0065726	12/2009	Zhong et al.	N/A	N/A
2010/0079841	12/2009	Levola	N/A	N/A
2010/0079865	12/2009	Saarikko et al.	N/A	N/A
2010/0084261	12/2009	Lee et al.	N/A	N/A
2010/0086256	12/2009	Ben Bakir et al.	N/A	N/A
2010/0092124	12/2009	Magnusson et al.	N/A	N/A
2010/0096562	12/2009	Klunder et al.	N/A	N/A
2010/0097674	12/2009	Kasazumi et al.	N/A	N/A
2010/0097820	12/2009	Owen et al.	N/A	N/A
2010/0103078	12/2009	Mukawa et al.	N/A	N/A
2010/0134534	12/2009	Seesselberg et al.	N/A	N/A
2010/0135615	12/2009	Ho et al.	N/A	N/A
2010/0136319	12/2009	Imai et al.	N/A	N/A
2010/0141555	12/2009	Rorberg et al.	N/A	N/A
2010/0141905	12/2009	Burke	N/A	N/A
2010/0149073	12/2009	Chaum et al.	N/A	N/A
2010/0165465	12/2009	Levola	N/A	N/A
2010/0165660	12/2009	Weber et al.	N/A	N/A
2010/0171680	12/2009	Lapidot et al.	N/A	N/A
2010/0177388	12/2009	Cohen et al.	N/A	N/A
2010/0202725	12/2009	Popovich et al.	N/A	N/A
2010/0214659	12/2009	Levola	N/A	N/A
2010/0220293	12/2009	Mizushima et al.	N/A	N/A
2010/0225834	12/2009	Li	N/A	N/A
2010/0225876	12/2009	Escuti et al.	N/A	N/A
2010/0231532	12/2009	Nho et al.	N/A	N/A
2010/0231693	12/2009	Levola	N/A	N/A
2010/0231705	12/2009	Yahav et al.	N/A	N/A

2010/0232003	12/2009	Baldy et al.	N/A	N/A
2010/0232016	12/2009	Landa et al.	N/A	N/A
2010/0245756	12/2009	Sugihara et al.	N/A	N/A
2010/0245757	12/2009	Sugihara et al.	N/A	N/A
2010/0246003	12/2009	Simmonds et al.	N/A	N/A
2010/0246004	12/2009	Simmonds	N/A	N/A
2010/0246993	12/2009	Rieger et al.	N/A	N/A
2010/0253987	12/2009	Leopold et al.	N/A	N/A
2010/0265117	12/2009	Weiss	N/A	N/A
2010/0277803	12/2009	Pockett et al.	N/A	N/A
2010/0284085	12/2009	Laakkonen	N/A	N/A
2010/0284090	12/2009	Simmonds	N/A	N/A
2010/0284180	12/2009	Popovich et al.	N/A	N/A
2010/0296163	12/2009	Saarikko	N/A	N/A
2010/0299814	12/2009	Celona et al.	N/A	N/A
2010/0315719	12/2009	Saarikko et al.	N/A	N/A
2010/0321781	12/2009	Levola et al.	N/A	N/A
2010/0322555	12/2009	Vermeulen et al.	N/A	N/A
2011/0001895	12/2010	Dahl	N/A	N/A
2011/0002143	12/2010	Saarikko et al.	N/A	N/A
2011/0013423	12/2010	Selbrede et al.	N/A	N/A
2011/0019250	12/2010	Aiki et al.	N/A	N/A
2011/0019874	12/2010	Jarvenpaa	382/103	A61B 3/113
2011/0026128	12/2010	Baker et al.	N/A	N/A
2011/0026774	12/2010	Flohr et al.	N/A	N/A
2011/0032602	12/2010	Rothenberg et al.	N/A	N/A
2011/0032618	12/2010	Handerek et al.	N/A	N/A
2011/0032706	12/2010	Mukawa	N/A	N/A
2011/0038024	12/2010	Wang et al.	N/A	N/A
2011/0050548	12/2010	Blumenfeld et al.	N/A	N/A
2011/0063604	12/2010	Hamre et al.	N/A	N/A
2011/0096401	12/2010	Levola	N/A	N/A
2011/0102711	12/2010	Sutherland et al.	N/A	N/A
2011/0109880	12/2010	Nummela	N/A	N/A
2011/0157707	12/2010	Tilleman et al.	N/A	N/A
2011/0164221	12/2010	Tilleman et al.	N/A	N/A
2011/0187293	12/2010	Travis et al.	N/A	N/A
2011/0211239	12/2010	Mukawa et al.	N/A	N/A
2011/0216255	12/2010	Miyauchi et al.	N/A	N/A
2011/0221656	12/2010	Haddick et al.	N/A	N/A
2011/0232211	12/2010	Farahi	N/A	N/A
2011/0235179	12/2010	Simmonds	N/A	N/A
2011/0235365	12/2010	McCollum et al.	N/A	N/A
2011/0236803	12/2010	Weiser et al.	N/A	N/A
2011/0238399	12/2010	Ophir et al.	N/A	N/A
2011/0242349	12/2010	Izuha et al.	N/A	N/A
2011/0242661	12/2010	Simmonds	N/A	N/A
2011/0242670	12/2010	Simmonds	N/A	N/A
2011/0249309	12/2010	McPheters et al.	N/A	N/A

2011/0274435	12/2010	Finì et al.	N/A	N/A
2011/0299075	12/2010	Meade et al.	N/A	N/A
2011/0310356	12/2010	Vallius	N/A	N/A
2012/0007979	12/2011	Schneider et al.	N/A	N/A
2012/0027347	12/2011	Mathal et al.	N/A	N/A
2012/0033306	12/2011	Valera et al.	N/A	N/A
2012/0044572	12/2011	Simmonds et al.	N/A	N/A
2012/0044573	12/2011	Simmonds et al.	N/A	N/A
2012/0062850	12/2011	Travis	N/A	N/A
2012/0062998	12/2011	Schultz et al.	N/A	N/A
2012/0067864	12/2011	Kusuda et al.	N/A	N/A
2012/0075168	12/2011	Osterhout et al.	N/A	N/A
2012/0081789	12/2011	Mukawa et al.	N/A	N/A
2012/0092632	12/2011	McLeod et al.	N/A	N/A
2012/0099203	12/2011	Boubis et al.	N/A	N/A
2012/0105634	12/2011	Meidan et al.	N/A	N/A
2012/0105740	12/2011	Jannard et al.	N/A	N/A
2012/0120493	12/2011	Simmonds et al.	N/A	N/A
2012/0127577	12/2011	Desserouer	N/A	N/A
2012/0162549	12/2011	Gao et al.	N/A	N/A
2012/0162764	12/2011	Shimizu	N/A	N/A
2012/0176665	12/2011	Song et al.	N/A	N/A
2012/0183888	12/2011	Oliveira et al.	N/A	N/A
2012/0194420	12/2011	Osterhout et al.	N/A	N/A
2012/0200532	12/2011	Powell et al.	N/A	N/A
2012/0206811	12/2011	Mukawa et al.	N/A	N/A
2012/0206937	12/2011	Travis et al.	N/A	N/A
2012/0207432	12/2011	Travis et al.	N/A	N/A
2012/0207434	12/2011	Large	N/A	N/A
2012/0214089	12/2011	Hönel et al.	N/A	N/A
2012/0214090	12/2011	Weiser et al.	N/A	N/A
2012/0218481	12/2011	Popovich et al.	N/A	N/A
2012/0224062	12/2011	Lacoste et al.	N/A	N/A
2012/0235884	12/2011	Miller et al.	N/A	N/A
2012/0235886	12/2011	Border et al.	N/A	N/A
2012/0235900	12/2011	Border et al.	N/A	N/A
2012/0242661	12/2011	Takagi et al.	N/A	N/A
2012/0280956	12/2011	Yamamoto et al.	N/A	N/A
2012/0281943	12/2011	Popovich et al.	N/A	N/A
2012/0290973	12/2011	Robertson et al.	N/A	N/A
2012/0294037	12/2011	Holman et al.	N/A	N/A
2012/0300311	12/2011	Simmonds et al.	N/A	N/A
2012/0320460	12/2011	Levola	N/A	N/A
2012/0326950	12/2011	Park et al.	N/A	N/A
2012/0328234	12/2011	Lu et al.	N/A	N/A
2013/0016324	12/2012	Travis	N/A	N/A
2013/0016362	12/2012	Gong et al.	N/A	N/A
2013/0021392	12/2012	Travis	N/A	N/A
2013/0021586	12/2012	Lippey	N/A	N/A
2013/0027006	12/2012	Holloway et al.	N/A	N/A

2013/0033485	12/2012	Kollin et al.	N/A	N/A
2013/0039619	12/2012	Laughlin	N/A	N/A
2013/0044376	12/2012	Valera et al.	N/A	N/A
2013/0051730	12/2012	Travers et al.	N/A	N/A
2013/0059233	12/2012	Askham	N/A	N/A
2013/0069850	12/2012	Mukawa et al.	N/A	N/A
2013/0077049	12/2012	Bohn	N/A	N/A
2013/0088637	12/2012	Duparre	N/A	N/A
2013/0093893	12/2012	Schofield et al.	N/A	N/A
2013/0101253	12/2012	Popovich et al.	N/A	N/A
2013/0107186	12/2012	Ando et al.	N/A	N/A
2013/0117377	12/2012	Miller	N/A	N/A
2013/0125027	12/2012	Abovitz et al.	N/A	N/A
2013/0128230	12/2012	Macnamara	N/A	N/A
2013/0138275	12/2012	Nauman et al.	N/A	N/A
2013/0141937	12/2012	Katsuta et al.	N/A	N/A
2013/0143336	12/2012	Jain	N/A	N/A
2013/0163089	12/2012	Bohn	N/A	N/A
2013/0163928	12/2012	Wang et al.	N/A	N/A
2013/0170031	12/2012	Bohn et al.	N/A	N/A
2013/0176704	12/2012	Lanman et al.	N/A	N/A
2013/0184904	12/2012	Gadzinski	N/A	N/A
2013/0200710	12/2012	Robbins	N/A	N/A
2013/0207887	12/2012	Raffle et al.	N/A	N/A
2013/0224634	12/2012	Berneth et al.	N/A	N/A
2013/0229717	12/2012	Amitai	N/A	N/A
2013/0249895	12/2012	Westerinen et al.	N/A	N/A
2013/0250207	12/2012	Bohn	N/A	N/A
2013/0250380	12/2012	Fujikawa et al.	N/A	N/A
2013/0250430	12/2012	Robbins et al.	N/A	N/A
2013/0250431	12/2012	Robbins et al.	N/A	N/A
2013/0257848	12/2012	Westerinen et al.	N/A	N/A
2013/0258701	12/2012	Westerinen et al.	N/A	N/A
2013/0267309	12/2012	Robbins et al.	N/A	N/A
2013/0271731	12/2012	Popovich et al.	N/A	N/A
2013/0277890	12/2012	Bowman et al.	N/A	N/A
2013/0286053	12/2012	Fleck et al.	N/A	N/A
2013/0300997	12/2012	Popovich et al.	N/A	N/A
2013/0301014	12/2012	DeJong et al.	N/A	N/A
2013/0305437	12/2012	Weller et al.	N/A	N/A
2013/0308185	12/2012	Robinson et al.	N/A	N/A
2013/0312811	12/2012	Aspnes et al.	N/A	N/A
2013/0314789	12/2012	Saarikko et al.	N/A	N/A
2013/0314793	12/2012	Robbins et al.	N/A	N/A
2013/0322810	12/2012	Robbins	N/A	N/A
2013/0328948	12/2012	Kunkel et al.	N/A	N/A
2013/0342525	12/2012	Benko et al.	N/A	N/A
2014/0002514	12/2013	Richards	N/A	N/A
2014/0003762	12/2013	Macnamara	N/A	N/A
2014/0009809	12/2013	Pyun et al.	N/A	N/A

2014/0022616	12/2013	Popovich et al.	N/A	N/A
2014/0024159	12/2013	Jain	N/A	N/A
2014/0027006	12/2013	Foley et al.	N/A	N/A
2014/0037242	12/2013	Popovich et al.	N/A	N/A
2014/0043672	12/2013	Clarke et al.	N/A	N/A
2014/0043689	12/2013	Mason	N/A	N/A
2014/0055845	12/2013	Jain	N/A	N/A
2014/0063055	12/2013	Osterhout et al.	N/A	N/A
2014/0064655	12/2013	Nguyen et al.	N/A	N/A
2014/0071538	12/2013	Muller	N/A	N/A
2014/0098010	12/2013	Travis	N/A	N/A
2014/0104665	12/2013	Popovich et al.	N/A	N/A
2014/0104685	12/2013	Bohn et al.	N/A	N/A
2014/0118647	12/2013	Momonoï et al.	N/A	N/A
2014/0126029	12/2013	Fuetterer	N/A	N/A
2014/0126175	12/2013	Amitai et al.	N/A	N/A
2014/0130132	12/2013	Cahill et al.	N/A	N/A
2014/0140653	12/2013	Brown et al.	N/A	N/A
2014/0140654	12/2013	Brown et al.	N/A	N/A
2014/0146394	12/2013	Tout et al.	N/A	N/A
2014/0152778	12/2013	Ihlenburg et al.	N/A	N/A
2014/0154614	12/2013	Xie et al.	N/A	N/A
2014/0160576	12/2013	Robbins et al.	N/A	N/A
2014/0168055	12/2013	Smith	N/A	N/A
2014/0168260	12/2013	O'Brien et al.	N/A	N/A
2014/0168735	12/2013	Yuan et al.	N/A	N/A
2014/0168783	12/2013	Luebke et al.	N/A	N/A
2014/0172296	12/2013	Shtukater	N/A	N/A
2014/0176528	12/2013	Robbins	N/A	N/A
2014/0177023	12/2013	Gao et al.	N/A	N/A
2014/0185286	12/2013	Popovich et al.	N/A	N/A
2014/0198128	12/2013	Hong et al.	N/A	N/A
2014/0198896	12/2013	Hemmendorff et al.	N/A	N/A
2014/0204455	12/2013	Popovich et al.	N/A	N/A
2014/0211322	12/2013	Bohn et al.	N/A	N/A
2014/0218468	12/2013	Gao et al.	N/A	N/A
2014/0218801	12/2013	Simmonds et al.	N/A	N/A
2014/0232759	12/2013	Simmonds et al.	N/A	N/A
2014/0240834	12/2013	Mason	N/A	N/A
2014/0240842	12/2013	Nguyen et al.	N/A	N/A
2014/0255662	12/2013	Enomoto et al.	N/A	N/A
2014/0267420	12/2013	Schowengerdt et al.	N/A	N/A
2014/0268017	12/2013	Sweis et al.	N/A	N/A
2014/0268353	12/2013	Fujimura et al.	N/A	N/A
2014/0300947	12/2013	Fattal et al.	N/A	N/A
2014/0300960	12/2013	Santori et al.	N/A	N/A
2014/0300966	12/2013	Travers et al.	N/A	N/A
2014/0327970	12/2013	Bohn et al.	N/A	N/A
2014/0330159	12/2013	Costa et al.	N/A	N/A
2014/0367719	12/2013	Jain	N/A	N/A

2014/0375542	12/2013	Robbins et al.	N/A	N/A
2014/0375789	12/2013	Lou et al.	N/A	N/A
2014/0375790	12/2013	Robbins et al.	N/A	N/A
2015/0001677	12/2014	Palumbo et al.	N/A	N/A
2015/0003796	12/2014	Bennett	N/A	N/A
2015/0010265	12/2014	Popovich et al.	N/A	N/A
2015/0015946	12/2014	Muller	N/A	N/A
2015/0016777	12/2014	Abovitz et al.	N/A	N/A
2015/0035744	12/2014	Robbins et al.	N/A	N/A
2015/0036068	12/2014	Fattal et al.	N/A	N/A
2015/0058791	12/2014	Robertson et al.	N/A	N/A
2015/0062675	12/2014	Ayres et al.	N/A	N/A
2015/0062707	12/2014	Simmonds et al.	N/A	N/A
2015/0086163	12/2014	Valera et al.	N/A	N/A
2015/0086907	12/2014	Mizuta et al.	N/A	N/A
2015/0107671	12/2014	Bodan et al.	N/A	N/A
2015/0109763	12/2014	Shinkai et al.	N/A	N/A
2015/0125109	12/2014	Robbins et al.	N/A	N/A
2015/0148728	12/2014	Sallum et al.	N/A	N/A
2015/0160529	12/2014	Popovich et al.	N/A	N/A
2015/0167868	12/2014	Boncha	N/A	N/A
2015/0177443	12/2014	Faecke et al.	N/A	N/A
2015/0177686	12/2014	Lee et al.	N/A	N/A
2015/0177688	12/2014	Popovich et al.	N/A	N/A
2015/0185475	12/2014	Saarikko et al.	N/A	N/A
2015/0211960	12/2014	Shimizu	N/A	N/A
2015/0219834	12/2014	Nichol et al.	N/A	N/A
2015/0235447	12/2014	Abovitz et al.	N/A	N/A
2015/0235448	12/2014	Schowengerdt et al.	N/A	N/A
2015/0243068	12/2014	Solomon	N/A	N/A
2015/0247975	12/2014	Abovitz et al.	N/A	N/A
2015/0260994	12/2014	Akutsu et al.	N/A	N/A
2015/0262424	12/2014	Tabaka et al.	N/A	N/A
2015/0268399	12/2014	Futterer	N/A	N/A
2015/0268415	12/2014	Schowengerdt et al.	N/A	N/A
2015/0277375	12/2014	Large et al.	N/A	N/A
2015/0285682	12/2014	Popovich et al.	N/A	N/A
2015/0288129	12/2014	Jain	N/A	N/A
2015/0289762	12/2014	Popovich et al.	N/A	N/A
2015/0309264	12/2014	Abovitz et al.	N/A	N/A
2015/0316768	12/2014	Simmonds	N/A	N/A
2015/0338689	12/2014	Min et al.	N/A	N/A
2015/0346490	12/2014	Tekolste et al.	N/A	N/A
2015/0346495	12/2014	Welch et al.	N/A	N/A
2015/0355394	12/2014	Leighton et al.	N/A	N/A
2016/0003847	12/2015	Ryan et al.	N/A	N/A
2016/0004090	12/2015	Popovich et al.	N/A	N/A
2016/0018673	12/2015	Wang	N/A	N/A
2016/0026253	12/2015	Bradski et al.	N/A	N/A
2016/0033705	12/2015	Fattal	N/A	N/A

2016/0033706	12/2015	Fattal et al.	N/A	N/A
2016/0038992	12/2015	Arthur et al.	N/A	N/A
2016/0041387	12/2015	Valera et al.	N/A	N/A
2016/0055822	12/2015	Bell	N/A	N/A
2016/0077338	12/2015	Robbins et al.	N/A	N/A
2016/0085300	12/2015	Robbins et al.	N/A	N/A
2016/0097959	12/2015	Bruizeman et al.	N/A	N/A
2016/0116739	12/2015	TeKolste et al.	N/A	N/A
2016/0124223	12/2015	Shinbo et al.	N/A	N/A
2016/0124241	12/2015	Popovich et al.	N/A	N/A
2016/0132025	12/2015	Taff et al.	N/A	N/A
2016/0147067	12/2015	Hua et al.	N/A	N/A
2016/0170226	12/2015	Popovich et al.	N/A	N/A
2016/0178901	12/2015	Ishikawa	N/A	N/A
2016/0195664	12/2015	Fattal et al.	N/A	N/A
2016/0209648	12/2015	Haddick et al.	N/A	N/A
2016/0209657	12/2015	Popovich et al.	N/A	N/A
2016/0231568	12/2015	Saarikko et al.	N/A	N/A
2016/0231570	12/2015	Levola et al.	N/A	N/A
2016/0238772	12/2015	Waldern et al.	N/A	N/A
2016/0266398	12/2015	Poon et al.	N/A	N/A
2016/0274356	12/2015	Mason	N/A	N/A
2016/0274362	12/2015	Tinch et al.	N/A	N/A
2016/0283773	12/2015	Popovich et al.	N/A	N/A
2016/0291328	12/2015	Popovich et al.	N/A	N/A
2016/0299344	12/2015	Dobschal et al.	N/A	N/A
2016/0320536	12/2015	Simmonds et al.	N/A	N/A
2016/0327705	12/2015	Simmonds et al.	N/A	N/A
2016/0336033	12/2015	Tanaka	N/A	N/A
2016/0341964	12/2015	Amitai	N/A	N/A
2016/0363840	12/2015	Mizoguchi et al.	N/A	N/A
2016/0377879	12/2015	Popovich et al.	N/A	N/A
2017/0003505	12/2016	Vallius et al.	N/A	N/A
2017/0010466	12/2016	Klug et al.	N/A	N/A
2017/0010488	12/2016	Klug et al.	N/A	N/A
2017/0030550	12/2016	Popovich et al.	N/A	N/A
2017/0031160	12/2016	Popovich et al.	N/A	N/A
2017/0031171	12/2016	Vallius et al.	N/A	N/A
2017/0032166	12/2016	Raguin et al.	N/A	N/A
2017/0034435	12/2016	Vallius	N/A	N/A
2017/0038579	12/2016	Yeoh et al.	N/A	N/A
2017/0052374	12/2016	Waldern et al.	N/A	N/A
2017/0052376	12/2016	Amitai et al.	N/A	N/A
2017/0059759	12/2016	Ayres et al.	N/A	N/A
2017/0059775	12/2016	Coles et al.	N/A	N/A
2017/0102543	12/2016	Vallius	N/A	N/A
2017/0115487	12/2016	Travis et al.	N/A	N/A
2017/0123208	12/2016	Vallius	N/A	N/A
2017/0131460	12/2016	Lin et al.	N/A	N/A
2017/0131545	12/2016	Wall et al.	N/A	N/A

2017/0131546	12/2016	Woltman et al.	N/A	N/A
2017/0131551	12/2016	Robbins et al.	N/A	N/A
2017/0138789	12/2016	Ivanov	N/A	N/A
2017/0160546	12/2016	Bull et al.	N/A	N/A
2017/0176747	12/2016	Vallius et al.	N/A	N/A
2017/0180404	12/2016	Bersch et al.	N/A	N/A
2017/0180408	12/2016	Yu et al.	N/A	N/A
2017/0192246	12/2016	Popovich et al.	N/A	N/A
2017/0192499	12/2016	Trail	N/A	N/A
2017/0199333	12/2016	Waldern et al.	N/A	N/A
2017/0212295	12/2016	Vasylyev	N/A	N/A
2017/0219841	12/2016	Popovich et al.	N/A	N/A
2017/0255257	12/2016	Tiana et al.	N/A	N/A
2017/0270637	12/2016	Perreault et al.	N/A	N/A
2017/0276940	12/2016	Popovich et al.	N/A	N/A
2017/0299793	12/2016	Fattal	N/A	N/A
2017/0299794	12/2016	Fattal	N/A	N/A
2017/0299860	12/2016	Wall et al.	N/A	N/A
2017/0307800	12/2016	Fattal	N/A	N/A
2017/0322426	12/2016	Tervo	N/A	N/A
2017/0329075	12/2016	Yeoh et al.	N/A	N/A
2017/0329140	12/2016	Yeoh et al.	N/A	N/A
2017/0356801	12/2016	Popovich et al.	N/A	N/A
2017/0357841	12/2016	Popovich et al.	N/A	N/A
2018/0003805	12/2017	Popovich	N/A	G06F 3/013
2018/0011324	12/2017	Popovich et al.	N/A	N/A
2018/0017801	12/2017	Chang et al.	N/A	N/A
2018/0052277	12/2017	Schowengerdt et al.	N/A	N/A
2018/0059305	12/2017	Popovich et al.	N/A	N/A
2018/0067251	12/2017	Baldwin et al.	N/A	N/A
2018/0067318	12/2017	St. Hilaire	N/A	N/A
2018/0074265	12/2017	Waldern et al.	N/A	N/A
2018/0074340	12/2017	Robbins et al.	N/A	N/A
2018/0074352	12/2017	Popovich et al.	N/A	N/A
2018/0081190	12/2017	Lee et al.	N/A	N/A
2018/0082644	12/2017	Bohn	N/A	N/A
2018/0095283	12/2017	Takeda et al.	N/A	N/A
2018/0107011	12/2017	Lu et al.	N/A	N/A
2018/0113303	12/2017	Popovich et al.	N/A	N/A
2018/0120669	12/2017	Popovich et al.	N/A	N/A
2018/0129060	12/2017	Lee et al.	N/A	N/A
2018/0143438	12/2017	Oh	N/A	N/A
2018/0143449	12/2017	Popovich et al.	N/A	N/A
2018/0172995	12/2017	Lee et al.	N/A	N/A
2018/0188542	12/2017	Waldern et al.	N/A	N/A
2018/0188691	12/2017	Fattal	N/A	N/A
2018/0203230	12/2017	Vallius et al.	N/A	N/A
2018/0210198	12/2017	Brown et al.	N/A	N/A
2018/0210205	12/2017	Grey et al.	N/A	N/A

2018/0210396	12/2017	Popovich et al.	N/A	N/A
2018/0232048	12/2017	Popovich et al.	N/A	N/A
2018/0246354	12/2017	Popovich et al.	N/A	N/A
2018/0252869	12/2017	Ayres et al.	N/A	N/A
2018/0275350	12/2017	Oh et al.	N/A	N/A
2018/0275402	12/2017	Popovich	N/A	G02B 6/0035
2018/0275410	12/2017	Yeoh et al.	N/A	N/A
2018/0284440	12/2017	Popovich et al.	N/A	N/A
2018/0299678	12/2017	Singer et al.	N/A	N/A
2018/0338131	12/2017	Robbins	N/A	N/A
2018/0348524	12/2017	Blum et al.	N/A	N/A
2018/0373115	12/2017	Brown et al.	N/A	N/A
2019/0041634	12/2018	Popovich et al.	N/A	N/A
2019/0042827	12/2018	Popovich et al.	N/A	N/A
2019/0064526	12/2018	Connor	N/A	N/A
2019/0064735	12/2018	Waldern et al.	N/A	N/A
2019/0072723	12/2018	Waldern et al.	N/A	N/A
2019/0094548	12/2018	Nicholson et al.	N/A	N/A
2019/0113751	12/2018	Waldern et al.	N/A	N/A
2019/0113829	12/2018	Waldern et al.	N/A	N/A
2019/0114484	12/2018	Keech et al.	N/A	N/A
2019/0121027	12/2018	Popovich et al.	N/A	N/A
2019/0129085	12/2018	Waldern et al.	N/A	N/A
2019/0162962	12/2018	Leighton et al.	N/A	N/A
2019/0162963	12/2018	Leighton et al.	N/A	N/A
2019/0171031	12/2018	Popovich et al.	N/A	N/A
2019/0179153	12/2018	Popovich et al.	N/A	N/A
2019/0187474	12/2018	Bhargava et al.	N/A	N/A
2019/0187538	12/2018	Popovich et al.	N/A	N/A
2019/0188471	12/2018	Osterhout et al.	N/A	N/A
2019/0212195	12/2018	Popovich et al.	N/A	N/A
2019/0212557	12/2018	Waldern et al.	N/A	N/A
2019/0212573	12/2018	Popovich et al.	N/A	N/A
2019/0212588	12/2018	Waldern et al.	N/A	N/A
2019/0212589	12/2018	Waldern et al.	N/A	N/A
2019/0212596	12/2018	Waldern et al.	N/A	N/A
2019/0212597	12/2018	Waldern et al.	N/A	N/A
2019/0212698	12/2018	Waldern et al.	N/A	N/A
2019/0212699	12/2018	Waldern et al.	N/A	N/A
2019/0219822	12/2018	Popovich et al.	N/A	N/A
2019/0226830	12/2018	Edwin et al.	N/A	N/A
2019/0243142	12/2018	Tekolste et al.	N/A	N/A
2019/0243209	12/2018	Perreault et al.	N/A	N/A
2019/0265486	12/2018	Hansotte et al.	N/A	N/A
2019/0278224	12/2018	Schlottau et al.	N/A	N/A
2019/0285796	12/2018	Waldern et al.	N/A	N/A
2019/0293880	12/2018	Nishiwaki et al.	N/A	N/A
2019/0319426	12/2018	Lu et al.	N/A	N/A
2019/0339558	12/2018	Waldern et al.	N/A	N/A

2020/0012839	12/2019	Popovich et al.	N/A	N/A
2020/0026074	12/2019	Waldern et al.	N/A	N/A
2020/0033190	12/2019	Popovich et al.	N/A	N/A
2020/0033801	12/2019	Waldern et al.	N/A	N/A
2020/0033802	12/2019	Popovich et al.	N/A	N/A
2020/0041791	12/2019	Shipton et al.	N/A	N/A
2020/0057353	12/2019	Popovich et al.	N/A	N/A
2020/0064637	12/2019	Popovich et al.	N/A	N/A
2020/0081317	12/2019	Popovich et al.	N/A	N/A
2020/0089319	12/2019	Popovich et al.	N/A	N/A
2020/0103661	12/2019	Kamakura	N/A	N/A
2020/0116997	12/2019	Lee et al.	N/A	N/A
2020/0142131	12/2019	Waldern et al.	N/A	N/A
2020/0150469	12/2019	Popovich et al.	N/A	N/A
2020/0159023	12/2019	Bhargava et al.	N/A	N/A
2020/0159026	12/2019	Waldern et al.	N/A	N/A
2020/0183163	12/2019	Waldern et al.	N/A	N/A
2020/0201042	12/2019	Wang et al.	N/A	N/A
2020/0201051	12/2019	Popovich et al.	N/A	N/A
2020/0209630	12/2019	Schultz et al.	N/A	N/A
2020/0225471	12/2019	Waldern et al.	N/A	N/A
2020/0241304	12/2019	Popovich et al.	N/A	N/A
2020/0247016	12/2019	Calafiore	N/A	N/A
2020/0247017	12/2019	Waldern et al.	N/A	N/A
2020/0249484	12/2019	Waldern et al.	N/A	N/A
2020/0249491	12/2019	Popovich et al.	N/A	N/A
2020/0249568	12/2019	Rao et al.	N/A	N/A
2020/0264378	12/2019	Grant	N/A	G02B 6/0016
2020/0271973	12/2019	Waldern et al.	N/A	N/A
2020/0292745	12/2019	Waldern et al.	N/A	N/A
2020/0292840	12/2019	Popovich et al.	N/A	N/A
2020/0333606	12/2019	Popovich et al.	N/A	N/A
2020/0341194	12/2019	Waldern et al.	N/A	N/A
2020/0348519	12/2019	Waldern et al.	N/A	N/A
2020/0348531	12/2019	Popovich et al.	N/A	N/A
2020/0363771	12/2019	Waldern et al.	N/A	N/A
2020/0372236	12/2019	Popovich et al.	N/A	N/A
2020/0386947	12/2019	Waldern et al.	N/A	N/A
2020/0400946	12/2019	Waldern et al.	N/A	N/A
2021/0026297	12/2020	Waldern et al.	N/A	N/A
2021/0033857	12/2020	Waldern et al.	N/A	N/A
2021/0055551	12/2020	Chi et al.	N/A	N/A
2021/0063634	12/2020	Waldern et al.	N/A	N/A
2021/0109285	12/2020	Jiang et al.	N/A	N/A
2021/0191122	12/2020	Yaroshchuk et al.	N/A	N/A
2021/0199873	12/2020	Shi et al.	N/A	N/A
2021/0199971	12/2020	Lee et al.	N/A	N/A
2021/0216040	12/2020	Waldern et al.	N/A	N/A
2021/0223585	12/2020	Waldern et al.	N/A	N/A

2021/0231874	12/2020	Popovich et al.	N/A	N/A
2021/0231955	12/2020	Waldern et al.	N/A	N/A
2021/0238374	12/2020	Ye et al.	N/A	N/A
2021/0239984	12/2020	Popovich et al.	N/A	N/A
2021/0247560	12/2020	Waldern et al.	N/A	N/A
2021/0247620	12/2020	Popovich et al.	N/A	N/A
2021/0247719	12/2020	Waldern et al.	N/A	N/A
2021/0278739	12/2020	Brown et al.	N/A	N/A
2021/0349328	12/2020	Popovich et al.	N/A	N/A
2021/0364803	12/2020	Schowengerdt et al.	N/A	N/A
2021/0364836	12/2020	Waldern et al.	N/A	N/A
2021/0405299	12/2020	Grant et al.	N/A	N/A
2021/0405365	12/2020	Popovich et al.	N/A	N/A
2021/0405514	12/2020	Waldern et al.	N/A	N/A
2022/0019015	12/2021	Calafiore et al.	N/A	N/A
2022/0082739	12/2021	Franke et al.	N/A	N/A
2022/0091323	12/2021	Yaroshchuk et al.	N/A	N/A
2022/0099898	12/2021	Waldern et al.	N/A	N/A
2022/0163728	12/2021	Waldern et al.	N/A	N/A
2022/0204790	12/2021	Zhang et al.	N/A	N/A
2022/0206232	12/2021	Zhang et al.	N/A	N/A
2022/0283377	12/2021	Popovich et al.	N/A	N/A
2023/0221493	12/2022	Grant et al.	N/A	N/A
2023/0290290	12/2022	Shams et al.	N/A	N/A
2023/0314810	12/2022	Delapp et al.	N/A	N/A
2024/0012242	12/2023	Waldern et al.	N/A	N/A
2024/0012247	12/2023	Popovich et al.	N/A	N/A
2024/0019640	12/2023	Waldern et al.	N/A	N/A
2024/0027670	12/2023	Waldern et al.	N/A	N/A
2024/0027689	12/2023	Waldern et al.	N/A	N/A
2024/0272724	12/2023	Bradski et al.	N/A	N/A
2024/0295703	12/2023	Grant et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
PI0720469	12/2013	BR	N/A
2889727	12/2013	CA	N/A
1320217	12/2000	CN	N/A
1886680	12/2005	CN	N/A
200944140	12/2006	CN	N/A
101103297	12/2007	CN	N/A
101151562	12/2007	CN	N/A
101263412	12/2007	CN	N/A
100492099	12/2008	CN	N/A
101589326	12/2008	CN	N/A
101688977	12/2009	CN	N/A
101726857	12/2009	CN	N/A
101793555	12/2009	CN	N/A
101881936	12/2009	CN	N/A

101910900	12/2009	CN	N/A
101945612	12/2010	CN	N/A
102314092	12/2011	CN	N/A
102498425	12/2011	CN	N/A
102608762	12/2011	CN	N/A
102782563	12/2011	CN	N/A
102928981	12/2012	CN	N/A
103000188	12/2012	CN	N/A
103185970	12/2012	CN	N/A
103562802	12/2013	CN	N/A
103777282	12/2013	CN	N/A
103823267	12/2013	CN	N/A
103959133	12/2013	CN	N/A
104040308	12/2013	CN	N/A
104040410	12/2013	CN	N/A
104136952	12/2013	CN	N/A
104204901	12/2013	CN	N/A
104246626	12/2013	CN	N/A
303019849	12/2013	CN	N/A
104520751	12/2014	CN	N/A
303217936	12/2014	CN	N/A
104956252	12/2014	CN	N/A
105074537	12/2014	CN	N/A
105074539	12/2014	CN	N/A
105137598	12/2014	CN	N/A
105190407	12/2014	CN	N/A
105229514	12/2015	CN	N/A
105393159	12/2015	CN	N/A
105408801	12/2015	CN	N/A
105408802	12/2015	CN	N/A
105408803	12/2015	CN	N/A
105531716	12/2015	CN	N/A
105705981	12/2015	CN	N/A
105940451	12/2015	CN	N/A
106125308	12/2015	CN	N/A
106443867	12/2016	CN	N/A
106575034	12/2016	CN	N/A
106716223	12/2016	CN	N/A
106842397	12/2016	CN	N/A
106950744	12/2016	CN	N/A
107015368	12/2016	CN	N/A
107466372	12/2016	CN	N/A
107533137	12/2017	CN	N/A
107873086	12/2017	CN	N/A
108107506	12/2017	CN	N/A
108351516	12/2017	CN	N/A
108474945	12/2017	CN	N/A
108549124	12/2017	CN	N/A
108681067	12/2017	CN	N/A
108780224	12/2017	CN	N/A

208092344	12/2017	CN	N/A
109073889	12/2017	CN	N/A
109154717	12/2018	CN	N/A
208621784	12/2018	CN	N/A
103823267	12/2018	CN	N/A
110383117	12/2018	CN	N/A
107873086	12/2019	CN	N/A
111025657	12/2019	CN	N/A
111323867	12/2019	CN	N/A
111386495	12/2019	CN	N/A
111566571	12/2019	CN	N/A
305973971	12/2019	CN	N/A
111615655	12/2019	CN	N/A
111684362	12/2019	CN	N/A
111902768	12/2019	CN	N/A
107466372	12/2020	CN	N/A
109073889	12/2020	CN	N/A
108780224	12/2020	CN	N/A
113424095	12/2020	CN	N/A
108474945	12/2020	CN	N/A
113692544	12/2020	CN	N/A
113728258	12/2020	CN	N/A
113759555	12/2020	CN	N/A
114207492	12/2021	CN	N/A
114341729	12/2021	CN	N/A
116149058	12/2022	CN	N/A
19751190	12/1998	DE	N/A
10221837	12/2002	DE	N/A
102006003785	12/2006	DE	N/A
102006036831	12/2007	DE	N/A
102012108424	12/2013	DE	N/A
102013209436	12/2013	DE	N/A
001747551-0002	12/2011	EM	N/A
007234190-0001	12/2018	EM	N/A
0795775	12/1996	EP	N/A
0822441	12/1997	EP	N/A
1347641	12/2002	EP	N/A
1413972	12/2003	EP	N/A
1526709	12/2004	EP	N/A
1748305	12/2006	EP	N/A
1938152	12/2007	EP	N/A
1413972	12/2007	EP	N/A
2110701	12/2008	EP	N/A
2196729	12/2009	EP	N/A
2225592	12/2009	EP	N/A
2244114	12/2009	EP	N/A
2326983	12/2010	EP	N/A
2381290	12/2010	EP	N/A
1828832	12/2012	EP	N/A
2733517	12/2013	EP	N/A

1573369	12/2013	EP	N/A
2748670	12/2013	EP	N/A
2634605	12/2014	EP	N/A
2929378	12/2014	EP	N/A
2748670	12/2014	EP	N/A
2995986	12/2015	EP	N/A
1402298	12/2015	EP	N/A
2995986	12/2016	EP	N/A
3198192	12/2016	EP	N/A
3245444	12/2016	EP	N/A
3245551	12/2016	EP	N/A
3256888	12/2016	EP	N/A
3359999	12/2017	EP	N/A
2494388	12/2017	EP	N/A
3398007	12/2017	EP	N/A
3433658	12/2018	EP	N/A
3433659	12/2018	EP	N/A
2842003	12/2018	EP	N/A
3245551	12/2018	EP	N/A
3548939	12/2018	EP	N/A
3698214	12/2019	EP	N/A
3710876	12/2019	EP	N/A
3710887	12/2019	EP	N/A
3710893	12/2019	EP	N/A
3710894	12/2019	EP	N/A
3894938	12/2020	EP	N/A
3924759	12/2020	EP	N/A
3927793	12/2020	EP	N/A
3938821	12/2021	EP	N/A
3980825	12/2021	EP	N/A
4004646	12/2021	EP	N/A
2831659	12/2022	EP	N/A
3256888	12/2023	EP	N/A
4350422	12/2023	EP	N/A
20176157	12/2018	FI	N/A
20176158	12/2018	FI	N/A
20176161	12/2018	FI	N/A
2677463	12/1991	FR	N/A
2975506	12/2011	FR	N/A
2115178	12/1982	GB	N/A
2140935	12/1983	GB	N/A
2500631	12/2012	GB	N/A
2508661	12/2013	GB	N/A
2509536	12/2013	GB	N/A
2512077	12/2013	GB	N/A
2514658	12/2013	GB	N/A
1204684	12/2014	HK	N/A
1205563	12/2014	HK	N/A
1205793	12/2014	HK	N/A
1206101	12/2014	HK	N/A

57089722	12/1981	JP	N/A
02186319	12/1989	JP	N/A
03239384	12/1990	JP	N/A
H04303812	12/1991	JP	N/A
06294952	12/1993	JP	N/A
07098439	12/1994	JP	N/A
0990312	12/1996	JP	N/A
H09185313	12/1996	JP	N/A
10096903	12/1997	JP	N/A
11109320	12/1998	JP	N/A
11142806	12/1998	JP	N/A
2953444	12/1998	JP	N/A
2000056259	12/1999	JP	N/A
2000511306	12/1999	JP	N/A
2000261706	12/1999	JP	N/A
2000267042	12/1999	JP	N/A
2001027739	12/2000	JP	N/A
2001296503	12/2000	JP	N/A
2002090858	12/2001	JP	N/A
2002122906	12/2001	JP	N/A
2002156617	12/2001	JP	N/A
2002162598	12/2001	JP	N/A
2002523802	12/2001	JP	N/A
2002529790	12/2001	JP	N/A
2002311379	12/2001	JP	N/A
2003066428	12/2002	JP	N/A
2003270419	12/2002	JP	N/A
2004157245	12/2003	JP	N/A
2006350129	12/2005	JP	N/A
2007011057	12/2006	JP	N/A
2007094175	12/2006	JP	N/A
2007219106	12/2006	JP	N/A
2008112187	12/2007	JP	N/A
2008145619	12/2007	JP	N/A
2008233226	12/2007	JP	N/A
2009036955	12/2008	JP	N/A
2009515225	12/2008	JP	N/A
2009132221	12/2008	JP	N/A
2009133999	12/2008	JP	N/A
2009211091	12/2008	JP	N/A
4367775	12/2008	JP	N/A
2010044326	12/2009	JP	N/A
2010256631	12/2009	JP	N/A
2011075681	12/2010	JP	N/A
2011164545	12/2010	JP	N/A
2011187108	12/2010	JP	N/A
2011232510	12/2010	JP	N/A
2012137616	12/2011	JP	N/A
2012533089	12/2011	JP	N/A
2013061480	12/2012	JP	N/A

5303928	12/2012	JP	N/A
2013235256	12/2012	JP	N/A
2014132328	12/2013	JP	N/A
5588794	12/2013	JP	N/A
5646748	12/2013	JP	N/A
2015053163	12/2014	JP	N/A
2015523586	12/2014	JP	N/A
2015172713	12/2014	JP	N/A
2016030503	12/2015	JP	N/A
2017156389	12/2016	JP	N/A
2017528739	12/2016	JP	N/A
2018508037	12/2017	JP	N/A
2018512562	12/2017	JP	N/A
2018524621	12/2017	JP	N/A
2018533069	12/2017	JP	N/A
2019512745	12/2018	JP	N/A
2019520595	12/2018	JP	N/A
6598269	12/2018	JP	N/A
6680793	12/2019	JP	N/A
2020512578	12/2019	JP	N/A
2020514783	12/2019	JP	N/A
1664536	12/2019	JP	N/A
6734933	12/2019	JP	N/A
2020-537187	12/2019	JP	N/A
2021509488	12/2020	JP	N/A
2021509736	12/2020	JP	N/A
2021509737	12/2020	JP	N/A
2021509739	12/2020	JP	N/A
6895451	12/2020	JP	N/A
2022509083	12/2021	JP	N/A
2022513896	12/2021	JP	N/A
2022-520472	12/2021	JP	N/A
2022-525165	12/2021	JP	N/A
2022535460	12/2021	JP	N/A
2022543571	12/2021	JP	N/A
7461357	12/2023	JP	N/A
7612592	12/2023	JP	N/A
20060132474	12/2005	KR	N/A
100803288	12/2007	KR	N/A
20100092059	12/2009	KR	N/A
20140140063	12/2013	KR	N/A
20140142337	12/2013	KR	N/A
20170031357	12/2016	KR	N/A
20170039655	12/2016	KR	N/A
30-1061010	12/2019	KR	N/A
10-2020-0106932	12/2019	KR	N/A
10-2020-0108030	12/2019	KR	N/A
2020-0106170	12/2019	KR	N/A
20200104402	12/2019	KR	N/A
20210100174	12/2020	KR	N/A

10-2021-0127237	12/2020	KR	N/A
0138609	12/2020	KR	N/A
20210134763	12/2020	KR	N/A
200535633	12/2004	TW	N/A
200801583	12/2007	TW	N/A
201314263	12/2012	TW	N/A
201600943	12/2015	TW	N/A
201604601	12/2015	TW	N/A
1997001133	12/1996	WO	N/A
1997027519	12/1996	WO	N/A
1998004650	12/1997	WO	N/A
1999009440	12/1998	WO	N/A
9931658	12/1998	WO	N/A
1999052002	12/1998	WO	N/A
2000016136	12/1999	WO	N/A
2000023830	12/1999	WO	N/A
2000023832	12/1999	WO	N/A
2000023847	12/1999	WO	N/A
2000028369	12/1999	WO	N/A
2000028369	12/1999	WO	N/A
2001050200	12/2000	WO	N/A
2001090822	12/2000	WO	N/A
2002082168	12/2001	WO	N/A
2003081320	12/2002	WO	N/A
2004023174	12/2003	WO	N/A
2004053531	12/2003	WO	N/A
2004102226	12/2003	WO	N/A
2004109349	12/2003	WO	N/A
2004109349	12/2004	WO	N/A
2005001753	12/2004	WO	N/A
2005006065	12/2004	WO	N/A
2005006065	12/2004	WO	N/A
2005073798	12/2004	WO	N/A
2005093493	12/2004	WO	N/A
2006002870	12/2005	WO	N/A
2006064301	12/2005	WO	N/A
2006064325	12/2005	WO	N/A
2006064334	12/2005	WO	N/A
2006102073	12/2005	WO	N/A
2006132614	12/2005	WO	N/A
2006102073	12/2006	WO	N/A
2007015141	12/2006	WO	N/A
2007029032	12/2006	WO	N/A
2007085682	12/2006	WO	N/A
2007130130	12/2006	WO	N/A
2007141587	12/2006	WO	N/A
2007141589	12/2006	WO	N/A
2008011066	12/2007	WO	N/A
2008011066	12/2007	WO	N/A
2008081070	12/2007	WO	N/A

2008100545	12/2007	WO	N/A
2008011066	12/2007	WO	N/A
2009013597	12/2008	WO	N/A
2009013597	12/2008	WO	N/A
2009077802	12/2008	WO	N/A
2009077803	12/2008	WO	N/A
2009101238	12/2008	WO	N/A
2007130130	12/2008	WO	N/A
2009155437	12/2008	WO	N/A
2009155437	12/2009	WO	N/A
2010023444	12/2009	WO	N/A
2010057219	12/2009	WO	N/A
2010067114	12/2009	WO	N/A
2010067117	12/2009	WO	N/A
2010078856	12/2009	WO	N/A
2010104692	12/2009	WO	N/A
2010122330	12/2009	WO	N/A
2010125337	12/2009	WO	N/A
2010125337	12/2009	WO	N/A
2010131046	12/2009	WO	N/A
2011012825	12/2010	WO	N/A
2011032005	12/2010	WO	N/A
2011042711	12/2010	WO	N/A
2011051660	12/2010	WO	N/A
2011055109	12/2010	WO	N/A
2011042711	12/2010	WO	N/A
2011073673	12/2010	WO	N/A
2011107831	12/2010	WO	N/A
2011110821	12/2010	WO	N/A
2011131978	12/2010	WO	N/A
2012052352	12/2011	WO	N/A
2012062658	12/2011	WO	N/A
2012136970	12/2011	WO	N/A
2012158950	12/2011	WO	N/A
2012172295	12/2011	WO	N/A
2013027004	12/2012	WO	N/A
2013027006	12/2012	WO	N/A
2013033274	12/2012	WO	N/A
2013034879	12/2012	WO	N/A
2013049012	12/2012	WO	N/A
2013054972	12/2012	WO	N/A
2013102759	12/2012	WO	N/A
2013163347	12/2012	WO	N/A
2013167864	12/2012	WO	N/A
2013190257	12/2012	WO	N/A
2014064427	12/2013	WO	N/A
2014080155	12/2013	WO	N/A
2014085734	12/2013	WO	N/A
2014090379	12/2013	WO	N/A
2014091200	12/2013	WO	N/A

2014093601	12/2013	WO	N/A
2014100182	12/2013	WO	N/A
2014113506	12/2013	WO	N/A
2014116615	12/2013	WO	N/A
2014130383	12/2013	WO	N/A
2014144526	12/2013	WO	N/A
2014159621	12/2013	WO	N/A
2014164901	12/2013	WO	N/A
2014176695	12/2013	WO	N/A
2014179632	12/2013	WO	N/A
2014188149	12/2013	WO	N/A
2014209733	12/2013	WO	N/A
2014209819	12/2013	WO	N/A
2014209820	12/2013	WO	N/A
2014209821	12/2013	WO	N/A
2014210349	12/2013	WO	N/A
2015006784	12/2014	WO	N/A
2015015138	12/2014	WO	N/A
2015017291	12/2014	WO	N/A
2015069553	12/2014	WO	N/A
2015081313	12/2014	WO	N/A
2015117039	12/2014	WO	N/A
2015145119	12/2014	WO	N/A
2016010289	12/2015	WO	N/A
2016020630	12/2015	WO	N/A
2016020643	12/2015	WO	N/A
2016025350	12/2015	WO	N/A
2016020630	12/2015	WO	N/A
2016042283	12/2015	WO	N/A
2016044193	12/2015	WO	N/A
2016046514	12/2015	WO	N/A
2016048729	12/2015	WO	N/A
2016054092	12/2015	WO	N/A
2016069606	12/2015	WO	N/A
2016087442	12/2015	WO	N/A
2016103263	12/2015	WO	N/A
2016111706	12/2015	WO	N/A
2016111707	12/2015	WO	N/A
2016111708	12/2015	WO	N/A
2016111709	12/2015	WO	N/A
2016113533	12/2015	WO	N/A
2016113534	12/2015	WO	N/A
2016116733	12/2015	WO	N/A
2016118107	12/2015	WO	N/A
2016122679	12/2015	WO	N/A
2016130509	12/2015	WO	N/A
2016135434	12/2015	WO	N/A
2016113533	12/2015	WO	N/A
2016156776	12/2015	WO	N/A
2016162606	12/2015	WO	N/A

2016181108	12/2015	WO	N/A
2016046514	12/2016	WO	N/A
2017060665	12/2016	WO	N/A
2017094129	12/2016	WO	N/A
2017120320	12/2016	WO	N/A
2017134412	12/2016	WO	N/A
2017160367	12/2016	WO	N/A
2017162999	12/2016	WO	N/A
2017176861	12/2016	WO	N/A
2017178781	12/2016	WO	N/A
2017180403	12/2016	WO	N/A
2017182771	12/2016	WO	N/A
2017203200	12/2016	WO	N/A
2017203201	12/2016	WO	N/A
2017207987	12/2016	WO	N/A
2018102834	12/2017	WO	N/A
2018102834	12/2017	WO	N/A
2018096359	12/2017	WO	N/A
2018129398	12/2017	WO	N/A
2017162999	12/2017	WO	N/A
2018150163	12/2017	WO	N/A
2018152337	12/2017	WO	N/A
2018175546	12/2017	WO	N/A
2018206487	12/2017	WO	N/A
2019046649	12/2018	WO	N/A
2019077307	12/2018	WO	N/A
2019079350	12/2018	WO	N/A
2019079350	12/2018	WO	N/A
2019046649	12/2018	WO	N/A
2019122806	12/2018	WO	N/A
2019135784	12/2018	WO	N/A
2019135796	12/2018	WO	N/A
2019135837	12/2018	WO	N/A
2019136470	12/2018	WO	N/A
2019136471	12/2018	WO	N/A
2019136473	12/2018	WO	N/A
2019171038	12/2018	WO	N/A
2019185973	12/2018	WO	N/A
2019185975	12/2018	WO	N/A
2019185976	12/2018	WO	N/A
2019185977	12/2018	WO	N/A
2019217453	12/2018	WO	N/A
2020023779	12/2019	WO	N/A
2020123506	12/2019	WO	N/A
2020149956	12/2019	WO	N/A
2020163524	12/2019	WO	N/A
2020168348	12/2019	WO	N/A
2020172681	12/2019	WO	N/A
2020186113	12/2019	WO	N/A
2020212682	12/2019	WO	N/A

2020219092	12/2019	WO	N/A
2020227236	12/2019	WO	N/A
2020247930	12/2019	WO	N/A
2021016371	12/2020	WO	N/A
2021021926	12/2020	WO	N/A
2021032982	12/2020	WO	N/A
2021032983	12/2020	WO	N/A
2021041949	12/2020	WO	N/A
2021044121	12/2020	WO	N/A
2021138607	12/2020	WO	N/A
2021262759	12/2020	WO	N/A
2022109615	12/2021	WO	N/A
2022150841	12/2021	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability for International Application

PCT/US2020/018686, Report issued Aug. 10, 2021, Mailed Aug. 26, 2021, 6 pgs. cited by applicant

International Preliminary Report on Patentability for International Application

PCT/US2020/019549, Report issued Aug. 10, 2021, Mailed Sep. 2, 2021, 7 pgs. cited by applicant

International Preliminary Report on Patentability for International Application

PCT/US2020/022482, issued Aug. 25, 2021, Mailed Sep. 23, 2021, 7 Pgs. cited by applicant

International Preliminary Report on Patentability for International Application

PCT/US2020/031363, issued Nov. 2, 2021, Mailed Nov. 18, 2021, 7 pgs. cited by applicant

International Preliminary Report on Patentability for International Application

PCT/US2020/036654, Report issued Dec. 7, 2021, Mailed Dec. 16, 2021, 9 pgs. cited by applicant

International Preliminary Report on Patentability for PCT Application No. PCT/US2013/038070, dated Oct. 28, 2014, 6 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2010/000835, completed Oct. 26, 2010, mailed Nov. 8, 2010, 12 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2010/001920, completed Mar. 29, 2011, mailed Apr. 6, 2011, 15 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2015/000228, Search completed May 4, 2011, Mailed Jul. 15, 2011, 15 Pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2016/000036, completed Jul. 4, 2016, mailed Jul. 13, 2016, 10 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2016/000065, completed Jul. 14, 2016, mailed Jul. 27, 2016, 10 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2017/000015, Search completed Apr. 25, 2017, Mailed May 8, 2017, 10 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/GB2017/000055, Search completed Jul. 19, 2017, Mailed Jul. 26, 2017, 12 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/IB2008/001909, Search completed Feb. 4, 2009, Mailed Feb. 17, 2009, 6 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.

PCT/US2013/038070, completed Aug. 12, 2013, mailed Aug. 14, 2013, 12 pgs. cited by applicant

International Search Report and Written Opinion for International Application No.
PCT/US2014/011736, completed Apr. 18, 2014, mailed May 8, 2014, 10 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/012227, Search completed Feb. 28, 2018, Mailed Mar. 14, 2018, 8 Pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/012691, completed Mar. 10, 2018, mailed Mar. 28, 2018, 16 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/015553, completed Aug. 6, 2018, Mailed Sep. 19, 2018, 12 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/037410, Search completed Aug. 16, 2018, Mailed Aug. 30, 2018, 11 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/048636, Search completed Nov. 1, 2018, Mailed Nov. 15, 2018, 16 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/048960, Search completed Dec. 14, 2018, Mailed Jan. 8, 2019, 14 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/056150, Search completed Dec. 4, 2018, Mailed Dec. 26, 2018, 10 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2018/062835, Search completed Jan. 14, 2019, Mailed Jan. 31, 2019, 14 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2019/012758, completed Mar. 12, 2019, mailed Mar. 27, 2019, 9 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2019/012764, completed Mar. 1, 2019, mailed Mar. 18, 2019, 9 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2019/031163, Search completed Jul. 9, 2019, Mailed Jul. 29, 2019, 11 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2019/043496, Search completed Sep. 28, 2019, Mailed Nov. 14, 2019, 12 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2019/064765, Search completed Feb. 3, 2020, Mailed Mar. 18, 2020, 11 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2019/065478, Search completed Jan. 29, 2020, Mailed Feb. 11, 2020, 14 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2020/018686, Search completed Apr. 25, 2020, Mailed May 22, 2020, 11 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2020/019549, Search completed Apr. 14, 2020, Mailed May 22, 2020, 12 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.
PCT/US2020/022482, Search completed May 12, 2020, Mailed Jun. 9, 2020, 11 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.

PCT/US2020/031363, completed May 28, 2020, Mailed Jun. 10, 2020, 8 pgs. cited by applicant
International Search Report and Written Opinion for International Application No.

PCT/US2020/036654, Search completed Aug. 21, 2020, Mailed Sep. 4, 2020, 14 pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2020/043107, Search completed Sep. 28, 2020, Mailed Oct. 15, 2020, 12 pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2020/044060, Search completed Oct. 9, 2020, Mailed Nov. 9, 2020, 12 pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2020/048590, Search completed Dec. 7, 2020, Mailed Jan. 11, 2021, 19 pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2020/067737, Search completed Mar. 3, 2021, Mailed Mar. 5, 2021, 12 pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2021/038542, search Completed Sep. 21, 2021, Mailed Oct. 20, 2021 16 pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2021/072548, Search completed Jan. 25, 2022, Mailed Feb. 8, 2022, 15 Pgs. cited by
applicant

International Search Report and Written Opinion for International Application No.

PCT/US2022/070095, Search completed Mar. 10, 2022, Mailed Mar. 22, 2022, 13 pgs. cited by
applicant

International Search Report and Written Opinion for International Application

PCT/GB2009/051676, completed May 10, 2010, mailed May 18, 2010, 7 pgs. cited by applicant

International Search Report and Written Opinion for International Application

PCT/GB2016/000181, completed Dec. 21, 2016, mailed Feb. 27, 2017, 21 pgs. cited by applicant

International Search Report and Written Opinion for International Application

PCT/US2016/017091, completed by the European Patent Office on Apr. 20, 2016, 7 pgs. cited by
applicant

International Search Report and Written Opinion for International Application

PCT/US2019/012759, completed Mar. 14, 2019, mailed Apr. 15, 2019, 12 pgs. cited by applicant

International Search Report and Written Opinion for International Application

PCT/US2019/047097, completed Nov. 22, 2015, mailed Dec. 16, 2019, 10 pgs. cited by applicant

International Search Report and Written Opinion for International Application

PCT/US2020/016875, Report Completed Apr. 9, 2020, Mailed Apr. 29, 2020, 11 pgs. cited by
applicant

International Search Report for International Application PCT/GB2014/000295, completed Nov.
18, 2014, mailed Jan. 5, 2015, 4 pgs. cited by applicant

International Search Report for International Application PCT/GB2017/000040, mailed Jul. 18,
2017, completed Jul. 10, 2017, 3 pgs. cited by applicant

International Search Report for PCT/GB2010/001982, completed by the European Patent Office on
Feb. 24, 2011, 4 pgs. cited by applicant

International Search Report for PCT/GB2011/000349, completed by the European Patent Office on
Aug. 17, 2011, 4 pgs. cited by applicant

International Search Report for PCT/GB2012/000331, completed by the European Patent Office on
Aug. 29, 2012, 4 pgs. cited by applicant

International Search Report for PCT/GB2012/000677, completed by the European Patent Office on

Dec. 10, 2012, 4 pgs. cited by applicant
International Search Report for PCT/GB2013/000005, completed by the European Patent Office on Jul. 16, 2013, 3 pgs. cited by applicant
International Search Report for PCT/GB2013/000210, completed by the European Patent Office on Aug. 12, 2013, 3 pgs. cited by applicant
International Search Report for PCT/GB2013/000273, completed by the European Patent Office on Aug. 30, 2013, 4 pgs. cited by applicant
International Search Report for PCT/GB2014/000197, Completed by the European Patent Office on Jul. 31, 2014, 3 pgs. cited by applicant
International Search Report for PCT/GB2015/000203, completed by the European Patent Office on Oct. 9, 2015, 4 pgs. cited by applicant
International Search Report for PCT/GB2015/000225, completed by the European Patent Office on Nov. 10, 2015, mailed Dec. 2, 2016, 5 pgs. cited by applicant
International Search Report for PCT/GB2015/000274, completed by the European Patent Office on Jan. 7, 2016, 4 pgs. cited by applicant
International Search Report for PCT/GB2016/000003, Completed by the European Patent Office May 31, 2016, 6 pgs. cited by applicant
International Search Report for PCT/GB2016/000005, completed by the European Patent Office on May 27, 2016, 4 pgs. cited by applicant
International Search Report for PCT/GB2016/000014, completed by the European Patent Office on Jun. 27, 2016, 4 pgs. cited by applicant
International Search Report for PCT/GB2016/000051, Completed Aug. 11, 2016, 3 Pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2010/001982, search completed Feb. 24, 2011, mailed Mar. 8, 2011, 6 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2011/000349, completed Aug. 17, 2011, mailed Aug. 25, 2011, 9 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2012/000331, completed Aug. 29, 2012, mailed Sep. 6, 2012, 7 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2012/000677, completed Dec. 10, 2012, mailed Dec. 17, 2012, 4 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2013/000005, search completed Jul. 16, 2013, mailed Jul. 24, 2013, 11 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2013/000273, completed Aug. 30, 2013, mailed Sep. 9, 2013, 7 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2014/000197, Search completed Jul. 31, 2014, Mailed Aug. 7, 2014, 6 Pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2014/000295, search completed Nov. 18, 2014, mailed Jan. 5, 2015, 3 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2015/000203, completed Oct. 29, 2015, mailed Nov. 16, 2015, 7 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2015/000225, search completed Nov. 10, 2015, mailed Feb. 4, 2016, 7 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2015/000274, search completed Jan. 7, 2016, mailed Jan. 19, 2016, 7 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2016/000014, search completed Jun. 27, 2016, mailed Jul. 7, 2016, 6 pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2016/000051, Search completed Aug. 11, 2016, Mailed Aug. 22, 2016, 6 Pgs. cited by applicant
Written Opinion for International Application No. PCT/GB2017/000040, search completed Jul. 10,

2017, mailed Jul. 18, 2017, 6 pgs. cited by applicant
Written Opinion for International Application PCT/GB2013/000210, completed Aug. 12, 2013, Mailed Aug. 20, 2013, 5 pgs. cited by applicant
Written Opinion for International Application PCT/GB2016/000003, completed May 31, 2016, mailed Aug. 12, 2016, 10 pgs. cited by applicant
Written Opinion for International Application PCT/GB2016/000005, search completed May 27, 2016, mailed Jun. 6, 2016, 6 pgs. cited by applicant
“Compact linear Fresnel reflector”, Wikipedia, Sep. 16, 2019, https://en.wikipedia.org/wiki/Compact_linear_Fresnel_reflector. cited by applicant
Digi Lens: Waveguides, announced unknown, [online], [site visited Nov. 6, 2020]. Available from Internet, <URL: <https://www.digilens.com/technology/waveguides/>> (Year: 2020). cited by applicant
Google search: “digilens waveguide” [site visited Sep. 14, 2020], https://www.google.com/search?q=digilens+waveguide&sxsrf=ALeKk02RFwZAZOvrlxVHOM_2fiXFkhW1FA:1604777621684&source=Inms&tbm=isch&sa=X&ved=2ah_UKEwjyNXAlvHsAh_U_Rh_HIEHTufCvsQ_AUoAnoECBwQBA&biw=1200&bih=1777.. cited by applicant
Google search: “eyewear display devices” [site visited Sep. 14, 2020], https://www.google.com/search?q=eyewear+display+devices&sxsrf=ALeKk01WWfnOAgSQR_bhydJaYK3e37r_J_Lg:1604779001617&source=Inms&tbm=isch&sa=X&ved=2ah_UKEwi99txSm_HsAhVaoHIEHawtD8QQ_AUoAnoECC8QBA&biw=1200&bih=1777.. cited by applicant
Google search: “smart glasses” [site visited Sep. 14, 2020], https://www.google.com/search?q=smart+glasses&sxsrf=ALeKk01KN1wj23-NqP-KCnrcsUpCgxyKA:1604779046920&source=Inms&tbm=isch&sa=X&ved=2ah_UKEwipkqPom_HsAhVKhXIEHQGFbP8Q_AUoBHoECCgQBg&biw=1200&bih=1777.. cited by applicant
“Agilent ADNS-2051 Optical Mouse Sensor: Data Sheet”, Agilent Technologies, Jan. 9, 2002, 40 pgs. cited by applicant
“Application Note—MOXTEK ProFlux Polarizer use with LCOS displays”, CRL Opto Limited, <http://www.crlopto.com>, 2003, 6 pgs. cited by applicant
“Application Note AN16: Optical Considerations for Bridgelux LED Arrays”, BridgeLux, Jul. 31, 2010, 23 pgs. cited by applicant
“Application Note: Variable Attenuator for Lasers”, Technology and Applications Center, Newport Corporation, www.newport.com, 2006, DS-08067, 6 pgs. cited by applicant
“Bae Systems to Unveil Q-Sight Family of Helmet-Mounted Display at AUSA Symposium”, Released on Tuesday, Oct. 9, 2007, 1 pg. cited by applicant
“Beam Steering Using Liquid Crystals”, Boulder Nonlinear Systems, Inc., info@bnonlinear.com, May 8, 2001, 4 pgs. cited by applicant
“BragGrate—Deflector: Transmitting vol. Bragg Grating for angular selection and magnification”, 2015, www.OptiGrate.com. cited by applicant
“Cree XLamp XP-E LEDs”, Cree, Inc., Retrieved from www.cree.com/Xlamp, CLD-DS18 Rev 17, 2013, 17 pgs. cited by applicant
“Desmodur N 3900”, Bayer MaterialScience AG, Mar. 18, 2013, www.bayercoatings.com, 4 pgs. cited by applicant
“Digilens—Innovative Augmented Reality Display and Sensor Solutions for OEMs”, Jun. 6, 2017, 31 pgs. cited by applicant
“Exotic Optical Components”, Building Electro-Optical Systems, Making It All Work, Chapter 7, John Wiley & Sons, Inc., pp. 233-261. cited by applicant
Lu et al., “Polarization switch using thick holographic polymer-dispersed liquid crystal grating”, *Journal of Applied Physics*, vol. 95, No. 3, Feb. 1, 2004, pp. 810-815. cited by applicant
Lu et al., “The Mechanism of electric-field-induced segregation of additives in a liquid-crystal

host”, Phys Rev E Stat Nonlin Soft Matter Phys., Nov. 27, 2012, 14 pgs. cited by applicant

Ma et al., “Holographic Reversed-Mode Polymer-Stabilized Liquid Crystal Grating”, Chinese Phys. Lett., 2005, vol. 22, No. 1, pp. 103-106. cited by applicant

Mach et al., “Switchable Bragg diffraction from liquid crystal in colloid-templated structures”, Europhysics Letters, Jun. 1, 2002, vol. 58, No. 5, pp. 679-685. cited by applicant

Magarinos et al., “Wide Angle Color Holographic infinity optics display”, Air Force Systems Command, Brooks Air Force Base, Texas, AFHRL-TR-80-53, Mar. 1981, 100 pgs. cited by applicant

Marino et al., “Dynamical Behaviour of Polycrystals Gratings”, Electronic-Liquid Crystal Communications, Feb. 5, 2004, 10 pgs. cited by applicant

Massenot et al., “Multiplexed holographic transmission gratings recorded in holographic polymer-dispersed liquid crystals: static and dynamic studies”, Applied Optics, 2005, vol. 44, Issue 25, pp. 5273-5280. cited by applicant

Matay et al., “Planarization of Microelectronic Structures by Using Polyimides”, Journal of Electrical Engineering, 2002, vol. 53, No. 3-4, pp. 86-90. cited by applicant

Mathews, “The LED FAQ Pages”, Jan. 31, 2002, 23 pgs. cited by applicant

Matic, “Blazed phase liquid crystal beam steering”, Proc. of the SPIE, 1994, vol. 2120, pp. 194-205. cited by applicant

Matsushima et al., “Thiol-Isocyanate-Acrylate Ternary Networks by Selective Thiol-Click Chemistry”, Journal of Polymer Science: Part A: Polymer Chemistry, Apr. 16, 2010, vol. 48, 3255-3264, doi: 10.1002/pola.24102. cited by applicant

McLeod, “Axicons and Their Uses”, Journal of the Optical Society of America, Feb. 1960, vol. 50, No. 2, pp. 166-169. cited by applicant

McManamon et al., “A Review of Phased Array Steering for Narrow-Band Electrooptical Systems”, Proceedings of the IEEE, Jun. 2009, vol. 97, No. 6, pp. 1078-1096. cited by applicant

McManamon et al., “Optical Phased Array Technology”, Proceedings of the IEEE, Feb. 1996, vol. 84, Issue 2, pp. 268-298. cited by applicant

Miller, “Coupled Wave Theory and Waveguide Applications”, The Bell System Technical Journal, Short Hills, NJ, Feb. 2, 1954, 166 pgs. cited by applicant

Missinne et al., “Flexible thin polymer waveguide Bragg grating sensor foils for strain sensing”, Proc. SPIE, 10101, Organic Photonic Materials and Devices, Feb. 16, 2017, <https://doi.org/10.1117/12.2250823>. cited by applicant

Moffitt, “Head-Mounted Display Image Configurations”, retrieved from the internet on Dec. 19, 2014, dated May 2008, 25 pgs. cited by applicant

Moharam et al., “Diffraction characteristics of photoresist surface-relief gratings”, Applied Optics, Sep. 15, 1984, vol. 23, pp. 3214-3220. cited by applicant

Mulik, “Adhesion Enhancement of Polymeric Films on Glass Surfaces by a Silane Derivative of Azobisisobutyronitrile (AIBN).”, Polymer Preprints, American Chemical Society (ACS), Jan. 2008, 3 pgs. cited by applicant

Nair et al., “Enhanced Two-Stage Reactive Polymer Network Forming Systems”, Polymer (Guildf). May 25, 2012, vol. 53, No. 12, pp. 2429-2434, doi: 10.1016/j.polymer.2012.04.007. cited by applicant

Nair et al., “Two-Stage Reactive Polymer Network Forming Systems”, Advanced Functional Materials, 2012, pp. 1-9, DOI: 10.1002/adfm.201102742. cited by applicant

Naqvi et al., “Concentration-dependent toxicity of iron oxide nanoparticles mediated by increased oxidative stress”, International Journal of Nanomedicine, Dovepress, Nov. 13, 2010, vol. 5, pp. 983-989. cited by applicant

Natarajan et al., “Electro Optical Switching Characteristics of vol. Holograms in Polymer Dispersed Liquid Crystals”, Journal of Nonlinear Optical Physics and Materials, 1997, vol. 5, No. 1, pp. 666-668. cited by applicant

Natarajan et al., "Electro-Optical Switching Characteristics of vol. Holograms in Polymer Dispersed Liquid Crystals", Journal of Nonlinear Optical Physics and Materials, Jan. 1996, vol. 5, No. 1, pp. 89-98. cited by applicant

Natarajan et al., "Holographic polymer dispersed liquid crystal reflection gratings formed by visible light initiated thiol-ene photopolymerization", Polymer, vol. 47, May 8, 2006, pp. 4411-4420. cited by applicant

Naydenova et al., "Low-scattering Volume Holographic Material", DIT PhD Project, <http://www.dit.ie/ieo/>, Oct. 2017, 2 pgs. cited by applicant

Neipp et al., "Non-local polymerization driven diffusion based model: general dependence of the polymerization rate to the exposure intensity", Optics Express, Aug. 11, 2003, vol. 11, No. 16, pp. 1876-1886. cited by applicant

Nielsen et al., "Grating Couplers for Fiber-to-Fiber Characterizations of Stand-Alone Dielectric Loaded Surface Plasmon Waveguide Components", Journal of Lightwave Technology, Oct. 1, 2012, vol. 30, No. 19, pp. 3118-3125, DOI: 10.1109/JLT.2012.2212418. cited by applicant

Nishikawa et al., "Mechanically and Light Induced Anchoring of Liquid Crystal on Polyimide Film", Mol. Cryst. Liq. Cryst., Aug. 1999, vol. 329, 8 pgs. cited by applicant

Nishikawa et al., "Mechanism of Unidirectional Liquid-Crystal Alignment on Polyimides with Linearly Polarized Ultraviolet Light Exposure", Applied Physics Letters, May 11, 1998, vol. 72, No. 19, 4 pgs. cited by applicant

Nordin et al., "Diffraction Properties of Stratified vol. Holographic Optical Elements", Journal of the Optical Society of America A. Dec. 1992, vol. 9, No. 12, , pp. 2206-2217. cited by applicant

Ogiwara et al., "Temperature Dependence of Anisotropic Diffraction in Holographic Polymer-Dispersed Liquid Crystal Memory", Applied Optics, Sep. 10, 2013, vol. 52, No. 26, pp. 6529-6536. cited by applicant

Ogiwara et al., "Thermo-Driven Light Controller by Using Thermal Modulation of Diffraction Wavelength in Holographic Polymer Dispersed Liquid Crystal Grating", Proc. SPIE, Feb. 19, 2014, 9004, Article 90040Q, 8 pgs., doi: 10.1117/12.2039104. cited by applicant

Oh et al., "Achromatic diffraction from polarization gratings with high efficiency", Optic Letters, Oct. 15, 2008, vol. 33, No. 20, pp. 2287-2289. cited by applicant

Olson et al., "Templating Nanoporous Polymers with Ordered Block Copolymers", Chemistry of Materials, Web publication Nov. 27, 2007, vol. 20, pp. 869-890. cited by applicant

Ondax, Inc., "vol. Holographic Gratings (VHG)", 2005, 7 pgs. cited by applicant

Orcutt, "Coming Soon: Smart Glasses That Look Like Regular Spectacles", Intelligent Machines, Jan. 9, 2014, 4 pgs. cited by applicant

Osredkar, "A study of the limits of spin-on-glass planarization process", Informacije MIDEM, 2001, vol. 31, 2, ISSN0352-9045, pp. 102-105. cited by applicant

Osredkar et al., "Planarization methods in IC fabrication technologies", Informacije MIDEM, 2002, vol. 32, 3, ISSN0352-9045, 5 pgs. cited by applicant

Ou et al., "A Simple LCOS Optical System (Late News)", Industrial Technology Research Institute/OES Lab. Q100/Q200, SID 2002, Boston, USA, 2 pgs. cited by applicant

Paolini et al., "High-Power LED Illuminators in Projection Displays", Lumileds, Aug. 7, 2001, 19 pgs. cited by applicant

Park et al., "Aligned Single-Wall Carbon Nanotube Polymer Composites Using an Electric Field", Journal of Polymer Science: Part B: Polymer Physics, Mar. 24, 2006, DOI 10.1002/polb.20823, pp. 1751-1762. cited by applicant

Park et al., "Fabrication of Reflective Holographic Gratings with Polyurethane Acrylates (PUA)", Current Applied Physics, Jun. 2002, vol. 2, pp. 249-252. cited by applicant

Peng et al., "Facile Image Patterning via Sequential Thiol-Michael/Thiol-Yne Click Reactions", Chemistry of Materials, Nov. 20, 2014, vol. 26, pp. 6819-6826, doi: 10.1021/cm5034436. cited by applicant

Peng et al., "High Performance Graded Rainbow Holograms via Two-Stage Sequential Orthogonal Thiol-Click Chemistry", *Macromolecules*, Mar. 28, 2014, vol. 47, pp. 2306-2315, doi: 10.1021/ma500167x. cited by applicant

Peng et al., "Low Voltage Driven and Highly Diffractive Holographic Polymer Dispersed Liquid Crystals with Spherical Morphology", *RSC Advances*, 2017, vol. 7, pp. 51847-51857, doi: 10.1039/c7ra08949. cited by applicant

Pierantoni et al., "Efficient modeling of 3-D photonic crystals for integrated optical devices", *IEEE Photonics Technology Letters*, Feb. 2006, vol. 18, No. 2, pp. 319-321, DOI: 10.1109/LPT.2005.861991. cited by applicant

Plawsky et al., "Engineered nanoporous and nanostructured films", *Materials Today*, Jun. 2009, vol. 12, No. 6, pp. 36-45. cited by applicant

Pogue et al., "Electrically Switchable Bragg Gratings from Liquid Crystal/Polymer Composites", *Applied Spectroscopy*, 2000, vol. 54, Issue 1, pp. 12A-28A. cited by applicant

Potenza, "These smart glasses automatically focus on what you're looking at", *The Verge*, Voc Media, Inc., Jan. 29, 2017, <https://www.theverge.com/2017/1/29/14403924/smart-glasses-automatic-focus-presbyopia-ces-2017>, 6 pgs. cited by applicant

Presnyakov et al., "Electrically tunable polymer stabilized liquid-crystal lens", *Journal of Applied Physics*, Apr. 29, 2005, vol. 97, pp. 103101-1-103101-6. cited by applicant

Prokop et al., "Air-Suspended SU-8 Polymer Waveguide Grating Couplers", *Journal of Lightwave Technology*, Sep. 1, 2016, vol. 34, No. 17, pp. 3966-3971, DOI: 10.1109/JLT.2016.2593025. cited by applicant

Qi et al., "P-111: Reflective Display Based on Total Internal Reflection and Grating-Grating Coupling", *Society for Information Display Digest*, May 2003, pp. 648-651, DOI: 10.1889/1.1832359. cited by applicant

Ramón, "Formation of 3D micro-and nanostructures using liquid crystals as a template", *Technische Universiteit Eindhoven*, Apr. 17, 2008, Thesis, 117 pgs., DOI:<http://dx.doi.org/10.6100/IR634422>. cited by applicant

Ramsey, "Holographic Patterning of Polymer Dispersed Liquid Crystal Materials for Diffractive Optical Elements", Thesis, The University of Texas at Arlington, Dec. 2006, 166 pgs. cited by applicant

Ramsey et al., "Holographically recorded reverse-mode transmission gratings in polymer-dispersed liquid crystal cells", *Applied Physics B: Laser and Optics*, Sep. 10, 2008, vol. 93, Nos. 2-3, pp. 481-489. cited by applicant

Reid, "Thin film silica nanocomposites for anti-reflection coatings", *Oxford Advance Surfaces*, www.oxfordsurfaces.com, Oct. 18, 2012, 23 pgs. cited by applicant

Riechert, "Speckle Reduction in Projection Systems", Dissertation, University Karlsruhe, 2009, 178 pgs. cited by applicant

Rossi et al., "Diffractive Optical Elements for Passive Infrared Detectors" Submitted to OSA Topical Meeting "Diffractive Optics and Micro-Optics", Quebec, Jun. 18-22, 2000, 3 pgs. cited by applicant

Roussel et al., "Photopolymerization Kinetics and Phase Behavior of Acrylate Based Polymers Dispersed Liquid Crystals", *Liquid Crystals*, 1998, vol. 24, Issue 4, pp. 555-561. cited by applicant

Saarikko, Extract of US 2010/0296163 ,Figure 2 as marked up by Examiner in Final Office Action for U.S. Appl. No. 14/044,676 dated Oct. 20, 2015, 1 page. cited by applicant

Sabel et al., "Simultaneous formation of holographic surface relief gratings and vol. phase gratings in photosensitive polymer", *Materials Research Letters*, May 30, 2019, vol. 7, No. 10, pp. 405-411, doi: 10.1080/21663831.2019.1621956. cited by applicant

Sagan et al., "Electrically Switchable Bragg Grating Technology for Projection Displays", *Proc. SPIE*. vol. 4294, Jan. 24, 2001, pp. 75-83. cited by applicant

Sakhno et al., "Deep surface relief grating in azobenzene-containing materials using a low-intensity

532 nm laser”, Optical Materials: X, Jan. 23, 2019, 100006, pp. 3-7, doi: 10.1016/j.omx.2019.100006. cited by applicant

Saleh et al., “Fourier Optics: 4.1 Propagation of light in free space, 4.2 Optical Fourier Transform, 4.3 Diffraction of Light, 4.4 Image Formation, 4.5 Holography”, Fundamentals of Photonics 1991, Chapter 4, pp. 108-143. cited by applicant

Saraswat, “Deposition & Planarization”, EE 311 Notes, Aug. 29, 2017, 28 pgs. cited by applicant

Schreiber et al., “Laser display with single-mirror MEMS scanner”, Journal of the SID 17/7, 2009, pp. 591-595. cited by applicant

Seiberle et al., “Photo-aligned anisotropic optical thin films”, Journal of the SID 12/1, 2004, 6 pgs. cited by applicant

Serebriakov et al., “Correction of the phase retardation caused by intrinsic birefringence in deep UV lithography”, Proc. of SPIE, May 21, 2010, vol. 5754, pp. 1780-1791. cited by applicant

Shechter et al., “Compact beam expander with linear gratings”, Applied Optics, vol. 41, No. 7, Mar. 1, 2002, pp. 1236-1240. cited by applicant

Shi et al., “Design considerations for high efficiency liquid crystal decentered microlens arrays for steering light”, Applied Optics, vol. 49, No. 3, Jan. 20, 2010, pp. 409-421. cited by applicant

Shriyan et al., “Analysis of effects of oxidized multiwalled carbon nanotubes on electro-optic polymer/liquid crystal thin film gratings”, Optics Express, Nov. 12, 2010, vol. 18, No. 24, pp. 24842-24852. cited by applicant

Simonite, “How Magic Leap's Augmented Reality Works”, Intelligent Machines, Oct. 23, 2014, 7 pgs. cited by applicant

Smith et al., “RM-PLUS—Overview”, Licrivue, Nov. 5, 2013, 16 pgs. cited by applicant

Sony Global, “Sony Releases the Transparent Lens Eyewear ‘SmartEyeglass Developer Edition’”, printed Oct. 19, 2017, Sony Global—News Releases, 5 pgs. cited by applicant

Steranka et al., “High-Power LEDs—Technology Status and Market Applications”, Lumileds, Jul. 2002, 23 pgs. cited by applicant

Stumpe et al., “Active and Passive LC Based Polarization Elements”, Mol. Cryst. Liq. Cryst., 2014, vol. 594: pp. 140-149. cited by applicant

Stumpe et al., “New type of polymer-LC electrically switchable diffractive devices—POLIPHEN”, May 19, 2015, p. 97. cited by applicant

Subbarayappa et al., “Bistable Nematic Liquid Crystal Device”, Jul. 30, 2009, 14 pgs. cited by applicant

Sun et al., “Effects of multiwalled carbon nanotube on holographic polymer dispersed liquid crystal”, Polymers Advanced Technologies, Feb. 19, 2010, DOI: 10.1002/pat. 1708, 8 pgs. cited by applicant

Sun et al., “Low-birefringence lens design for polarization sensitive optical systems”, Proceedings of SPIE, 2006, vol. 6289, pp. 6289DH-1 6289DH-10, doi: 10.1117/12.679416. cited by applicant

Sun et al., “Transflective multiplexing of holographic polymer dispersed liquid crystal using Si additives”, eXPRESS Polymer Letters, 2011, vol. 5, No. 1, pp. 73-81. cited by applicant

Sutherland et al., “Bragg Gratings in an Acrylate Polymer Consisting of Periodic Polymer-Dispersed Liquid-Crystal Planes”, Chem. Mater., 1993, vol. 5, pp. 1533-1538. cited by applicant

Sutherland et al., “Electrically switchable volume gratings in polymer-dispersed liquid crystals”, Applied Physics Letters, Feb. 28, 1994, vol. 64, No. 9, pp. 1074-1076. cited by applicant

Sutherland et al., “Enhancing the electro-optical properties of liquid crystal nanodroplets for switchable Bragg gratings”, Proc. of SPIE, 2008, vol. 7050, pp. 705003-1-705003-9, doi: 10.1117/12.792629. cited by applicant

Sutherland et al., “Liquid crystal bragg gratings: dynamic optical elements for spatial light modulators”, Hardened Materials Branch, Hardened Materials Branch, AFRL-ML-WP-TP-2007-514, Jan. 2007, Wright-Patterson Air Force Base, OH, 18 pgs. cited by applicant

Sutherland et al., “Phenomenological model of anisotropic vol. hologram formation in liquid-

crystal-photopolymer mixtures”, *Journal of Applied Physics*, Jun. 30, 2004, vol. 96, No. 2, pp. 951-965, <https://doi.org/10.1063/1.1762713>. cited by applicant

Sutherland et al., “The physics of photopolymer liquid crystal composite holographic gratings”, presented at SPIE: Diffractive and Holographic Optics Technology San Jose, CA, 1996, SPIE, vol. 2689, pp. 158-169. cited by applicant

Sweatt, “Achromatic triplet using holographic optical elements”, *Applied Optics*, May 1977, vol. 16, No. 5, pp. 1390-1391. cited by applicant

Tahata et al., “Effects of Polymer Matrix on Electro-Optic Properties of Liquid Crystal Mixed With Polymer”, *Proc SPIE*, Mar. 11, 1996, vol. 2651, pp. 101-106, doi: 10.1117/12.235342. cited by applicant

Talukdar, “Technology Forecast: Augmented reality”, *Changing the economics of Smartglasses*, Issue 2, 2016, 5 pgs. cited by applicant

Tao et al., “TiO₂ nanocomposites with high refractive index and transparency”, *J. Mater. Chem.*, Oct. 4, 2011, vol. 21, pp. 18623-18629. cited by applicant

Titus et al., “Efficient, Accurate Liquid Crystal Digital Light Deflector”, *Proc. SPIE* 3633, *Diffractive and Holographic Technologies, Systems, and Spatial Light Modulators VI*, 1 Jun. 1, 1999, doi: 10.1117/12.349334, 10 pgs. cited by applicant

Tiziani, “Physical Properties of Speckles”, *Speckle Metrology*, Chapter 2, Academic Press, Inc., 1978, pp. 5-9. cited by applicant

Tominaga et al., “Fabrication of holographic polymer dispersed liquid crystals doped with gold nanoparticles”, 2010 Japanese Liquid Crystal Society Annual Meeting, 2 pgs. cited by applicant

Tomita, “Holographic assembly of nanoparticles in photopolymers for photonic applications”, *The International Society for Optical Engineering*, SPIE Newsroom, 2006, 3 pgs., doi: 10.1117/2.1200612.0475. cited by applicant

Tondiglia et al., “Holographic Formation of Electro-Optical Polymer-Liquid Crystal Photonic Crystals”, *Advanced Materials*, 2002, Published Online Nov. 8, 2001, vol. 14, No. 3, pp. 187-191. cited by applicant

Trisnadi, “Hadamard Speckle Contrast Reduction”, *Optics Letters*, Jan. 1, 2004, vol. 29, No. 1, pp. 11-13. cited by applicant

Ducharme, “Microlens diffusers for efficient laser speckle generation”, *Optics Express*, Oct. 29, 2007, vol. 15, No. 22, pp. 14573-14579. cited by applicant

Duong et al., “Centrifugal Deposition of Iron Oxide Magnetic Nanorods for Hyperthermia Application”, *Journal of Thermal Engineering*, Yildiz Technical University Press, Istanbul, Turkey, Apr. 2015, vol. 1, No. 2, pp. 99-103. cited by applicant

Escuti et al., “Holographic photonic crystals”, *Society of Photo-Optical Instrumentation Engineers*, Sep. 2004, vol. 43, No. 9, pp. 1973-1987, DOI: 10.1117/1.1773773. cited by applicant

Fattal et al., “A multi directional backlight for a wide-angle glasses-free three-dimensional display”, *Nature*, Mar. 21, 2012, vol. 495, pp. 348-351. cited by applicant

Flory, “Molecular size distribution in three-dimensional polymers. I. Gelation”, *J. Am. Chem. Soc.*, Nov. 1941, vol. 63, pp. 3083-3090. cited by applicant

Fontecchio et al., “Spatially Pixelated Reflective Arrays from Holographic Polymer Dispersed Liquid Crystals”, *SID 00 Digest*, May 2000, pp. 774-776. cited by applicant

Forman et al., “Materials development for PhotoINhibited SuperResolution (PINSR) lithography”, *Proc. of SPIE*, 2012, vol. 8249, pp. 824904-1-824904-9, doi: 10.1117/12.908512. cited by applicant

Forman et al., “Radical diffusion limits to photoinhibited superresolution lithography”, *Phys. Chem. Chem. Phys.*, May 31, 2013, vol. 15, pp. 14862-14867, DOI: 10.1039/c3cp51512. cited by applicant

Friedrich-Schiller, “Spatial Noise and Speckle”, Version 1.12.2011, Dec. 2011, Abbe School of Photonics, Jena, Germany, 27 pgs. cited by applicant

Fries et al., “Real-time beam shaping without additional optical elements”, *Light Science &*

Applications, Jun. 20, 2018, vol. 7, No. 18, doi: 10.1038/s41377-018-0014-0. cited by applicant

Fuh et al., "Thermally and Electrically Switchable Gratings Based Upon the Polymer-Balls Type Polymer-Dispersed Liquid Crystal Films", Appl. Phys. Vol. 41, No. 22, Aug. 1, 2002, pp. 4585-4589. cited by applicant

Fujii et al., "Nanoparticle-polymer-composite volume gratings incorporating chain-transfer agents for holography and slow-neutron optics", Optics Letters, Apr. 25, 2014, vol. 39, Issue 12, 5 pgs. cited by applicant

Funayama et al., "Proposal of a new type thin film light-waveguide display device using", The International Conference on Electrical Engineering, 2008, No. P-044, 5 pgs. cited by applicant

Gabor, "Laser Speckle and its Elimination", BM Research and Development, Eliminating Speckle Noise, Sep. 1970, vol. 14, No. 5, pp. 509-514. cited by applicant

Gardiner et al., "Bistable liquid-crystals reduce power consumption for high-efficiency smart glazing", SPIE, 2009, 10.1117/2.1200904. 1596, 2 pgs. cited by applicant

Gerritsen et al., "Application of Kogelnik's two-wave theory to deep, slanted, highly efficient, relief transmission gratings", Applied Optics, Mar. 1, 1991, vol. 30; No. 7, pp. 807-814. cited by applicant

Giancola, "Holographic Diffuser, Makes Light Work of Screen Tests", Photonics Spectra, 1996, vol. 30, No. 8, pp. 121-122. cited by applicant

Golub et al., "Bragg properties of efficient surface relief gratings in the resonance domain", Optics Communications, Feb. 24, 2004, vol. 235, pp. 261-267, doi: 10.1016/j.optcom.2004.02.069. cited by applicant

Goodman, "Introduction to Fourier Optics", Second Edition, Jan. 1996, 457 pgs. cited by applicant

Goodman, "Some fundamental properties of speckle", J. Opt. Soc. Am., Nov. 1976, vol. 66, No. 11, pp. 1145-1150. cited by applicant

Goodman, "Statistical Properties of Laser Speckle Patterns", Applied Physics, 1975, vol. 9, Chapter 2, Laser Speckle and Related Phenomena, pp. 9-75. cited by applicant

Goodman et al., "Speckle Reduction by a Moving Diffuser in Laser Projection Displays", The Optical Society of America, 2000, 15 pgs. cited by applicant

Guldin et al., "Self-Cleaning Antireflective Optical Coatings", Nano Letters, Oct. 14, 2013, vol. 13, pp. 5329-5335. cited by applicant

Guo et al., "Analysis of the effects of viscosity, volume and temperature in photopolymer material for holographic applications", Proc. SPIE, May 2013, vol. 8776, pp. 87760J-1-87760-J15, DOI: 10.1117/12.2018330. cited by applicant

Guo et al., "Review Article: A Review of the Optimisation of Photopolymer Materials for Holographic Data Storage", Physics Research International, vol. 2012, Article ID 803439, Academic Editor: Sergi Gallego, 16 pages, <http://dx.doi.org/10.1155/2012/803439>, May 4, 2012. cited by applicant

Han et al., "Study of Holographic Waveguide Display System", Advanced Photonics for Communications, 2014, 4 pgs. cited by applicant

Harbers et al., "I-15.3: LED Backlighting for LCD-HDTV", Journal of the Society for Information Display, 2002, vol. 10, No. 4, pp. 347-350. cited by applicant

Harbers et al., "Performance of High Power LED Illuminators in Color Sequential Projection Displays", Lumileds Lighting, 2007, 4 pgs. cited by applicant

Harbers et al., "Performance of High Power LED Illuminators in Color Sequential Projection Displays", Lumileds, Aug. 7, 2001, 11 pgs. cited by applicant

Harbers et al., "Performance of High-Power LED illuminators in Projection Displays", Proc. Int. Disp. Workshops, Japan. Vol. 10, 2003, pp. 1585-1588. cited by applicant

Harding et al., "Reactive Liquid Crystal Materials for Optically Anisotropic Patterned Retarders", Merck, licrivue, 2008, ME-GR-RH-08-010, 20 pgs. cited by applicant

Harding et al., "Reactive Liquid Crystal Materials for Optically Anisotropic Patterned Retarders",

SPIE Lithography Asia—Taiwan, 2008, Proceedings vol. 7140, Lithography Asia 2008; 71402J, doi: 10.1117/12.805378. cited by applicant

Hariharan, “Optical Holography: Principles, techniques and applications”, Cambridge University Press, 1996, pp. 231-233. cited by applicant

Harris, “Photonic Devices”, EE 216 Principals and Models of Semiconductor Devices, Autumn 2002, 20 pgs. cited by applicant

Harrold et al., “3D Display Systems Hardware Research at Sharp Laboratories of Europe: an update”, Sharp Laboratories of Europe, Ltd., received May 21, 1999, 7 pgs. cited by applicant

Harthong et al., “Speckle phase averaging in high-resolution color holography”, J. Opt. Soc. Am. A, vol. 14, No. 2, Feb. 1997, pp. 405-409. cited by applicant

Hasan et al., “Tunable-focus lens for adaptive eyeglasses”, Optics Express, Jan. 23, 2017, vol. 25, No. 2, 1221, 13 pgs. cited by applicant

Hasman et al., “Diffraction Optics: Design, Realization, and Applications”, Fiber and Integrated Optics, vol. 16, 1997, pp. 1-25. cited by applicant

Hata et al., “Holographic nanoparticle-polymer composites based on step-growth thiol-ene photopolymerization”, Optical Materials Express, vol. 1, No. 2, Jun. 1, 2011, pp. 207-222. cited by applicant

He et al., “Dynamics of peristrophic multiplexing in holographic polymer-dispersed liquid crystal”, Liquid Crystals, Mar. 26, 2014, vol. 41, No. 5, pp. 673-684. cited by applicant

He et al., “Holographic 3D display based on polymer-dispersed liquid-crystal thin films”, Proceedings of China Display/Asia Display 2011, pp. 158-160. cited by applicant

He et al., “Properties of vol. Holograms Recording in Photopolymer Films with Various Pulse Exposures Repetition Frequencies”, Proceedings of SPIE vol. 5636, Bellingham, WA, 2005, pp. 842-848, doi: 10.1117/12.580978. cited by applicant

He et al., “Transmission Holographic Gratings Using Siloxane Containing Liquid Crystalline Compounds, Importance of Chemical Structure of Polymer Matrix Components”, Polymer Journal, Jun. 9, 2006, vol. 38, No. 7, pp. 678-685. cited by applicant

Herman et al., “Production and Uses of Diffractionless Beams”, J. Opt. Soc. Am. A., Jun. 1991, vol. 8, No. 6, pp. 932-942. cited by applicant

Hisano, “Alignment layer-free molecular ordering induced by masked photopolymerization with nonpolarized light”, Appl. Phys. Express 9, Jun. 6, 2016, p. 072601-1 -072601-4. cited by applicant

Hoepfner et al., “LED Front Projection Goes Mainstream”, Luminus Devices, Inc., Projection Summit, 2008, 18 pgs. cited by applicant

Holmes et al., “Controlling the Anisotropy of Holographic Polymer- Dispersed Liquid-Crystal Gratings”, Physical Review E, Jun. 11, 2002, vol. 65, p. 066603-1 -066603-4. cited by applicant

Hoyle et al., “Advances in the Polymerization of Thiol-Ene Formulations”, Heraeus Noblelight Fusion UV Inc., 2003 Conference, 6 pgs. cited by applicant

Hua, “Sunglass-like displays become a reality with free-form optical technology”, Illumination & Displays 3D Visualization and Imaging Systems Laboratory (3DVIS) College of Optical Sciences University of Arizona Tucson, AZ. 2014, 3 pgs. cited by applicant

Huang et al., “Diffraction properties of substrate guided-wave holograms”, Optical Engineering, Oct. 1995, vol. 34, No. 10, pp. 2891-2899. cited by applicant

Huang et al., “Theory and characteristics of holographic polymer dispersed liquid crystal transmission grating with scaffolding morphology”, Applied Optics, Jun. 20, 2012, vol. 51, No. 18, pp. 4013-4020. cited by applicant

Iannacchione et al., “Deuterium NMR and morphology study of copolymer-dispersed liquid-crystal Bragg gratings”, Europhysics Letters, 1996, vol. 36, No. 6, pp. 425-430. cited by applicant

Irie, “Photochromic diarylethenes for photonic devices”, Pure and Applied Chemistry, 1996, pp. 1367-1371, vol. 68, No. 7, IUPAC. cited by applicant

Jang et al., “Low Driving Voltage Holographic Polymer Dispersed Liquid Crystals with Chemically

Incorporated Graphene Oxide”, *Journal of Materials Chemistry*, 2011, vol. 21, pp. 19226-19232, doi. 10.1039/1jm13827h. cited by applicant

Jeng et al., “Aligning liquid crystal molecules”, *SPIE*, 2012, 10.1117/2.1201203.004148, 2 pgs. cited by applicant

Jeong et al., “Memory Effect of Polymer Dispersed Liquid Crystal by Hybridization with Nanoclay”, *express Polymer Letters*, vol. 4, No. 1, 2010, pp. 39-46, DOI: 10.3144/expresspolymlett.2010.7. cited by applicant

Jo et al., “Control of Liquid Crystal Pretilt Angle using Polymerization of Reactive Mesogen”, *IMID 2009 Digest*, P1-25, 2009, pp. 604-606. cited by applicant

Juhl, “Interference Lithography for Optical Devices and Coatings”, *Dissertation*, University of Illinois at Urbana-Champaign, 2010. cited by applicant

Juhl et al., “Holographically Directed Assembly of Polymer Nanocomposites”, *ACS Nano*, Oct. 7, 2010, vol. 4, No. 10, pp. 5953-5961. cited by applicant

Jurbergs et al., “New recording materials for the holographic industry”, *Proc. of SPIE*, 2009 Vol. 7233, pp. 72330K-1-72330L-10, doi: 10.1117/12.809579. cited by applicant

Kahn et al., “Private Line Report on Large Area Display”, *Kahn International*, Jan. 7, 2003, vol. 8, No. 10, 9 pgs. cited by applicant

Kakiuchida et al., “Multiple Bragg Diffractions with Different Wavelengths and Polarizations Composed of Liquid Crystal/Polymer Periodic Phases”, *ACS Omega*, Sep. 22, 2017, pp. 6081-6090, doi: 10.1021/acsomega.7b01149. cited by applicant

Karasawa et al., “Effects of Material Systems on the Polarization Behavior of Holographic Polymer Dispersed Liquid Crystal Gratings”, *Japanese Journal of Applied Physics*, Oct. 1997, vol. 36, No. 10, pp. 6388-6392. cited by applicant

Karp et al., “Planar micro-optic solar concentration using multiple imaging lenses into a common slab waveguide”, *Proc. of SPIE* vol. 7407, 2009 *SPIE*, pp. 74070D-1-74070D-11, CCC code: 0277-786X/09, doi: 10.1117/12.826531. cited by applicant

Karp et al., “Planar micro-optic solar concentrator”, *Optics Express*, Jan. 18, 2010, vol. 18, No. 2, pp. 1122-1133. cited by applicant

Kato et al., “Alignment-Controlled Holographic Polymer Dispersed Liquid Crystal (HPDLC) for Reflective Display Devices”, *SPIE*, 1998, vol. 3297, pp. 52-57. cited by applicant

Kessler, “Optics of Near to Eye Displays (NEDs)”, *Oasis 2013*, Tel Aviv, Feb. 19, 2013, 37 pgs. cited by applicant

Keuper et al., “26.1: RGB LED Illuminator for Pocket-Sized Projectors”, *SID 04 Digest*, 2004, ISSN/0004-0966X/04/3502, pp. 943-945. cited by applicant

Keuper et al., “P-126: Ultra-Compact LED based Image Projector for Portable Applications”, *SID 03 Digest*, 2003, ISSN/0003-0966X/03/3401-0713, pp. 713-715. cited by applicant

Kim et al., “Effect of Polymer Structure on the Morphology and Electro optic Properties of UV Curable PNLCS”, *Polymer*, Feb. 2000, vol. 41, pp. 1325-1335. cited by applicant

Kim et al., “Enhancement of electro-optical properties in holographic polymer-dispersed liquid crystal films by incorporation of multiwalled carbon nanotubes into a polyurethane acrylate matrix”, *Polym. Int.*, Jun. 16, 2010, vol. 59, pp. 1289-1295. cited by applicant

Kim et al., “Fabrication of Reflective Holographic PDLC for Blue”, *Molecular Crystals and Liquid Crystals Science*, 2001, vol. 368, pp. 3845-3853. cited by applicant

Kim et al., “Optimization of Holographic PDLC for Green”, *Mol. Cryst. Liq. Cryst.*, vol. 368, 2001, pp. 3855-3864. cited by applicant

Klein, “Optical Efficiency for Different Liquid Crystal Colour Displays”, *Digital Media Department*, HPL-2000-83, Jun. 29, 2000, 18 pgs. cited by applicant

Kogelnik, “Coupled Wave Theory for Thick Hologram Gratings”, *The Bell System Technical Journal*, vol. 48, No. 9, Nov. 1969, pp. 2909-2945. cited by applicant

Konuray et al., “State of the Art in Dual-Curing Acrylate Systems”, *Polymers*, Feb. 12, 2018, vol.

10, No. 178, 24 pgs, doi: 10.3390/polym10020178. cited by applicant

Kotakonda et al., “Electro-optical Switching of the Holographic Polymer-dispersed Liquid Crystal Diffraction Gratings”, *Journal of Optics A: Pure and Applied Optics*, Jan. 1, 2009, vol. 11, No. 2, 11 pgs. cited by applicant

Kress et al., “Diffraction and Holographic Optics as Optical Combiners in Head Mounted Displays”, *UbiComp '13*, Sep. 9-12, 2013, Session: Wearable Systems for Industrial Augmented Reality Applications, pp. 1479-1482. cited by applicant

Kwon et al., “Polymer waveguide notch filter using two stacked thermo-optic long-period gratings”, *IEEE Photonics Technology Letters*, Apr. 4, 2005, vol. 17, Issue 4, pp. 792-794, DOI: 10.1109/LPT.2005.844008. cited by applicant

Lauret et al., “Solving the Optics Equation for Effective LED Applications”, *Gaggione North America, LFFY System Design Workshop 2010*, Oct. 28, 2010, 26 pgs. cited by applicant

Lee, “Patents Shows Widespread Augmented Reality Innovation”, *PatentVue*, May 26, 2015, 5 pgs. cited by applicant

Levin et al., “A Closed Form Solution To Natural Image Matting”, *Illumination & Displays 3D Visualization and Imaging Systems Laboratory (3DVIS) College of Optical Sciences University of Arizona Tucson*, 2014, 8 pgs. cited by applicant

Levola, “Diffraction optics for virtual reality displays”, *Journal of the SID*, 2006, 14/5, pp. 467-475. cited by applicant

Levola et al., “Near-to-eye display with diffraction exit pupil expander having chevron design”, *Journal of the SID*, 2008, 16/8, pp. 857-862. cited by applicant

Levola et al., “Replicated slanted gratings with a high refractive index material for in and outcoupling of light”, *Optics Express*, vol. 15, Issue 5, 2007, pp. 2067-2074. cited by applicant

Li et al., “A low cost, label-free biosensor based on a novel double-sided grating waveguide coupler with sub-surface cavities”, *Sensors and Actuators B: Chemical*, Jan. 2015, vol. 206, pp. 371-380, <https://doi.org/10.1016/j.snb.2014.09.065>. cited by applicant

Li et al., “Design and Optimization of Tapered Light Pipes”, *Proceedings vol. 5529, Nonimaging Optics and Efficient Illumination Systems*, Sep. 29, 2004, doi: 10.1117/12.559844, 10 pgs. cited by applicant

Li et al., “Dual Paraboloid Reflector and Polarization Recycling Systems for Projection Display”, *Proceedings vol. 5002, Projection Displays IX*, Mar. 28, 2003, doi: 10.1117/12.479585, 12 pgs. cited by applicant

Li et al., “Light Pipe Based Optical Train and its Applications”, *Proceedings vol. 5524, Novel Optical Systems Design and Optimization VII*, Oct. 24, 2004, doi: 10.1117/12.559833, 10 pgs. cited by applicant

Li et al., “Novel Projection Engine with Dual Paraboloid Reflector and Polarization Recovery Systems”, *Wavien Inc., SPIE EI 5289-38*, Jan. 21, 2004, 49 pgs. cited by applicant

Li et al., “Polymer crystallization/melting induced thermal switching in a series of holographically patterned Bragg reflectors”, *Soft Matter*, Jul. 11, 2005, vol. 1, pp. 238-242. cited by applicant

Lin et al., “Ionic Liquids in Photopolymerizable Holographic Materials”, in book: *Holograms—Recording Materials and Applications*, Nov. 9, 2011, 21 pgs. cited by applicant

Liu et al., “Effect of Surfactant on the Electro-Optical Properties of Holographic Polymer Dispersed Liquid Crystal Bragg Gratings”, *Optical Materials*, 2005, vol. 27, pp. 1451-1455, available online Dec. 25, 2004, doi: 10.1016/j.optmat.2004.10.010. cited by applicant

Liu et al., “Holographic Polymer Dispersed Liquid Crystals Materials, Formation and Applications”, *Advances in OptoElectronics*, Nov. 30, 2008, vol. 2008, Article ID 684349, 52 pgs. cited by applicant

Liu et al., “Realization and Optimization of Holographic Waveguide Display System”, *Acta Optica Sinica*, vol. 37, Issue 5, Issuing date—May 10, 2017, pp. 310-317. cited by applicant

Lorek, “Experts Say Mass Adoption of augmented and Virtual Reality is Many Years Away”,

Siliconhills, Sep. 9, 2017, 4 pgs. cited by applicant

Loungnot et al., “Polymers for holographic recording: VI. Some basic ideas for modelling the kinetics of the recording process”, Pure and Applied Optics: Journal of the European Optical Society Part A, 1997, vol. 6, No. 2, pp. 225-245, <https://doi.org/10.1088/0963-9659/6/2/007>. cited by applicant

Lowenthal et al., “Speckle Removal by a Slowly Moving Diffuser Associated with a Motionless Diffuser”, Journal of the Optical Society of America, Jul. 1971, vol. 61, No. 7, pp. 847-851. cited by applicant

“FHS Lenses Series”, Fraen Corporation, www.fraen.com, Jun. 16, 2003, 10 pgs. cited by applicant

“FLP Lens Series for Luxeon™ Rebel and Rebel ES LEDs”, Fraen Corporation, www.fraensrl.com, Aug. 7, 2015, 8 pgs. cited by applicant

“Head-up Displays, See-through display for military aviation”, BAE Systems, 2016, 3 pgs. cited by applicant

“Holder for LUXEON Rebel—Part No. 180”, Polymer Optics Ltd., 2008, 12 pgs. cited by applicant

“LED 7-Segment Displays”, Lumex, uk.digikey.com, 2003, UK031, 36 pgs. cited by applicant

“LED325W UVTOP UV LED with Window”, Thorlabs, Specifications and Documentation, 21978-S01 Rev. A, Apr. 8, 2011, 5 pgs. cited by applicant

“Liquid Crystal Phases”, Phases of Liquid Crystals, <http://plc.cwru.edu/tutorial/enhanced/files/lc/phase>, Retrieved on Sep. 21, 2004, 6 pgs. cited by applicant

“LiteHUD Head-up display”, BAE Systems, 2016, 2 pgs. cited by applicant

“LiteHUD Head-up display infographic”, BAE Systems, 2017, 2 pgs. cited by applicant

“Luxeon C: Power Light Source”, Philips Lumileds, www.philipslumileds.com, 2012, 18 pgs. cited by applicant

“Luxeon Rebel ES: Leading efficacy and light output, maximum design flexibility”, Luxeon Rebel ES Datasheet DS61 Feb. 21, 2013, www.philipslumileds.com, 2013, 33 pgs. cited by applicant

“Mobile Display Report”, Insight Media, LLC, Apr. 2012, vol. 7, No. 4, 72 pgs. cited by applicant

“Molecular Imprints Imprio 55”, Engineering at Illinois, Micro + Nanotechnology Lab, Retrieved from <https://mntl.illinois.edu/facilities/cleanrooms/equipment/Nano-Imprint.asp>, Dec. 28, 2015, 2 pgs. cited by applicant

“Navy awards SGB Labs a contract for HMDs for simulation and training”, Press releases, DigiLens, Oct. 2012, pp. 1-2. cited by applicant

“Optical measurements of retinal flow”, Industrial Research Limited, Feb. 2012, 18 pgs. cited by applicant

“Osterhout Design Group Develops Next-Generation, Fully-integrated Smart Glasses Using Qualcomm Technologies”, ODG, www.osterhoutgroup.com, Sep. 18, 2014, 2 pgs. cited by applicant

“Plastic has replaced glass in photochromic lens”, www.plastemart.com, 2003, 1 page. cited by applicant

“Range Finding Using Pulse Lasers”, OSRAM, Opto Semiconductors, Sep. 10, 2004, 7 pgs. cited by applicant

“Response time in Liquid-Crystal Variable Retarders”, Meadowlark Optics, Inc., 2005, 4 pgs. cited by applicant

“Secondary Optics Design Considerations for SuperFlux LEDs”, Lumileds, application brief AB20-5, Sep. 2002, 23 pgs. cited by applicant

“Solid-State Optical Mouse Sensor with Quadrature Outputs”, IC Datasheet, UniquelCs, Jul. 15, 2004, 11 pgs. cited by applicant

“SVGA TransparentVLSITM Microdisplay Evaluation Kit”, Radiant Images, Inc., Product Data Sheet, 2003, 3 pgs. cited by applicant

“Technical Data Sheet LPR1”, Luminus Devices, Inc., Luminus Projection Chipset, Release 1, Preliminary, Revision B, Sep. 21, 2004, 9 pgs. cited by applicant

“The Next Generation of TV”, SID Information Display, Nov./Dec. 2014, vol. 30, No. 6, 56 pgs. cited by applicant

“Thermal Management Considerations for SuperFlux LEDs”, Lumileds, application brief AB20-4, Sep. 2002, 14 pgs. cited by applicant

“USAF Awards SBG Labs an SBIR Contract for Wide Field of View HUD”, Press Release, SBG Labs DigiLens, Apr. 2014, 2 pgs. cited by applicant

“UVTOP240”, Roithner LaserTechnik GmbH, v 2.0, Jun. 24, 2013, 6 pgs. cited by applicant

“UVTOP310”, Roithner LaserTechnik GmbH, v 2.0, Jun. 24, 2013, 6 pgs. cited by applicant

“Velodyne's HDL-64E: A High Definition Lidar Sensor for 3-D Applications”, High Definition Lidar, white paper, Oct. 2007, 7 pgs. cited by applicant

“VerLASE Gets Patent for Breakthrough Color Conversion Technology That Enables Full Color MicroLED Arrays for Near Eye Displays”, Cision PRweb, Apr. 28, 2015, Retrieved from the Internet <http://www.prweb.com/releases/2015/04/prweb12681038.htm>, 3 pgs. cited by applicant

“X-Cubes—Revisited for LCOS”, BASID, RAF Electronics Corp. Rawson Optics, Inc., Oct. 24, 2002, 16 pgs. cited by applicant

Aachen, “Design of plastic optics for LED applications”, Optics Colloquium 2009, Mar. 19, 2009, 30 pgs. cited by applicant

Abbate et al., “Characterization of LC-polymer composites for opto-electronic application”, Proceedings of OPTOEL'03, Leganes-Madrid, Spain, Jul. 14-16, 2003, 4 pgs. cited by applicant

Al-Kalbani et al., “Ocular Microtremor laser speckle metrology”, Proc. of SPIE, 2009, vol. 7176 717606-1, 12 pgs., doi: 10.1117/12.808855. cited by applicant

Almanza-Workman et al., “Planarization coating for polyimide substrates used in roll-to-roll fabrication of active matrix backplanes for flexible displays”, HP Laboratories, HPL-2012-23, Feb. 6, 2012, 12 pgs. cited by applicant

Amitai et al., “Visor-display design based on planar holographic optics”, Applied Optics, vol. 34, No. 8, Mar. 10, 1995, pp. 1352-1356. cited by applicant

Amundson et al., “Morphology and electro-optic properties of polymer- dispersed liquid-crystal films”, Physical Review E, Feb. 1997, vol. 55. No. 2, pp. 1646-1654. cited by applicant

An et al., “Speckle suppression in laser display using several partially coherent beams”, Optics Express, Jan. 5, 2009, vol. 17, No. 1, pp. 92-103, first published Dec. 22, 2008. cited by applicant

Apter et al., “Electrooptical Wide-Angle Beam Deflector Based on Fringing-Field-Induced Refractive Inhomogeneity in a Liquid Crystal Layer”, 23rd IEEE Convention of Electrical and Electronics Engineers in Israel, Sep. 6-7, 2004, pp. 240-243. cited by applicant

Arnold et al., “52.3: An Improved Polarizing Beamsplitter LCOS Projection Display Based on Wire-Grid Polarizers”, Society for Information Display, Jun. 2001, pp. 1282-1285. cited by applicant

Ayras et al., “Exit pupil expander with a large field of view based on diffractive optics”, Journal of the Society for Information Display, May 18, 2009, vol. 17, No. 8, pp. 659-664, DOI: 10.1889/JSID17.8.659. cited by applicant

Baets et al., “Resonant-Cavity Light-Emitting Diodes: a review”, Proceedings of SPIE, 2003, vol. 4996, pp. 74-86. cited by applicant

Bayer et al., “Introduction to Helmet-Mounted Displays”, 2016, pp. 47-108. cited by applicant

Beckel et al., “Electro-optic properties of thiol-ene polymer stabilized ferroelectric liquid crystals”, Liquid Crystals, vol. 30, No. 11, Nov. 2003, pp. 1343-1350, DOI: 10.1080/02678290310001605910. cited by applicant

Bergkvist, “Biospeckle-based Study of the Line Profile of Light Scattered in Strawberries”, Master Thesis, Lund Reports on Atomic Physics, LRAP-220, Lund 1997, pp. 1-62. cited by applicant

Bernards et al., “Nanoscale porosity in polymer films: fabrication and therapeutic applications”,

Soft Matter, Jan. 1, 2010, vol. 6, No. 8, pp. 1621-1631, doi: 10.1039/B922303G. cited by applicant

Bhuvaneshwaran et al., "Spectral response of Bragg gratings in multimode polymer waveguides", Applied Optics, Dec. 1, 2017, vol. 56. No. 34, pp. 9573-9582, doi: 10.1364/AO.56.009573. cited by applicant

Bleha et al., "Binocular Holographic Waveguide Visor Display", SID Symposium Digest of Technical Papers, Holoeye Systems Inc., Jun. 2014, San Diego, CA, 4 pgs. cited by applicant

Bleha et al., "D-ILA Technology for High Resolution Projection Displays", Sep. 10, 2003, Proceedings, vol. 5080, 11 pgs., doi:10.1117/12.497532. cited by applicant

Bone, "Design Obstacles for LCOS Displays in Projection Applications Optics architectures for LCOS are still evolving", Aurora Systems Inc., Bay Area SID Seminar, Mar. 27, 2001, 22 pgs. cited by applicant

Born et al., "Optics of Crystals", Principles of Optics 5th Edition 1975, pp. 705-707. cited by applicant

Bourzac, "Magic Leap Needs to Engineer a Miracle", Intelligent Machines, Jun. 11, 2015, 7 pgs. cited by applicant

Bowen et al., "Optimisation of interdigitated electrodes for piezoelectric actuators and active fibre composites", J Electroceram, Jul. 2006, vol. 16, pp. 263-269, DOI 10.1007/s10832-006-9862-8. cited by applicant

Bowley et al., "Variable-wavelength switchable Bragg gratings formed in polymer-dispersed liquid crystals", Applied Physics Letters, Jul. 2, 2001, vol. 79, No. 1, pp. 9-11, DOI: 10.1063/1.1383566. cited by applicant

Bronnikov et al., "Polymer-Dispersed Liquid Crystals: Progress in Preparation, Investigation and Application", Journal of Macromolecular Science Part B, published online Sep. 30, 2013, vol. 52, pp. 1718-1738, DOI: 10.1080/00222348.2013.808926. cited by applicant

Brown, "Waveguide Displays", Rockwell Collins, 2015, 11 pgs. cited by applicant

Bruzzo et al., "Compact, high-brightness LED illumination for projection systems", Journal of the Society for Information Display, vol. 17, No. 12, Dec. 2009, pp. 1043-1049, Doi: 10.1189/JSID17.12. 1043. cited by applicant

Buckley, "Colour holographic laser projection technology for heads-up and instrument cluster displays", Conference: Proc. SID Conference 14th Annual Symposium on Vehicle Displays, Jan. 2007, 5 pgs. cited by applicant

Buckley, "Pixtronix DMS technology for head-up displays", Pixtronix, Inc., Jan. 2011, 4 pgs. cited by applicant

Buckley et al., "Full colour holographic laser projector HUD", Light Blue Optics Ltd., Aug. 10, 2015, 5 pgs. cited by applicant

Buckley et al., "Rear-view virtual image displays", in Proc. SID Conference 16th Annual Symposium on Vehicle Displays, Jan. 2009, 5 pgs. cited by applicant

Bunning et al., "Effect of gel-point versus conversion on the real-time dynamics of holographic polymer-dispersed liquid crystal (HPDLC) formation", Proceedings of SPIE—vol. 5213, Liquid Crystals VII, Iam-Choon Khoo, Editor, Dec. 2003, pp. 123-129. cited by applicant

Bunning et al., "Electro-optical photonic crystals formed in H-PDLCs by thiol-ene photopolymerization", American Physical Society, Annual APS, Mar. 3-7, 2003, abstract #R1.135. cited by applicant

Bunning et al., "Holographic Polymer-Dispersed Liquid Crystals (H-PDLCs)1", Annual Review of Material Science, 2000, vol. 30, pp. 83-115. cited by applicant

Bunning et al., "Morphology of Anisotropic Polymer Dispersed Liquid Crystals and the Effect of Monomer Functionality", Journal of Polymer Science: Part B: Polymer Physics, Jul. 30, 1997, vol. 35, pp. 2825-2833. cited by applicant

Busbee et al., "SiO₂ Nanoparticle Sequestration via Reactive Functionalization in Holographic Polymer-Dispersed Liquid Crystals", Advanced Materials, Sep. 2009, vol. 21, pp. 3659-3662, DOI:

10.1002/adma.200900298. cited by applicant

Butler et al., "Diffractive Properties of Highly Birefringent Volume Gratings: Investigation", *Journal of Optical Society of America*, Feb. 2002, vol. 19, No. 2, pp. 183-189. cited by applicant

Cai et al., "Recent advances in antireflective surfaces based on nanostructure arrays", *Materials Horizons*, 2015, vol. 2, pp. 37-53, DOI: 10.1038/c4mh00140k. cited by applicant

Cameron, "Optical Waveguide Technology & Its Application In Head Mounted Displays", *Proc. of SPIE*, May 22, 2012, vol. 8383, pp. 83830E-1-83830E-11, doi: 10.1117/12.923660. cited by applicant

Cameron, "The Application of Holographic Optical Waveguide Technology to Q-Sight™ Family of Helmet Mounted Displays", *Proc. of SPIE*, 2009, vol. 7326, 11 pages, doi: 10.1117/12.818581. cited by applicant

Caputo et al., "POLICRYPS Composite Materials: Features and Applications", *Advances in Composite Materials—Analysis of Natural and Man-Made Materials*, www.intechopen.com, Sep. 2011, pp. 93-118. cited by applicant

Caputo et al., "POLICRYPS Switchable Holographic Grating: A Promising Grating Electro-Optical Pixel for High Resolution Display Application", *Journal of Display Technology*, Mar. 2006, vol. 2, No. 1, pp. 38-51, Doi: 10.1109/JDT.2005.864156. cited by applicant

Caputo et al., "POLICRYPS: a liquid crystal composed nano/microstructure with a wide range of optical and electro-optical applications", *Journal of Optics A: Pure and Applied Optics*, Jan. 15, 2009, vol. 11, No. 2, 13 pgs., doi: 10.1088/1464-4258/11/2/024017. cited by applicant

Carclo Optics, "Guide to choosing secondary optics", Carclo Optics, Dec. 15, 2014, www.carclo-optics.com, 48 pgs. cited by applicant

Carothers, "Polymers and polyfunctionality", *Transactions of the Faraday Society*, Jan. 1, 1936, vol. 32, pp. 39-49, doi: 10.1039/TF9363200039. cited by applicant

Chen et al., "Polarization rotators fabricated by thermally-switched liquid crystal alignments based on rubbed poly(N-vinyl carbazole) films", *Optics Express*, Apr. 11, 2011, vol. 19, No. 8, pp. 7553-7558, first published Apr. 5, 2011. cited by applicant

Cheng et al., "Design of an ultra-thin near-eye display with geometrical waveguide and freeform optics", *Optics Express*, Aug. 2014, 16 pgs., DOI:10.1364/OE.22.020705. cited by applicant

Chi et al., "Ultralow-refractive-index optical thin films through nanoscale etching of ordered mesoporous silica films", *Optic Letters*, May 1, 2012, vol. 37, No. 9, pp. 1406-1408, first published Apr. 19, 2012. cited by applicant

Chigrinov et al., "Photo-aligning by azo-dyes: Physics and applications", *Liquid Crystals Today*, Sep. 6, 2006, <http://www.tandfonline.com/action/journalInformation?journalCode=tlcy20>, 15 pgs. cited by applicant

Cho et al., "Electro-optic Properties of CO₂ Fixed Polymer/Nematic LC Composite Films", *Journal of Applied Polymer Science*, Nov. 5, 2000, vol. 81, Issue 11, pp. 2744-2753. cited by applicant

Cho et al., "Optimization of Holographic Polymer Dispersed Liquid Crystals for Ternary Monomers", *Polymer International*, Nov. 1999, vol. 48, pp. 1085-1090. cited by applicant

Colegrove et al., "p. 59: Technology of Stacking HPDLC for Higher Reflectance", *SID 00 Digest*, May 2000, pp. 770-773. cited by applicant

Crawford, "Electrically Switchable Bragg Gratings", *Optics & Photonics News*, Apr. 2003, pp. 54-59. cited by applicant

Cruz-Arreola et al., "Diffraction of beams by infinite or finite amplitude-phase gratings", *Investigacio' n Revista Mexicana De Fi'sica*, Feb. 2011, vol. 57, No. 1, pp. 6-16. cited by applicant

Dabrowski, "High Birefringence Liquid Crystals", *Crystals*, Sep. 3, 2013, vol. 3, No. 3, pp. 443-482, doi: 10.3390/cryst3030443. cited by applicant

Dainty, "Some statistical properties of random speckle patterns in coherent and partially coherent illumination", *Optica Acta*, Mar. 12, 1970, vol. 17, No. 10, pp. 761-772. cited by applicant

Date, "Alignment Control in Holographic Polymer Dispersed Liquid Crystal", *Journal of*

Photopolymer Science and Technology, Nov. 2, 2000, vol. 13, No. 2, pp. 289-294. cited by applicant

Date et al., "52.3: Direct-viewing Display Using Alignment-controlled PDLC and Holographic PDLC", Society for Information Display Digest, May 2000, pp. 1184-1187, DOI: 10.1889/1.1832877. cited by applicant

Date et al., "Full-color reflective display device using holographically fabricated polymer-dispersed liquid crystal (HPDLC)", Journal of the SID, 1999, vol. 7, No. 1, pp. 17-22. cited by applicant

De Bitetto, "White light viewing of surface holograms by simple dispersion compensation", Applied Physics Letters, Dec. 15, 1966, vol. 9, No. 12, pp. 417-418. cited by applicant

De Sarkar et al., "Effect of Monomer Functionality on the Morphology and Performance of Holographic Transmission Gratings Recorded on Polymer Dispersed Liquid Crystals", Macromolecules, 2003, vol. 36, No. 3, pp. 630-638. cited by applicant

Developer World, "Create customized augmented reality solutions", printed Oct. 19, 2017, LMX-001 holographic waveguide display, Sony Developer World, 3 pgs. cited by applicant

Dhar et al., "Recording media that exhibit high dynamic range for digital holographic data storage", Optics Letters, Apr. 1, 1999, vol. 24, No. 7, pp. 487-489. cited by applicant

Domash et al., "Applications of switchable Polaroid holograms", SPIE Proceedings, vol. 2152, Diffractive and Holographic Optics Technology, Jan. 23-29, 1994, Los Angeles, CA, pp. 127-138, ISBN: 0-8194-1447-6. cited by applicant

Doolittle, "Studies in Newtonian Flow. II. The Dependence of the Viscosity of Liquids on Free-Space", Journal of Applied Physics, 1951, vol. 22, Issue 12, pp. 1471-1475, published online Apr. 29, 2004, <https://doi.org/10.1063/1.1699894>. cited by applicant

Drake et al., "Waveguide Hologram Fingerprint Entry Device", Optical Engineering, Sep. 1996, vol. 35, No. 9, pp. 2499-2505. cited by applicant

Drevensek-Olenik et al., "In-Plane Switching of Holographic Polymer-Dispersed Liquid Crystal Transmission Gratings", Mol. Cryst. Liq. Cryst., 2008, vol. 495, pp. 177/[529]-185/[537], DOI: 10.1080/15421400802432584. cited by applicant

Drevensek-Olenik et al., "Optical diffraction gratings from polymer-dispersed liquid crystals switched by interdigitated electrodes", Journal of Applied Physics, Dec. 1, 2004, vol. 96, No. 11, pp. 6207-6212, DOI: 10.1063/1.1807027. cited by applicant

European Office Action for European Patent Application No. 13192383.1 dated Oct. 16, 2017, 5 pgs. cited by applicant

Extended European Search Report for EP Application No. 13192383.1, dated Apr. 2, 2014, 7 pgs. cited by applicant

Extended European Search Report for European Application No. 13765610.4 dated Feb. 16, 2016, 6 pgs. cited by applicant

Extended European Search Report for European Application No. 15187491.4, search completed Jan. 15, 2016, mailed Jan. 28, 2016, 5 pgs. cited by applicant

Extended European Search Report for European Application No. 18867522.7, Search completed Sep. 15, 2021, Mailed Sep. 24, 2021, 9 Pgs. cited by applicant

Extended European Search Report for European Application No. 18898154.2, Search completed Aug. 13, 2021, Mailed Aug. 23, 2021, 7 pgs. cited by applicant

Extended European Search Report for European Application No. 19736108.2, Search completed Sep. 15, 2021, Mailed Sep. 27, 2021, 8 pgs. cited by applicant

Extended European Search Report for European Application No. 19897355.4, Search completed Jul. 12, 2022, Mailed Jul. 21, 2022, 8 Pgs. cited by applicant

Extended European Search Report for European Application No. 18727645.6, Search completed Oct. 14, 2020, Mailed Oct. 23, 2020, 13 pgs. cited by applicant

Extended Search Report for European Application No. 18898841.4, Search completed Mar. 18, 2021, Mailed Mar. 26, 2021, 10 pgs. cited by applicant

International Preliminary Report on Patentability for International Application No.
PCT/GB2010/000835, issued Nov. 1, 2011, mailed Nov. 10, 2011, 9 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2010/001920, issued Apr. 11, 2012, mailed Apr. 19, 2012, 10 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2010/001982, report issued May 1, 2012, mailed May 10, 2012, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2013/000273, issued Dec. 23, 2014, mailed Dec. 31, 2014, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2015/000203, issued Mar. 21, 2017, mailed Mar. 30, 2017, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2016/000005, Report issued Jul. 18, 2017, Mailed Jul. 27, 2017, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2016/000036, issued Aug. 29, 2017, mailed Sep. 8, 2017, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2016/000051, Report issued Sep. 19, 2017, Mailed Sep. 28, 2017, 7 Pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/GB2016/000065, issued Oct. 3, 2017, mailed Oct. 12, 2017, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/IB2008/001909, Report issued Jan. 26, 2010, Mailed Jan. 26, 2010, 5 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/US2018/012227, Report issued Jul. 30, 2019, Mailed Aug. 8, 2019, 7 Pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/US2018/037410, Report issued Jul. 14, 2020, Mailed Jul. 23, 2020, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/US2018/048636, Report issued Jul. 14, 2020, Mailed Jul. 23, 2020, 9 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/US2018/062835, Report issued Jul. 14, 2020, Mailed Jul. 23, 2020, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/US2019/012758, Report issued Jul. 14, 2020, Mailed Jul. 23, 2020, 4 Pgs. cited by applicant
International Preliminary Report on Patentability for International Application No.
PCT/US2019/012759, Report issued Jul. 14, 2020, Mailed Jul. 23, 2020, 6 Pgs. cited by applicant
International Preliminary Report on Patentability for International Application PCT
/US2018/015553, Report issued Jun. 4, 2019, Mailed Jun. 13, 2019, 6 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2009/051676, issued Jun. 14, 2011, mailed Jun. 23, 2011, 6 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2011/000349, issued Sep. 18, 2012, mailed Sep. 27, 2012, 10 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2012/000331, issued Oct. 8, 2013, mailed Oct. 17, 2013, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2012/000677, issued Feb. 25, 2014, mailed Mar. 6, 2014, 5 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2013/000005, issued Jul. 8, 2014, mailed Jul. 17, 2014, 12 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2013/000210, issued Nov. 11, 2014, Mailed Nov. 20, 2014, 6 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2014/000197, issued Nov. 24, 2015, mailed Dec. 3, 2015, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2014/000295, issued Feb. 2, 2016, mailed Feb. 11, 2016, 4 pgs. cited by applicant

International Preliminary Report on Patentability for International Application
PCT/GB2015/000225, issued Feb. 14, 2017, mailed Feb. 23, 2017, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2015/000228, issued Feb. 14, 2017, mailed Feb. 23, 2017, 11 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2015/000274, Issued Mar. 28, 2017, mailed Apr. 6, 2017, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2016/000003, issued Jul. 18, 2017, mailed Jul. 27, 2017, 11 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2016/000014, issued Jul. 25, 2017, mailed Aug. 3, 2017, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2017/000015, Report Completed Aug. 7, 2018, Mailed Aug. 16, 2018, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2017/000040, Report issued Sep. 25, 2018, Mailed Oct. 4, 2018, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/GB2017/000055, issued Oct. 16, 2018, Mailed Oct. 25, 2018, 9 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2014/011736, issued Jul. 21, 2015, mailed Jul. 30, 2015, 9 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2016/017091, issued Aug. 15, 2017, mailed Aug. 24, 2017, 5 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2018/012691, issued Jul. 9, 2019, Mailed Jul. 18, 2019, 10 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2018/048960, Report issued on Mar. 3, 2020, Mailed on Mar. 12, 2020, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2018/056150, Report Issued on Apr. 21, 2020, Mailed on Apr. 30, 2020, 6 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2019/012764, Report issued Jul. 14, 2020, Mailed Jul. 23, 2020, 5 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2019/031163, Report issued Nov. 10, 2020, Mailed Nov. 19, 2020, 6 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2019/043496 Report issued Jan. 26, 2021, Mailed Feb. 4, 2021, 5 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2019/047097 issued Sep. 28, 2021, Mailed on Nov. 4, 2021, 7 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2019/064765, Report issued Oct. 19, 2020, Mailed Oct. 28, 2020, 27 pgs. cited by applicant
International Preliminary Report on Patentability for International Application
PCT/US2019/065478, Report issued Jun. 8, 2021, mailed on Jun. 24, 2021, 9 pgs. cited by applicant
Trisnadi, "Speckle contrast reduction in laser projection displays", Proc. SPIE 4657, 2002, 7 pgs. cited by applicant
Tzeng et al., "Axially symmetric polarization converters based on photo-aligned liquid crystal films", Optics Express, Mar. 17, 2008, vol. 16, No. 6, pp. 3768-3775. cited by applicant
Upatnieks et al., "Color Holograms for white light reconstruction", Applied Physics Letters, Jun. 1, 1996, vol. 8, No. 11, pp. 286-287. cited by applicant

Urey, "Diffractive exit pupil expander for display applications", *Applied Optics*, Nov. 10, 2001, vol. 40, Issue 32, pp. 5840-5851. cited by applicant

Ushenko, "The Vector Structure of Laser Biospeckle Fields and Polarization Diagnostics of Collagen Skin Structures", *Laser Physics*, 2000, vol. 10, No. 5, pp. 1143-1149. cited by applicant

Valoriani, "Mixed Reality: Dalle demo a un prodotto", *Disruptive Technologies Conference*, Sep. 23, 2016, 67 pgs. cited by applicant

Van Gerwen et al., "Nanoscaled interdigitated electrode arrays for biochemical sensors", *Sensors and Actuators*, Mar. 3, 1998, vol. B 49, pp. 73-80. cited by applicant

Vecchi, "Studi ESR Di Sistemi Complessi Basati Su Cristalli Liquidi", Thesis, University of Bologna, Department of Physical and Inorganic Chemistry, 2004-2006, 110 pgs. cited by applicant

Veltri et al., "Model for the photoinduced formation of diffraction gratings in liquid-crystalline composite materials", *Applied Physics Letters*, May 3, 2004, vol. 84, No. 18, pp. 3492-3494. cited by applicant

Vita, "Switchable Bragg Gratings", Thesis, Università degli Studi di Napoli Federico II, Nov. 2005, 103 pgs. cited by applicant

Vuzix, "M3000 Smart Glasses, Advanced Waveguide Optics", brochure, Jan. 1, 2017, 2 pgs. cited by applicant

Waldern et al., "Waveguide Optics for All Day Wearable Displays", Jun. 20, 2017, 35 pgs. cited by applicant

Wang et al., "Liquid-crystal blazed-grating beam deflector", *Applied Optics*, Dec. 10, 2000, vol. 39, No. 35, pp. 6545-6555. cited by applicant

Wang et al., "Optical Design of Waveguide Holographic Binocular Display for Machine Vision", *Applied Mechanics and Materials*, Sep. 27, 2013, vols. 427-429, pp. 763-769. cited by applicant

Wang et al., "Speckle reduction in laser projection systems by diffractive optical elements", *Applied Optics*, Apr. 1, 1998, vol. 37, No. 10, pp. 1770-1775. cited by applicant

Weber et al., "Giant Birefringent Optics in Multilayer Polymer Mirrors", *Science*, Mar. 31, 2000, vol. 287, pp. 2451-2456. cited by applicant

Webster, "Webster's Third New International Dictionary 433", (1986), 3 pages. cited by applicant

Wei An, "Industrial Applications of Speckle Techniques", Doctoral Thesis, Royal Institute of Technology, Department of Production Engineering, Chair of Industrial Metrology & Optics, Stockholm, Sweden 2002, 76 pgs. cited by applicant

Welde et al., "Investigation of methods for speckle contrast reduction", Master of Science in Electronics, Jul. 2010, Norwegian University of Science and Technology, Department of Electronics and Telecommunications, 127 pgs. cited by applicant

White, "Influence of thiol-ene polymer evolution on the formation and performance of holographic polymer dispersed liquid crystals", The 232nd ACS National Meeting, San Francisco, CA, Sep. 10-14, 2006, 1 pg. cited by applicant

Wicht et al., "Nanoporous Films with Low Refractive Index for Large-Surface Broad-Band Anti-Reflection Coatings", *Macromol. Mater. Eng.*, 2010, 295, DOI: 10.1002/mame.201000045, 9 pgs. cited by applicant

Wilderbeek et al., "Photoinitiated Bulk Polymerization of Liquid Crystalline Thiolene Monomers", *Macromolecules*, 2002, vol. 35, pp. 8962-8969. cited by applicant

Wilderbeek et al., "Photo-Initiated Polymerization of Liquid Crystalline Thiol-Ene Monomers in Isotropic and Anisotropic Solvents", *J. Phys. Chem. B*, 2002, vol. 106, No. 50, pp. 12874-12883. cited by applicant

Wisely, "Head up and head mounted display performance improvements through advanced techniques in the manipulation of light", *Proc. of SPIE*, 2009, 10 pages, vol. 7327. cited by applicant

Wofford et al., "Liquid crystal bragg gratings: dynamic optical elements for spatial light modulators", Hardened Materials Branch, Survivability and Sensor Materials Division, AFRL-ML-

WP-TP-2007-551, Air Force Research Laboratory, Jan. 2007, Wright-Patterson Air Force Base, OH, 17 pgs. cited by applicant

Yang et al., “Robust and Accurate Surface Measurement Using Structured Light”, IEEE, Apr. 30, 2008, vol. 57, Issue 6, pp. 1275-1280, DOI:10.1109/TIM.2007.915103. cited by applicant

Yaqoob et al., “High-speed two-dimensional laser scanner based on Bragg grating stored in photothermorefractive glass”, Applied Optics, Sep. 10, 2003, vol. 42, No. 26, pp. 5251-5262. cited by applicant

Yaroshchuk et al., “Stabilization of liquid crystal photoaligning layers by reactive mesogens”, Applied Physics Letters, Jul. 14, 2009, vol. 95, pp. 021902-1-021902-3. cited by applicant

Ye, “Three-dimensional Gradient Index Optics Fabricated in Diffusive Photopolymers”, Thesis, Department of Electrical, Computer and Energy Engineering, University of Colorado, 2012, 224 pgs. cited by applicant

Yemtsova et al., “Determination of liquid crystal orientation in holographic polymer dispersed liquid crystals by linear and nonlinear optics”, Journal of Applied Physics, Oct. 13, 2008, vol. 104, p. 073115-1-073115-4. cited by applicant

Yeralan et al., “Switchable Bragg grating devices for telecommunications applications”, Opt. Eng., Aug. 2012, vol. 41, No. 8, pp. 1774-1779. cited by applicant

Yokomori, “Dielectric surface-relief gratings with high diffraction efficiency”, Applied Optics, Jul. 15, 1984, vol. 23; No. 14, pp. 2303-2310. cited by applicant

Yoshida et al., “Nanoparticle-Dispersed Liquid Crystals Fabricated by Sputter Doping”, Adv. Mater., 2010, vol. 22, pp. 622-626. cited by applicant

Youcef et al., “Phase Behavior Of Poly(N-Butyl Acrylate) And Poly(2-Ethylhexyl Acrylate) in Nematic Liquid Crystal E7”, Macromol. Symp. 2011, vol. 303, pp. 10-16, doi: 10.1002/masy.201150502. cited by applicant

Zeller et al., “Laminated Air Structured and Fluid Infiltrated Polymer Waveguides”, in IEEE Photonics Technology Letters, Nov. 2, 2011, vol. 23, Issue: 21, pp. 1564-1566, first published Aug. 12, 2011, DOI: 10.1109/LPT.2011.2164396. cited by applicant

Zhang et al., “Dynamic Holographic Gratings Recorded by Photopolymerization of Liquid Crystalline Monomers”, J. Am. Chem. Soc., 1994, vol. 116, pp. 7055-7063. cited by applicant

Zhang et al., “Switchable Liquid Crystalline Photopolymer Media for Holography”, J. Am. Chem. Soc., 1992, vol. 114, pp. 1506-1507. cited by applicant

Zhao et al., “Designing Nanostructures by Glancing Angle Deposition”, Proc. of SPIE, Oct. 27, 2003, vol. 5219, pp. 59-73. cited by applicant

Zhao et al., “Diffusion Model of Hologram Formation in Dry Photopolymer Materials”, Journal of Modern Optics, 1994. Vol. 41, No. 10, pp. 1929-1939, <https://doi.org/10.1080/09500349414551831>. cited by applicant

Zhao et al., “Extension of a diffusion model for holographic photopolymers”, Journal of Modern Optics, 1995, vol. 42, No. 12, pp. 2571-2573, <https://doi.org/10.1080/713824349>. cited by applicant

Zheng et al., “Holographic Polymer-Dispersed Liquid Crystal Grating with Low Scattering Losses”, Liquid Crystals, Mar. 2012, vol. 39, Issue 3, pp. 387-391, , <http://dx.doi.org/10.1080/02678292.2012.656716>. cited by applicant

Ziębacz, “Dynamics of nano and micro objects in complex liquids”, Ph.D. dissertation, Institute of Physical Chemistry of the Polish Academy of Sciences, Warsaw 2011, 133 pgs. cited by applicant

Zou et al., “Functionalized nano interdigitated electrodes arrays on polymer with integrated microfluidics for direct bio-affinity sensing using impedimetric measurement”, Sensors and Actuators A, Jan. 16, 2007, vol. 136, pp. 518-526, doi:10.1016/j.sna.2006.12.006. cited by applicant

Zyga, “Liquid crystals controlled by magnetic fields may lead to new optical applications”, Nanotechnology, Nanophysics, Retrieved from <http://phys.org/news/2014-07-liquid-crystals->

magnetic-fields-optical.html, Jul. 9, 2014, 3 pgs. cited by applicant
Extended European Search Report for European Application No. 20755316.5, Search completed Nov. 17, 2022, Mailed Nov. 25, 2022, 09 pgs. cited by applicant
Extended European Search Report for European Application No. 20818304.6, Search completed Mar. 23, 2023, Mailed Mar. 31, 2023, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application PCT/US2021/038542, Report issued Dec. 13, 2022, Mailed on Jan. 5, 2023, 8 pgs. cited by applicant
International Preliminary Report on Patentability for International Application PCT/US2021/072548, Report issued May 16, 2023, Mailed on Jun. 1, 2023, 05 pgs. cited by applicant
International Preliminary Report on Patentability for International Application PCT/US2022/070095, Report issued Jul. 4, 2023, Mailed on Jul. 20, 2023, 08 pgs. cited by applicant
Supplementary Partial European Search Report for European Application No. 18727645.6, Search completed Jul. 2, 2020, Mailed Jul. 13, 2020, 13 pgs. cited by applicant
Extended European Search Report for European Application No. 24158294.9, Search completed Apr. 25, 2024, Mailed May 6, 2024, 7 pgs. cited by applicant

Primary Examiner: Smith; Chad H

Attorney, Agent or Firm: KPPB LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of and priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/134,898, entitled “Grating Structures for Color Waveguides” and filed Jan. 7, 2021, which is incorporated herein by reference in its entirety for all purposes.

FIELD OF THE INVENTION

(1) The present invention generally relates to grating structures for providing beam expansion and extracting light from a waveguide.

BACKGROUND

(2) Waveguides can be referred to as structures with the capability of confining and guiding waves (e.g., restricting the spatial region in which waves can propagate). One subclass includes optical waveguides, which are structures that can guide electromagnetic waves, typically those in the visible spectrum. Waveguide structures can be designed to control the propagation path of waves using a number of different mechanisms. For example, planar waveguides can be designed to utilize diffraction gratings to diffract and couple incident light into the waveguide structure such that the incoupled light can proceed to travel within the planar structure via total internal reflection (“TIR”).

(3) Fabrication of waveguides can include the use of material systems that allow for the recording of holographic optical elements within the waveguides. One class of such material includes polymer dispersed liquid crystal (“PDLC”) mixtures, which are mixtures containing photopolymerizable monomers and liquid crystals. A further subclass of such mixtures includes holographic polymer dispersed liquid crystal (“HPDLC”) mixtures. Holographic optical elements, such as volume phase gratings, can be recorded in such a liquid mixture by illuminating the material with two mutually coherent laser beams. During the recording process, the monomers

polymerize, and the mixture undergoes a photopolymerization-induced phase separation, creating regions densely populated by liquid crystal (LC) micro-droplets, interspersed with regions of clear polymer. The alternating liquid crystal-rich and liquid crystal-depleted regions form the fringe planes of the grating.

(4) Waveguide optics, such as those described above, can be considered for a range of display and sensor applications. In many applications, waveguides containing one or more grating layers encoding multiple optical functions can be realized using various waveguide architectures and material systems, enabling new innovations in near-eye displays for Augmented Reality (“AR”) and Virtual Reality (“VR”), compact Heads Up Displays (“HUDs”) for aviation and road transport, and sensors for biometric and laser radar (“LIDAR”) applications. As many of these applications are directed at consumer products, there is a growing requirement for efficient low cost means for manufacturing holographic waveguides in large volumes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.
- (2) The description will be more fully understood with reference to the following figures and data graphs, which are presented as various embodiment of the disclosure and should not be construed as a complete recitation of the scope of the disclosure, wherein:
- (3) FIG. 1A illustrates a profile view of a waveguide-based display in accordance with an embodiment of the invention.
- (4) FIG. 1B shows a plan view of the waveguide shown in FIG. 1A.
- (5) FIG. 1C shows a profile view of a waveguide-based display in accordance with an embodiment of the invention.
- (6) FIG. 2A illustrates a portion of a surface relief grating in accordance with an embodiment of the invention.
- (7) FIG. 2B illustrates a cross-sectional view of a photonic crystal in accordance with an embodiment of the invention.
- (8) FIG. 3 conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention.
- (9) FIG. 4 conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention.
- (10) FIG. 5 conceptually illustrates a plan view of the waveguide-based display described in connection with FIG. 3.
- (11) FIG. 6 conceptually illustrates in cross section a waveguide-based display in accordance with an embodiment of the invention.
- (12) FIG. 7 conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention.
- (13) FIGS. 8A-8D illustrate a processing apparatus that can be used in a method for fabricating deep SRGs or EBGs in accordance with an embodiment.
- (14) FIG. 9 conceptually illustrates a method for forming a surface relief grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention.
- (15) FIG. 10A conceptually illustrates an apparatus that can be used in a step of a method for fabricating a hybrid surface relief/Bragg grating in accordance with an embodiment of the invention.

(16) FIG. 10B conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in accordance with an embodiment of the invention.

(17) FIG. 10C conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in accordance with an embodiment of the invention.

(18) FIG. 10D conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in which the surface relief grating formed in the step illustrated in FIG. 10C is partially refilled with liquid crystal in accordance with an embodiment of the invention.

(19) FIG. 10E conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in which the hybrid surface relief Bragg grating formed in the step illustrated in FIG. 10D is covered with a protective layer, in accordance with an embodiment of the invention.

(20) FIG. 11 is a flowchart showing an exemplary method for forming a hybrid surface relief-Bragg grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention.

(21) FIG. 12 is a graph showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 1-micron thickness deep surface relief grating demonstrating that in this case high S and P response can be achieved.

(22) FIG. 13 is a graph showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 2-micron thickness deep surface relief grating, demonstrating that in this case the S-polarization response is dominant over most of the angular range of the grating.

(23) FIG. 14 is a graph showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 3-micron thickness, demonstrating that in this case the P-polarization response is dominant over a substantial portion of the angular range of the grating.

(24) FIGS. 15A-15B shows a plot of calculated angular and spectral diffraction efficiency characteristics for a reflection structure formed from a HPDLC in accordance with an embodiment of the invention.

(25) FIGS. 16A-16B shows the corresponding diffraction efficiency characteristics of a grating formed from a polymer of index 1.8 (refractive index modulation 0.4) for the same beam angles in accordance with an embodiment of the invention.

(26) FIGS. 17A-17E illustrate the step in fabricating such a reflection Bragg grating incorporating polymer scaffolding in accordance with an embodiment of the invention.

(27) FIG. 18 conceptually illustrates a reflection grating in accordance with an embodiment of the invention.

(28) FIG. 19 conceptually illustrates a reflection grating in accordance with an embodiment of the invention.

(29) FIG. 20 conceptually illustrates a combination reflection and transmission grating in accordance with an embodiment of the invention.

(30) FIG. 21 conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

(31) FIG. 22 illustrates a plot of different materials that have different dispersion ($n-A$) curves which overlap at two or more points.

(32) FIG. 23 conceptually illustrates a portion of a Bragg grating formed from different dispersion materials in accordance with an embodiment of the invention.

(33) FIGS. 24-27 conceptually illustrate waveguide gratings with spatially varying diffraction efficiency in accordance with an embodiment of the invention.

(34) FIG. 28 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

(35) FIG. 29 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

(36) FIG. **30** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

(37) FIG. **31** conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention.

(38) FIG. **32** conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention.

(39) FIG. **33** conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention.

(40) FIG. **34** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

(41) FIG. **35** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

(42) FIG. **36A** conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

(43) FIG. **36B** conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

(44) FIG. **36C** conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

(45) FIG. **37** conceptually illustrates a side elevation view of a waveguide-based display based on the embodiments of FIGS. **36A, 36B, 36C**.

(46) FIG. **38** illustrates a plan view of the waveguide-based display described in connection with FIG. **37**.

(47) FIG. **39** conceptually illustrates an apparatus for recording a grating structure including a grating array in accordance with an embodiment of the invention.

(48) FIG. **40** illustrates a multiplexed grating structure in accordance with an embodiment of the invention.

(49) FIG. **41A** conceptually illustrates a portion of a grating element pattern including rectangular elements of differing size and aspect ratio for use in an emissive display panel in accordance with an embodiment of the invention.

(50) FIG. **41B** conceptually illustrates a portion of a grating element pattern including Penrose tiles in accordance with an embodiment of the invention.

(51) FIG. **41C** conceptually illustrates a portion of a grating element pattern including hexagons in accordance with an embodiment of the invention.

(52) FIG. **41D** conceptually illustrates a portion of a grating element pattern including squares in accordance with an embodiment of the invention.

(53) FIG. **41E** conceptually illustrates a portion of a grating element pattern comprising diamond-shaped elements in accordance with an embodiment of the invention.

(54) FIG. **41F** conceptually illustrates a portion of a grating element pattern including isosceles triangles in accordance with an embodiment of the invention.

(55) FIG. **41G** conceptually illustrates a portion of a grating element pattern including hexagons of horizontally biased aspect ratio in one embodiment.

(56) FIG. **41H** conceptually illustrates a portion of a grating element pattern including rectangles of horizontally biased aspect ratio in accordance with an embodiment of the invention.

(57) FIG. **41I** conceptually illustrates a portion of a grating element pattern including diamond shaped elements of horizontally biased aspect ratio in accordance with an embodiment of the invention.

(58) FIG. **41J** conceptually illustrates a portion of a grating element pattern including triangles of horizontally biased aspect ratio in accordance with an embodiment of the invention.

(59) FIG. **42** conceptually illustrates a method for forming a waveguide color grating structure in accordance with an embodiment of the invention.

SUMMARY OF THE DISCLOSURE

(60) In many of the embodiments to be described there is provided A waveguide-based display device including: a waveguide; a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection (TIR) path within the waveguide; and a grating structure for providing beam expansion of the TIR light in at least one direction and extracting the TIR light from the waveguide, wherein the grating structure comprises a plurality of grating elements having at least two different grating prescriptions and at least one surface relief grating formed by a phase separation process.

(61) In many embodiments, the plurality of grating elements are tiled in a repeating pattern. In many embodiments, the plurality of grating elements are tiled in a non-repeating pattern.

(62) In many embodiments, the grating structure further includes a spatial refractive index modulation.

(63) In many embodiments, the plurality of grating elements includes at least two different spectral responses.

(64) In many embodiments, the plurality of grating elements includes at least two different angular responses.

(65) In many embodiments, the plurality of grating elements includes at least two different birefringence characteristics.

(66) In many embodiments, the plurality of grating elements includes at least two different polarization responses.

(67) In many embodiments, the plurality of grating elements includes at least two different refractive index modulations.

(68) In many embodiments, the plurality of grating elements includes an average refractive index including at least two different refractive indices.

(69) In many embodiments, the plurality of grating elements includes at least two different grating thicknesses.

(70) In many embodiments, the grating structure can comprise a plurality of grating elements having grating prescriptions selected from a group containing at least two different grating prescriptions.

(71) In many embodiments, the plurality of grating elements includes at least two different K-vectors.

(72) In many embodiments, the plurality of grating elements includes diffracting grating elements arrayed with non-diffracting grating elements.

(73) In many embodiments, the plurality of grating elements is recorded in material containing at least two different materials.

(74) In many embodiments, the plurality of grating elements is recorded in material having dispersion curves selected from a group containing at least two different dispersion curves.

(75) In many embodiments, the plurality of grating elements is recorded in material having at least two different wavelength sensitivities.

(76) In many embodiments, the plurality of grating elements is recorded in material having at least two different holographic exposure times.

(77) In many embodiments, the plurality of grating elements is recorded in material having at least two different holographic exposure energies.

(78) In many embodiments, the grating structure is configured to enable a plurality of ray path lengths within the grating structure that differ in length by a distance shorter than the coherence length of the source of image modulated light.

(79) In many embodiments, the plurality of grating elements includes at least one grating pitch greater than a wavelength of the TIR light.

(80) In many embodiments, the plurality of grating elements includes a spatial variation of at least one selected from the group consisting of: spatial variation of grating thickness, average

refractive index, refractive index modulation, and birefringence.

(81) In many embodiments, the grating structure includes at least one selected from a group consisting of: a rolled K-vector grating, a dual interaction grating, a multiplexed grating, a lossy grating, a sub wavelength grating, and a chirped grating.

(82) In many embodiments, the at least one surface relief grating is formed by liquid crystal extraction from a grating recorded in holographic polymer dispersed liquid crystal (HPDLC).

(83) In many embodiments, the at least one surface relief grating is formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a holographic polymer dispersed liquid crystal (HPDLC) recorded grating with another material having a refractive index higher than that of the extracted liquid crystal.

(84) In many embodiments, the at least one surface relief grating is formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a holographic polymer dispersed liquid crystal (HPDLC) recorded grating with another material having a refractive index lower than that of the extracted liquid crystal.

(85) In many embodiments, the grating structure includes at least one reflection grating. In many embodiments, the grating structure includes at least one transmission grating.

(86) In many embodiments, the grating structure can be a switching grating.

(87) In many embodiments, the grating structure can further comprise at least one selected from the group of a beamsplitter layer, an anti-reflection coating, an optical bandpass filter, a polarization modification layer, and/or an alignment layer.

(88) In many embodiments, the grating structure provides beam expansion and extraction of red, green and blue image modulated light over a field of view. In many embodiments, the image modulated light is monochromatic.

(89) In many embodiments, the grating structure is recorded into a material selected from the group of a mixture including at least one liquid crystal and at least one monomer, a mixture including at least one blue phase liquid crystal and at least one monomer.

(90) In many embodiments, the input coupler can be a grating or a prism. In many embodiments the input coupler can include a plurality of gratings. In many embodiments, the input coupler can be a switching grating.

(91) In many embodiments, the plurality of grating elements are spatially distributed elements.

(92) In many embodiments, the at least one surface relief grating includes a grating modulation depth greater than a grating pitch.

(93) In many of the embodiments to be described a waveguide can support at least one photonic crystal. For the purposes of explaining the invention a photonic crystal can be considered as a periodic optical nanostructure that affects the motion of photons. Photonic crystals can be fabricated for one, two, or three dimensions. An example of a one-dimensional photonic crystal is a grating structure formed from alternating layers of high refractive index and low refractive index materials. Such gratings are commonly referred to as Bragg or volume gratings. A two-dimensional photonic crystal is formed by a two-dimensional array of elements of a first refractive index immersed in a material of a second refractive index. Two-dimensional photonic crystals can be fabricated by photolithography, or by drilling holes in a suitable substrate. Fabrication methods for three-dimensional photonic crystal include drilling under different angles, stacking multiple 2-D layers on top of each other and direct laser writing. Another approach is forming a matrix of spheres or instigating self-assembly of spheres in a matrix and dissolving the sphere. In many cases, the regions of low refractive index are provided by air. Two-dimensional photonic crystals can be any of the 5 2-D Bravais lattices. Typically, photonic crystals have periodicity of around half the wavelength of the light to be diffracted. The low dielectric can be provided by air. Three-dimensional photonic crystals can provide any of the fourteen 3D Bravais lattices.

(94) In many of the embodiments of the invention to be described below, a photonic crystal including a grating structure immersed at least partially in air can be formed from a mixture of

liquid crystal and monomer materials using a phase separation process taking place under holographic exposure. After exposure is complete liquid crystal can be removed from the surface relief grating. In many embodiments, the grating structure can be refilled with a different material such as a LC of a different index and different other properties. In many embodiments, the grating structure can be partially backfilled to provide a hybrid surface relief and volume grating structure. In many embodiments, the grating structure can be refilled with an organic or inorganic material with a high refractive index. In many embodiments, the grating can have material properties varying spatially can varying depth. In many embodiments, the backfilling can be a diffusion process. In many embodiments, the grating structure can be backfilled with chemical components that are phase separated under a laser exposure process. In many embodiments, backfilling can be carried out in the presence of thermal, mechanical, chemical or electromagnetic stimuli for influencing annealing and alignment of the grating structure.

(95) In some embodiments, the diffractive surface can be a metasurface, where for the purposes of discussing the invention a metasurface is defined as a surface structure supporting subwavelength in-plane features that are used to realize a desirable functionality by locally managing the interaction between the metasurface and incident electromagnetic fields. Metasurfaces are typically formed on sub-wavelength thickness substrates. However, much thicker substrates may be used. Metasurfaces may have diffractive features of nanometre-scale. This enables a much greater degree of wavefront phase and amplitude control than can be achieved using conventional diffractive optical elements (DOEs). In many applications, diffraction limited performance is a realistic goal for optical components (including lenses) made using metasurfaces. Conventional DOEs operate at the micron scale with feature heights that are significant fractions of visible band wavelengths, resulting in more limited capacity for wavefront amplitude and phase control, especially when manufacturing tolerances are taken into account.

(96) In many embodiments, a photonic crystal provides an input grating. As will be discussed in the following paragraphs a photonic crystal formed by liquid crystal extraction offers potential benefits in terms of improving the angular bandwidth of a waveguide and can be used to control the polarization characteristics of waveguided light. The various embodiments to be discussed can be applied in HUDs for automotive applications, near eye displays and other waveguide display applications. In many embodiments, photonic crystal according to the principles of the invention can be used in single axis or in dual expansion waveguides. In some embodiments, photonic crystals can be used to provide beam expansion gratings. In some embodiments, photonic crystals can be used to provide output gratings. In some embodiments, photonic crystals can be used to diffract more than one primary color. In some embodiments, waveguides incorporating photonic crystals can be arranged in stacks of waveguides each having grating prescription for diffracting a unique spectral bandwidth.

DETAILED DESCRIPTION

(97) FIG. 1A illustrates a profile view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **100** includes a waveguide **102** including an input coupler **104** and an output coupler **106**. Image containing light is incoupled by the input coupler **104** into total internal reflection (TIR) in the waveguide **102**. The TIR light is then outcoupled by the output coupler **106** into outcoupled light **110**. FIG. 1B shows a plan view of the waveguide shown in FIG. 1A.

(98) FIG. 1C shows a profile view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **100a** shares many components identically labelled with the waveguide-based display **100** of FIGS. 1A and 1B. The description of those components is applicable here and will not be repeated in detail. The waveguide-based display **100a** includes an input prism **104a** which operates to incouple image containing light **108** into TIR in the waveguide **102**.

(99) FIG. 2A illustrates a portion of a surface relief grating in accordance with an embodiment of

the invention. The surface relief grating **110A** includes an optical substrate **101** supporting grating elements **102** separated by air gaps **103**.

(100) FIG. **2B** illustrates a cross-sectional view of a photonic crystal in accordance with an embodiment of the invention. The photonic crystal **110B** includes a three-dimensional lattice elements including grating features **111** separated by air spaces **112**. In some embodiments, the lattice elements can include air regions surround by an optical material. In some embodiments, the lattice can be one of the Bravais lattices. Photonic crystals are described in detail in PCT App. No. PCT/US2021/072548, entitled “Photonic Crystals and Methods for Fabricating the Same” and filed on Nov. 22, 2021, which is hereby incorporated by reference in its entirety.

(101) FIG. **3** conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **120** includes a waveguide **121** including a reflective input grating **122** and an output grating **123**. Input light **124** from a picture generation unit (PGU) is coupled into the waveguide by the input grating **122** and propagates along a total internal reflection path indicated by the rays **125-127** before being extracted by the output grating **123** which provides one dimensional beam expansion into output light **128**. In some embodiments, the waveguide further supports a fold grating.

(102) FIG. **4** conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **130** includes a waveguide **131** supporting an input grating **132** and an output grating **133**. Input light **134** from a picture generation unit (PGU) is coupled into the waveguide **131** by the input grating **132** and propagates along a total internal reflection path indicated by the rays **135-136** before being extracted by the output grating **133** which provides one dimensional beam expansion. In many embodiments, at least one of the gratings is a photonic crystal formed using a liquid crystal extraction process.

(103) FIG. **5** conceptually illustrates a plan view of the waveguide-based display **120** described in connection with FIG. **3**. FIG. **5** further shows the PGU **141** which outputs image containing light **124** into the input grating **122**.

(104) FIG. **6** conceptually illustrates in cross section a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **150** includes a waveguide **151** supporting a reflection input grating **122**, a fold grating **152** which provides a first direction beam expansion, and an output grating **153**. Input light **145** from a PGU **141** is coupled into the waveguide **151** by the input grating **122** and propagates along total internal reflection paths indicated by the rays **154,155** before being extracted **156** by the output grating **153** which provides a second beam expansion orthogonal to the first beam expansion. Rays in FIGS. 5-7 (such as rays **154,155**) are in TIR between the reflection faces of the waveguide. The ray bounces illustrated would only be seen in the waveguide cross section.

(105) In many embodiments, at least one of the gratings **122,152,153** is a photonic crystal formed using a liquid crystal extraction process.

(106) FIG. **7** conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **160** includes a waveguide **161** which supports an input grating **162**, and a pair of overlapping or multiplexed fold gratings **163,164**. A PGU **141** outputs image containing light **145** into the input grating **162**. The multiplexed fold gratings **163,164** provide beam expansion in two orthogonal dimensions. The total internal reflection path from the input grating **162** to the fold gratings **163,164** and the pupil-expanded light extracted from the waveguide **161** by the overlapping fold gratings **163,164** are represented by the rays **165,166**. In many embodiments at least one of the gratings **162,163,164** is a photonic crystal formed using a liquid crystal extraction process.

(107) In some embodiments, LC can be extracted or evacuated from the SBG to provide an evacuated periodic grating (EPG). EPGs can be characterized as a surface relief grating (SRG) that has properties very similar to a Bragg grating due to the depth of the SRG structure (which is much

greater than that practically achievable using surface etching and other conventional processes commonly used to fabricate SRGs). The LC can be extracted using a variety of different methods, including but not limited to flushing with isopropyl alcohol and solvents. In many embodiments, one of the transparent substrates of the SBG is removed, and the LC is extracted. In further embodiments, the removed substrate is replaced. The SRG can be at least partially backfilled with a material of higher or lower refractive index. Such gratings offer scope for tailoring the efficiency, angular/spectral response, polarization, and other properties to suit various waveguide applications. Examples of EPGs and methods for manufacturing EPGs are discussed in US Pat. Pub. No. 2021/0063634, entitled "Evacuating Bragg Gratings and Methods of Manufacturing" and filed Aug. 28, 2020 which is hereby incorporated by reference in its entirety for all purposes.

(108) In many embodiments a photonic crystal formed by liquid crystal extraction provide an EPG which may be a deep surface relief grating (SRG). In many embodiments, a deep SRG formed using a liquid crystal extraction process can typically have a thickness in the range 1-3 micron with a Bragg fringe spacing 0.35 micron to 0.80 micron. In many embodiments, the condition for a deep SRG is characterized by a high grating depth to fringe spacing ratio. In some embodiments, the condition for the formation of a deep SRG is that the grating depth can be approximately twice the grating period. Modelling such SRGs using the Kogelnik theory can give reasonably accurate estimates of diffraction efficiency avoiding the need for more advanced modelling which typically entails the numerical solution of Maxwell's equations. The grating depths that can be achieved using liquid crystal removal from HPDLC gratings greatly surpass those possible using conventional nanoimprint lithographic methods, which do not achieve the condition for a deep SRG, typically providing only 250-300 nm. depth for grating periods 350-460 nm. (Pekka Äyräs, Pasi Saarikko, Tapani Levola, "Exit pupil expander with a large field of view based on diffractive optics", Journal of the SID 17/8, (2009), pp 659-664). Although deep SRGs may diffract S-polarization, deep SRGs can provide a range of polarization response characteristics depending on the thickness of the grating prescription and, in particular, the grating depth. Deep SRGs can also be used in conjunction with conventional Bragg gratings to enhance the color, uniformity and other properties of waveguide displays.

(109) SRGs have been fabricated in glassy monomeric azobenzene materials using laser holographic exposure (O. Sakhno, L. M. Goldenberg, M. Wegener, J. Stumpe, "Deep surface relief grating in azobenzene-containing materials using a low intensity 532 nm laser", Optical Materials: X, 1, (2019), 100006, pp 3-7. The Sakhno reference also discloses how SRGs can be recorded in a holographic photopolymer using two linearly orthogonally polarized laser beams.

(110) Deep SRG and Bragg gratings are also disclosed in Kiyoshi Yokomori, "Dielectric surface-relief gratings with high diffraction efficiency" Applied Optics; Vol. 23; Issue 14; (1984); pp. 2303-2310 which discloses the investigation of the diffraction properties of dielectric surface-relief gratings by solving Maxwell's equations numerically. The diffraction efficiency of a grating with a groove depth about twice as deep as the grating period was found to be comparable with the efficiency of a volume phase grating. The modelling by Yokomori predicted that dielectric surface-relief gratings interferometrically recorded in photoresist can possess a high diffraction efficiency of up to 94% (throughput efficiency 85%). The equivalence of deep SRGs and Bragg gratings is also discussed in another article by Golub (M. A. Golub, A. A. Friesem, L. Eisen "Bragg properties of efficient surface relief gratings in the resonance domain", Optics Communications; 235; (2004); pp 261-267). A further article by Gerritsen discusses the formation of Bragg-like SRGs in photoresist (Gerritsen H J, Thornton D K, Bolton S R; "Application of Kogelnik's two-wave theory to deep, slanted, highly efficient, relief transmission gratings" Applied Optics; Vol. 30; Issue 7; (1991); pp 807-814).

(111) The invention provides a method for making a surface relief grating that can offer very significant advantages over nanoimprint lithographic process particle for slanted gratings. Bragg gratings of any complexity can be made using interference or master and contact copy replication.

In some embodiments after removing the LC the SRG can be back filled with a material with different properties to the LC. This allows a Bragg grating with modulation properties that are not limited by the grating chemistry needed for grating formation. In some embodiments the deep SRG can be partially backfilled with LC to provide a hybrid SRG/Bragg grating. Alternatively, in some embodiments, the refill step can be avoided by removing just a portion of the LC from the LC rich regions of the HPDLC to provide a hybrid SRG/Bragg grating. The refill approach has the advantage that a different LC can be used to form the hybrid grating. In some embodiments, the backfill materials can be deposited using inkjet process.

(112) FIGS. **8A-8D** illustrate a processing apparatus that can be used in a method for fabricating deep SRGs or EBGs in accordance with an embodiment. FIG. **8A** conceptually illustrates an apparatus **170A** that can be used in a step of a method for fabricating a surface relief grating in which a mixture **171** of monomer and liquid crystal deposited on a transparent substrate **172** is exposed to holographic exposure beams **173,174**, in accordance with an embodiment of the invention. In some examples, the mixture may also include at least one of a photoinitiator, a coinitiator, a multifunctional thiol, adhesion promoter, surfactant, and/or additional additives. In some embodiments, the monomer may be isocyanate-acrylate based or thiolene based. In some embodiments, the liquid crystal may be a full liquid crystal mixture or a liquid crystal single only including a portion of a full liquid crystal mixture. Various examples of liquid crystal singles include one or both of cyanobiphenyls or pentylcyanobiphenyls. In some embodiments, liquid crystal may be replaced with another substance that phase separates with the monomer during exposure to create polymer rich regions and substance rich regions. Advantageously, the substance and liquid crystal singles may be a cost-effective substitute to full liquid crystal mixtures which are removed at a later step as described below.

(113) FIG. **8B** conceptually illustrates an apparatus **170B** that can be used in a step of a method for fabricating a surface relief grating from an HPDLC Bragg grating **175** formed on a transparent substrate using the holographic exposure beams, in accordance with an embodiment of the invention. The holographic exposure beams may transform the monomer into a polymer in some areas. The holographic exposure beams may include intersecting recording beams and include alternating bright and dark illumination regions. A polymerization-driven diffusion process may cause the diffusion of monomers and LC in opposite directions, with the monomers undergoing gelation to form polymer-rich regions (in the bright regions) and the liquid crystal becoming trapped in a polymer matrix to form liquid crystal rich regions (in the dark regions).

(114) FIG. **8C** conceptually illustrates an apparatus **170C** that can be used in a step of a method for fabricating a deep polymer surface relief grating **176** or EBG in which liquid crystal is removed from an HPDLC Bragg grating of FIG. **1B** to form a polymer surface relief grating in accordance with an embodiment of the invention. Advantageously, a polymer surface relief grating **176** may include a large depth with a comparatively small grating period in order to form a deep SRG. The liquid crystal may be removed by washing with a solvent such as isopropyl alcohol (IPA). The solvent should be strong enough to wash away the liquid crystal but weak enough to maintain the polymer. In some embodiments, the solvent may be chilled below room temperature before washing the grating. FIG. **8D** conceptually illustrates an apparatus **170D** that can be used in a step of a method for fabricating a polymer surface relief grating in which the polymer surface relief grating is covered with a protective layer **177** in accordance with an embodiment of the invention.

(115) FIG. **9** conceptually illustrates a method for forming a surface relief grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention. As shown, the method **180** of forming a surface relief grating is provided. Referring to the flow diagram, method **180** includes providing (**181**) a mixture of at least one monomer and at least one liquid crystal. A transparent substrate can be provided (**182**). A layer of the mixture can be deposited (**183**) onto a surface of the substrate. Holographic recording beams can be applied (**184**) to the mixture layer. A HPDLC grating comprising alternating polymer rich and liquid crystal rich

regions can be formed (185). The liquid crystal in the liquid crystal rich regions can be removed (186) to form a polymer surface relief grating.

(116) As discussed above, in many the embodiments, the invention also provides a method for fabricating a hybrid surface relief/Bragg grating. FIG. 10A conceptually illustrates apparatus 190A that can be used in a step of a method for fabricating a hybrid surface relief/Bragg grating in which a mixture 191 of monomer and liquid crystal deposited on a transparent substrate 212 is exposed to holographic exposure beams 193,194, in accordance with an embodiment of the invention. FIG. 10B conceptually illustrates apparatus 190B that can be used in a step of a method for fabricating a surface relief grating from a HPDLC Bragg grating 195 formed on the transparent substrate using the holographic exposure beams in accordance with an embodiment of the invention. FIG. 10C conceptually illustrates apparatus 190C that can be used in a step of a method for fabricating a surface relief grating in which liquid crystal is removed from the HPDLC Bragg grating to form a surface relief grating 196 in accordance with an embodiment of the invention. FIG. 10D conceptually illustrates apparatus 190D that can be used in a step of a method for fabricating a surface relief grating in which the surface relief grating formed in the step illustrated in FIG. 10C is partially refilled with liquid crystal to form a hybrid surface relief/Bragg grating 197 in accordance with an embodiment of the invention. FIG. 10E conceptually illustrates apparatus 190E that can be used in a step of a method for fabricating a surface relief grating in which the hybrid surface relief Bragg grating formed in the step illustrated in FIG. 10D is covered with a protective layer 198, in accordance with an embodiment of the invention.

(117) FIG. 11 is a flowchart showing an exemplary method for forming a hybrid surface relief-Bragg grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention. As shown, the method 220 of forming hybrid surface relief-Bragg grating is provided. Referring to the flow diagram, method 220 includes providing (221) a mixture of at least one monomer and at least one liquid crystal. The at least one monomer may include an isocyanate-acrylate monomer. Providing the mixture of the monomer and the liquid crystal may also include mixing one or more of the following with the at least one monomer and the liquid crystal: photoinitiator, coinitiator, multifunctional thiol, and/or additional additives. This mixture may be allowed to rest in order to allow the coinitiator to catalyze a reaction between the monomer and the thiol. The rest period may occur in a dark space or a space with red light (e.g. infrared light) at a cold temperature (e.g. 20° C.) for a period of approximately 8 hours. After resting, additional monomers may be mixed into the monomer. This mixture may be then strained or filtered through a filter with a small pore size (e.g. 0.45 µm pore size). After straining this mixture may be stored at room temperature in a dark space or a space with red light before coating.

(118) Next, a transparent substrate can be provided (222). In certain embodiments, the transparent substrate may be a glass substrate or a plastic substrate. A non-stick coating may be applied to the transparent substrate before the mixture is coated on the substrate. A layer of the mixture can be deposited (223) onto a surface of the substrate. In some embodiments, the mixture is sandwiched between the transparent substrate and another substrate using glass spacers to maintain internal dimensions. Holographic recording beams can be applied (224) to the mixture layer. The holographic recording beams may be a two-beam interference pattern which may cause phase separation of the LC and the polymer. After applying the holographic recording beams, the mixture may be cured. The curing process may include leaving the mixture under low-intensity white light for a period of time under the mixture fully cures. The low intensity white light may also cause a photo-bleach dye process to occur. Thus, an HPDLC grating having alternating polymer rich and liquid crystal rich regions can be formed (225). In some embodiments, the curing process may occur in 2 hours or less. After curing, one of the substrates may be removed exposing the HPDLC grating.

(119) HPDLC grating may include alternating sections of liquid crystal rich regions and polymer regions. The liquid crystal in the liquid crystal rich regions can be removed (226) to form polymer

surface relief gratings or EBGs which is a form of deep SRGs. The liquid crystal may be removed by gently immersing the grating into a solvent such as isopropyl alcohol (IPA). The IPA may be kept at a lower temperature while the grating is immersed in the IPA. The grating is then removed from the solvent and dried. In some embodiments, the grating is dried using a high flow air source such as compressed air. After the LC is removed from the grating, a polymer-air surface relief Bragg grating is formed. The steps **221-226** of FIG. **11** roughly correspond to the steps described in connection with FIG. **10** in creating a polymer-air SRG and thus these descriptions are applicable to FIG. **11**.

(120) Further, method **220** includes at least partially refilling (**227**) cleared liquid crystal rich regions with liquid crystal to form hybrid SRGs. The refilled liquid crystal may be of different consistency to the previously removed liquid crystal that was previously removed in step **226**. Further, it is appreciated that the liquid crystal removed in step **226** may only be partially removed in an alternative method to forming hybrid SRGs. Advantageously, hybrid SRGs may provide the ability to tailor specific beneficial characteristics of the SRGs. One particular characteristic that may be improved by the inclusion of at least some liquid crystal within the SRGs is a decrease in haze properties.

(121) As shown in FIG. **10E**, the formed surface relief grating can further be covered with a protective layer. In some instances, the protective layer may be a moisture and oxygen barrier with scratch resistance capabilities. In some instances, the protective layer may be a coating that does not fill in air gap regions where LC that was removed once existed. The coating may be deposited using a low temperature process. In some implementations, the protective layer may have anti-reflective (AR) properties. The coating may be a silicate or silicon nitride. The coating process may be performed by a plasma assisted chemical vapor deposition (CVD) process such as a plasmatreat nanocoating process. The coating may be a parylene coating. The protective layer may be a glass layer. A vacuum or inert gas may fill the gaps where LC that was removed once existed before the protective layer is implemented. In some embodiments, the coating process may be integrated with the LC removal process (**226**). For example, a coating material may be mixed with the solvent which is used to wash the LC from the grating. In some implementations, the coating material may be a material with a lower or higher refractive index than the polymer and fill the spaces between adjacent polymer portions. The refractive index difference between the polymer and the coating material may allow the polymer SRGs to continue to diffract. Hybrid SRG/Bragg gratings suffer from the shallowness of the SRG structure which leads to low SRG diffraction efficiencies. The method disclosed in the present application allows more effective SRG structures to be formed by optimised the depth of the liquid crystal in the liquid crystal rich regions such that the SRG has a high depth to grating pitch ratio while allowing the Bragg grating to be sufficiently thick for efficient diffraction. In many embodiments the Bragg grating component of the hybrid grating can have a thickness in the range 1-3 micron. In many embodiments, the SRG component of the hybrid grating can have a thickness in the range 0.25-3 micron. The initial HPDLC grating would have a thickness of equal to the sum of the final SRG and Bragg grating components. The thickness ratio of the two grating components depends on the waveguide application.

(122) In many embodiments, the refill depth of the liquid crystal regions of the grating can be varied across the grating to provides spatially varying relative SRG/Bragg grating strengths. In many embodiments, as an alternative to liquid crystal removal and refill, the liquid crystal in the liquid crystal rich grating regions can be totally or partially removed. In many embodiments, the liquid crystal used to refill or partially refill the liquid crystal-cleared regions can have a different chemical composition to the liquid crystal used to form the HPDLC grating. In some embodiments, a first liquid crystal with phase separation properties compatible with the monomer can be specified to provide a HPDLC grating with optimal modulation and grating definitions while a second refill liquid crystal can be specified to provide desired index modulation properties in the final hybrid grating. In many embodiments, the Bragg portion of the hybrid grating can be switchable with

electrodes applied to surfaces of the substrate and the cover layer. In many embodiments the refill liquid crystals can contain additives for improving switching voltage, switching time, polarization, transparency, and other parameters. A hybrid grating formed using a refill process would have the further advantages that the LC would be form a continuum (rather than an assembly of LC droplets) thereby reducing haze.

(123) In many embodiments, a deep SRG can provide a means for controlling polarization in a waveguide. SBGs are normally P-polarization selective leading to a 50% efficiency loss with unpolarized light sources such as OLEDs and LEDs. Hence combining S-polarization diffracting and P-polarization diffracting gratings as discussed above can provide a theoretical $2\times$ improvement over waveguides using P-diffracting gratings only. In some embodiments a S-polarization diffracting grating can be provided by a Bragg grating formed in a conventional holographic photopolymer. In some embodiments an S-polarization diffracting grating can be provided by a Bragg grating formed in a HPDLC with birefringence altered using an alignment layer or other process for realigning the liquid crystal directors. In some embodiments an S-polarization diffracting grating can be formed using liquid crystals, monomers and other additives that naturally organize into S-diffracting gratings under phase separation. In many embodiments, an S-polarization diffracting grating can be provided by a surface relief grating (SRG). The inventors have discovered by experimentation that a deep SRG exhibiting high S-diffraction efficiency (up to 99%) and low P-diffraction efficiency can be formed by removing the liquid crystal from an SBG formed from holographic phase separation of a liquid crystal and monomer mixture.

(124) Deep SRGs can also provide other polarization response characteristics. Several prior art theoretical studies such as an article by Moharam (Moharam M. G. et al. "Diffraction characteristics of photoresist surface-relief gratings", Applied Optics, Vol. 23, page 3214, Sep. 15, 1984) point to deep surface relief gratings having both S and P sensitivity with S being dominant. In many embodiments, the thickness of the SRG can be adjusted to provide a variety of S and P diffraction characteristics. In some embodiments, diffraction efficiency can be high for P across a spectral bandwidth and angular bandwidth and low for S across the same spectral bandwidth and angular bandwidth. In some embodiments, diffraction efficiency can be high for S across the spectral bandwidth and angular bandwidth and low for P across the same spectral bandwidth and angular bandwidth. In some embodiments, high efficiency for both S and P polarized light can be provided. Analysis of a SRG of refractive index 1.6 immersed in air (hence providing an average grating index of 1.3) of period 0.48 micron, with a 0 degrees incidence angle and 45 degree diffracted angle for a wavelength of 0.532 micron have been performed. FIG. 12 is a graph 210 showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 1-micron thickness deep surface relief grating demonstrating that in this case high S and P response can be achieved. FIG. 13 is a graph 220 showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 2-micron thickness deep surface relief grating, demonstrating that in this case the S-polarization response is dominant over most of the angular range of the grating. FIG. 14 is a graph 230 showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 3-micron thickness, demonstrating that in this case the P-polarization response is dominant over a substantial portion of the angular range of the grating.

(125) In many embodiments, the photonic crystal can be a reflection Bragg grating formed by a LC extraction process. A reflection Bragg grating made using phase separation followed by LC subtraction can enable wide angular and spectral bandwidth. In many embodiments replacing the current input SBG with a reflection photonic crystal can be used to reduce the optical path from the PGU to the waveguide. In some embodiments, the PGU pupil and the waveguide can be in contact. In many embodiments, the reflection Bragg grating can be approximately 3 microns in thickness. The diffracting properties of an LC extracted Bragg grating mainly result from the index gap between the polymer and air (not from the depth of the grating as in the case of a SRG).

(126) FIGS. **15A-15B** shows a plot of calculated angular and spectral diffraction efficiency characteristics for a reflection structure formed from a HPDLC in accordance with an embodiment of the invention. The input and diffracted beam angles are 0° and 45° and the grating thickness is 3 microns. The refractive index of the polymer component of the grating is 1.6 and the refractive index modulation (polymer/air) is 0.3. The average index is obtained by taking the average of the refractive indices of the polymer and air ($1.6+1.0/2=1.3$). FIG. **15A** shows the diffraction efficiency versus input angle in air. FIG. **15B** shows the diffractive efficiency versus wavelength.

(127) FIGS. **16A-16B** shows the corresponding diffraction efficiency characteristics of a grating formed from a polymer of index 1.8 (refractive index modulation 0.4) for the same beam angles in accordance with an embodiment of the invention. The higher index polymer results in an increase in the spectral bandwidth of the grating. The angular bandwidth (near 100%) covers all waveguiding angles. From consideration of the diffraction efficiency plots it should be apparent that the spectral bandwidth of the grating covers most of visible band. By considering the diffraction efficiency obtained for S and P polarized light the inventors have found that the DE characteristics do not vary significantly with polarization. Calculations also indicate that, in contrast to LC-extracted transmission gratings, neither the angular bandwidth nor the spectral bandwidth are affected by grating thickness.

(128) Reflection Bragg gratings with K-vectors substantially normal to the waveguide substrates may present problems in the flushing out of LC since the extraction will need to take place through the edges of the grating. Such a grating will also be structurally unstable due the polymer regions not being supported. In many embodiments directed at providing waveguide-based displays, the reflection grating may be slanted allowing LC extraction to take place through the upper and lower faces of the grating. In some embodiments with K-vectors substantially normal to the waveguide substrates, the reflection Bragg grating can incorporate polymer scaffolding. FIGS. **17A-17E** illustrate the step in fabricating such a reflection Bragg grating incorporating polymer scaffolding in accordance with an embodiment of the invention. In a first step conceptually illustrated in FIG. **17A**, a grating structure **250A** including alternating LC regions **252** and polymer regions **251** supported by a substrate **253** may be fabricated using a holographic exposure process as discussed above. Alternatively, a mask exposure process can be used. In a second step conceptually illustrated in FIG. **17B**, the LC is extracted to provide the surface relief grating structure **250B** in which the LC regions are now air-filled regions **254**. In a third step conceptually illustrated in FIG. **17C**, the grating can be refilled with a liquid crystal and monomer mixture **254a** to produce a hybrid grating. In a fourth step conceptually illustrated in FIG. **17C**, a multiplexed grating **250C** combining a reflection grating (having K-vectors substantially normal to the substrate) and a transmission grating (having K-vectors substantially parallel to the plane of the substrate) is recorded into to the mixture through an upper mask **255A** and a lower mask **255B** mask. The exposure illumination modulated by the masks is indicated by **256**. Other arrangements of masks and illumination profiles can be used depending on the grating structures to be recorded. The exposed grating **250D** conceptually illustrated in FIG. **17D** includes horizontal LC regions **256A**, horizontal polymer regions **256B**, vertical LC regions **256C**, and vertical polymer regions **256D** which provide scaffolding for the horizontal polymer regions. In a final step conceptually illustrated in FIG. **17E**, the LC is flushed out of the grating structure to form the finished grating **250E** including the horizontal polymer grating elements **257A** and vertical polymer grating elements **257B** that have principal optical surfaces in contact with air and are supported by vertical polymer scaffolding elements **257C**. In many embodiments, the reflection grating can have a thickness in the range 1-3 micron.

(129) The process described in connection with FIGS. **17A-17E** may produce various grating structures as illustrated in FIGS. **18-20**.

(130) FIG. **18** conceptually illustrates a reflection grating in accordance with an embodiment of the invention. The reflection grating **260** includes alternating first refractive index regions **261** and

second refractive index regions **262**.

(131) FIG. **19** conceptually illustrates a reflection grating in accordance with an embodiment of the invention. The reflection grating **270** includes alternating first refractive index regions **271** and second refractive index regions **272**. The reflection grating **270** further includes substantially vertical regions **273** of the second refractive index material.

(132) FIG. **20** conceptually illustrates a combination reflection and transmission grating in accordance with an embodiment of the invention. The combination reflection and transmission grating **280** includes a reflection grating and a transmission grating. The combination grating **280** includes alternating first refractive index regions **281** and second refractive index regions **282**. The combination grating **280** further includes approximately vertical regions **283** of a third refractive index material. The transmission grating is formed by the average of the first refractive index regions **281**, second refractive index regions **282**, vertical regions **283** of a third refractive index material. The diffraction of the transmission grating is represented by the rays **284,285**. The reflection grating formed by the first refractive index regions **281** and second refractive index regions **282**. The diffraction of the reflection grating is represented by the rays **286,287**.

(133) In many of the embodiments to be described below there is provided a waveguide-based display device including: a waveguide; a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection path within said waveguide; and a grating structure for providing beam expansion in at least one direction. The waveguide display can incorporate many of the embodiments discussed in the preceding paragraphs.

(134) FIG. **21** conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **290** includes a waveguide **291** supporting an input grating **292** and a grating structure **293**. Input light **294** having a wavelength less than or equal to a predefined threshold value is diffracted into a TIR path **295** while input light **296** having wavelength greater than the threshold value is not diffracted and emergence from the input grating without substantial deviation or attenuations as the zero-order light **297**. In some embodiments, the grating structure **293** can be designed to only diffract light below a threshold wavelength. The input grating **292** may include a short grating period that enables coupling of light below a threshold wavelength. In some embodiments, such as for example waveguide-based displays including layers operating on different primary colors, this feature can be used to limit color crosstalk between layers.

(135) In many embodiments, a waveguide can support Bragg gratings including alternating regions formed from two different materials. FIG. **22** illustrates a plot **300** of different materials that have different dispersion (n - λ) curves **301,302** which overlap at two or more points **303,304**. FIG. **23** conceptually illustrates a portion of a Bragg grating **310** formed from different dispersion materials **311,312** in accordance with an embodiment of the invention. The Bragg gratings may be used as the input grating **292** described in connection with FIG. **21** to provide waveguide selectivity. In some embodiments, the wavelength selectivity may be implemented in other gratings (e.g. fold gratings and output gratings) used in the waveguide to minimise the effects of cross talk between stacked RGB waveguides.

(136) FIGS. **24-27** conceptually illustrate waveguide gratings with spatially varying diffraction efficiency in accordance with an embodiment of the invention. The gratings can be implemented as Bragg gratings, as SRGs, or as hybrid Bragg/SRG structures according to the principles discussed above. In many embodiments, the diffraction efficiency can be controlled by varying the grating depth and fill factor as a function of wavelength. FIG. **24** conceptually illustrates a grating in accordance with an embodiment of the invention. The grating **320** may include spatially varying modulation **321,322** at different positions.

(137) FIG. **25** conceptually illustrates a SRG in accordance with an embodiment of the invention. The SRG **330** includes spatially varying modulation **321,322** as the grating **320** described in

connection with FIG. 24. The SRG 330 further includes at least partial backfilling with a backfilling material 331 of refractive index different to that of the previous grating 320. As discussed above the SRG 330 can be formed by extracting liquid crystal from a HPDLC grating and then refilling with the backfill material 331.

(138) FIGS. 26-27 conceptually illustrates various plots disclosing the spatial variation of the refractive index modulation: $\delta n(X)$, $\delta n(Y)$ 340,350, along orthogonal directions (X,Y) for a grating designed to provide continuous spatial variation of index modulation 341,351.

(139) FIG. 28 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus 360 including a waveguide 361 supporting an input grating 362 and a grating structure including a fold grating 363 configured such that ray paths 364,365 within the fold grating 363 have substantially the same geometrical path length in some embodiments. By designing the grating so that the path differences between the ray paths 364,365 are greater than the coherence length and the rays 364,365 can be summed incoherently to eliminate fringing and other illumination artefacts such as might occur in laser illuminated waveguides.

(140) FIG. 29 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus 370 includes a waveguide 371 supporting an input grating 372 having a spatially varying distribution of K-vectors 373 and a grating structure 374 having a spatially varying distribution of K-vectors 375 in some embodiments.

(141) FIG. 30 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus 380 comprising a waveguide 381 supporting an input grating 382, a grating structure 383 (e.g. a fold grating), and at least one partially reflective coating 384. In many embodiments the coating 384 can at least partially overlap the input grating 382. In many embodiments, the coating 384 can at least partially overlap the grating structure 383. In other embodiments, the coating 384 can at least partially overlap both of the input grating 382 and the grating structure 383. One purpose of the coating 384 may be to overcome interference fringe formation in a laser illuminated waveguide.

(142) FIG. 31 conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention. The grating structure 390A includes non-diffracting regions 391A, gratings of a first prescription 392A, and gratings of a second prescription 393A disposed in a non-repeating pattern.

(143) FIG. 32 conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention. The grating structure 390B includes a non-diffracting regions 391B, gratings 392B of a first prescription, and gratings 393B of a second prescription disposed in a repeating pattern. The grating structures of FIGS. 31-32 can be used to provide two-dimensional beam expansion and uniform light extraction from a waveguide.

(144) FIG. 33 conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention. The grating structure 390 includes alternating high index regions 391 and low index regions 392 with a grating pitch (p). As illustrated, the grating pitch p may be the distance between the peaks or centres of successive low index and high index regions. In many embodiments, the grating pitch p can be much greater than a wavelength λ to enable a wide field of view and low dispersion. In many embodiments, the grating pitch p can be in the range 5-1000 microns. The large pitch value results in many diffracted orders which can be used for optimising field of view and color dispersion, enabling single layer wide FOV full color waveguides. In many embodiments the high index regions and low index regions may have different widths.

(145) FIG. 34 conceptually illustrates a waveguide apparatus 400 in accordance with an embodiment of the invention. The waveguide apparatus 400 includes a ray multiplying element 404 disposed before an input coupler for debanding in accordance with an embodiment of the invention. In many embodiments, the input coupler 402 is a grating. In some embodiments, the ray multiplying element 404 can be disposed after the input coupler 402. The waveguide apparatus 400

includes a waveguide **401** supporting an input grating **402**, a grating structure **403**, and a ray multiplying element **404** for controlling beam width. An optical path through the waveguide in some embodiments is illustrated by the rays **405-407**. In many embodiments, the ray multiplying element **404** can include mirrors, gratings and can have angle dependence and different beam width multiplying factors matched to different beam incidence angles and source exit pupils.

(146) FIG. **35** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus **410** uses a ray multiplying element **414** for debanding and an input prism coupler **412**. The waveguide apparatus **410** includes a waveguide **411**, a prism coupler **412**, a grating structure **413**, and a ray multiplying element **414**. As illustrated, the ray multiplying element **414** may be positioned below the input prism coupler **412**. An optical path through the waveguide **411** is illustrated by the rays **415-417**.

(147) FIG. **36A** conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The display **420** uses a grating structure **423** including an array of grating elements **423A,423B,423C**. The display **420** includes a waveguide **421** supporting an input coupler **422** and a grating structure **423** for beam expansion and extraction including an array of grating elements **423A,423B**. The array of grating elements include elements **423A** for diffracting a first wavelength, elements **423B** for diffracting a second wavelength and elements **423C** for diffracting a third wavelength. In many embodiments, the grating elements **423A,423B,423C** of each wavelength can have different K-vectors. In many embodiments, the K-vectors can be selected from a group of two different K-vectors. In many embodiments, the K-vectors can be selected from a group of two opposing K-vectors. In some embodiments, in which the grating structure operates on red, green and blue light and requires two different K-vectors for performing two-dimensional beam expansion and extraction, a total of six different grating prescriptions are required. In several embodiments, the array may include non-diffracting regions. Input light can be directed into the waveguide **421** at different angle **81, 82**. Examples of optical paths through the waveguide are illustrated by the rays **424A-426A** and **424B-426B**. The ray paths shown can, in some embodiments, represent light of different wavelengths. Light of a given input direction may undergo multiple grating interactions within the grating structure **423**, undergoing diffraction only at gratings for which the incident ray lies within the spectral or angular bandwidth of the grating. Examples of optical paths through the grating structure is illustrated by the TIR ray **427,428**.

(148) In some embodiments, the red, green and blue gratings can be exposed using a spatial light modulator array mask, illuminating the separate RGB array pattern on the patterned mask time sequentially to avoid crosstalk during exposure.

(149) In many embodiments, angle and wavelength Bragg selectivity, spatial variations of modulation, thickness etc can be used to mitigate cross talk, control uniformity and minimise the effects of grating edge interactions.

(150) In many embodiments the grating elements can be approximately 30 microns in size.

(151) FIG. **36B** conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **430** uses a grating structure including an array of grating elements **433A,433B,433C** in accordance with an embodiment of the invention. The waveguide-based display **430** includes a waveguide substrate **431** supporting three input coupler gratings **434A,434B,434C** and a grating structure **433** for beam expansion and extraction including an array of grating elements. The array of grating elements includes elements **433A** for diffracting a first wavelength, elements **433B** for diffracting a second wavelength and elements **433C** for diffracting a third wavelength. In many embodiments, the input couplers **434A,434B,434C** diffract said first, second and third wavelengths respectively. An example of an optical path through the grating structure is illustrated by the TIR ray **437**. For example, the optical path of light coupled into the waveguide by the coupler **434A** is illustrated by the rays **434-436**.

(152) FIG. 36C conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **440** includes a grating structure **443** including an array of grating elements **443A, 443B, 443C**. The waveguide-based display **440** includes a waveguide substrate **441** supporting an input coupler grating array **442** including grating elements **444A, 444B, 444C** for coupling first, second and third wavelength light into TIR paths within the waveguide, and a grating structure **443** for beam expansion and extraction including an array of grating elements including, in many embodiments, elements **443A** for diffracting the first wavelength, elements **443B** for diffracting the second wavelength and elements **443C** for diffracting the third wavelength. An example of an optical paths through the grating structure is illustrated by the TIR ray **447**. The optical path of light coupled into the waveguide is illustrated by the rays **445-446**.

(153) FIG. 37 conceptually illustrates a side elevation view of a waveguide-based display **450** based on the embodiments of FIGS. 36A, 36B, 36C. The waveguide-based display **450** includes a waveguide **451** with an input coupler provided by a prism **454** in accordance with an embodiment of the invention. The waveguide-based display **450** further includes a grating structure **453**. The prism **454** couples incident light **455** into TIR paths within the waveguide. The grating structure **453** performs beam expansion and extracts the light **456**. FIG. 38 illustrates a plan view of the waveguide-based display described in connection with FIG. 37.

(154) FIG. 39 conceptually illustrates an apparatus **470** for recording a grating structure including a grating array in accordance with an embodiment of the invention. The apparatus **470** includes a substrate **471** supporting a layer of holographic recording material, a master grating **472**, red, green and blue lasers **473A, 473B, 473C** emitting beams **474A, 474B, 474C**, which can be combined by a stack of dichroic filters **475A, 475B, 475C** into the common beam direction into the collimating/beam expansion optics **476**. The expanded beam may be diffracted by the master grating **472** to form a diffracted beam **477** and a zero-order beam **478** used for copying the master grating **472** into the holographic material layer. Many other configurations for combining, collimating and beam expanding the laser beams should be apparent to those skilled in the art. For example, holographic recording material with red, green, or blue sensitized dyes can be deposited onto areas designated the red, green and blue gratings respectively. In many embodiments, the holographic recording material is a HPDLC material system. In many embodiments, the grating regions are coated using an inkjet printer. In many embodiments, the grating regions can be coated with a spatial resolution of 30 microns. In many embodiments the gratings have rolled K-vectors. In many embodiments, the gratings are recorded use rapid pulse laser exposure so that complete red green and blue gratings exposure occurs within substantially the same time window. In many embodiments, the master grating **472** supports a hard coat to allow it to be positioned close to the recording material layer. In many embodiments, a 10 nm hard coat can be used for this purpose. In many embodiments, the master grating **472** is coated with a 50-micron cover glass or polymer overcoating. In many embodiments, the principles of the embodiment of FIG. 39 can be applied in a two-color band (blue-green and red, for example) waveguide.

(155) In many embodiments, the grating structure can include a plurality of grating elements tiled in a repeating pattern. In many embodiments, the grating structure can include a plurality of grating elements tiled in a non-repeating pattern. In many embodiments, the grating structure can include grating elements of non-rectangular shape. In many embodiments, the grating structure can have a spatially varying refractive index modulation. In many embodiments, the grating structure can include a plurality of grating elements with spectral response selected from a group containing at least two different spectral responses. In many embodiments, the grating structure can include a plurality of grating elements with angular response selected from a group containing at least two different angular responses. In many embodiments, the grating structure can include a plurality of grating elements with birefringence including at least two different birefringence characteristics. In many embodiments, the grating structure can include a plurality of grating elements with

polarization response including at least two different polarization responses. In many embodiments, the grating structure can include a plurality of grating elements with refractive index modulation including at least two different refractive index modulations. In many embodiments, the grating structure can include a plurality of grating elements with average refractive index including at least two different refractive indices. In many embodiments, the grating structure can include a plurality of grating elements with grating thickness including at least two different grating thicknesses. In many embodiments, the grating structure can include a plurality of grating elements having grating prescriptions including at least two different grating prescriptions. In many embodiments, the grating structure can include a plurality of grating elements having K-vectors including at least two different K-vectors. In many embodiments, the grating structure can include a plurality of grating elements including diffracting grating elements arrayed with non-diffracting grating elements. In many embodiments, the grating structure can include a plurality of grating elements recorded in material including at least two different materials. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having dispersion curves including at least two different dispersion curves. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having wavelength sensitivity selected from a group including at least two different wavelength sensitivities. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having holographic exposure time including at least two different holographic exposure times. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having holographic exposure energy including at least two different holographic exposure energies. In many embodiments, the grating structure is configured to support ray paths differing by a distance shorter than the coherence length of the source. In many embodiments, the grating structure can include at least one grating having a grating pitch much greater than a wavelength of light. In many embodiments, the grating structure can include at least one grating having a spatial variation of: spatial variation of grating thickness; average refractive index; refractive index modulation, or birefringence. In many embodiments, the grating structure can include rolled K-vector grating, a dual interaction grating, a multiplexed grating, a lossy grating, a sub wavelength grating, and/or a chirped grating. In many embodiments, the grating structure can include at least one surface relief grating formed by liquid crystal extraction from a grating recorded in HPDLC. In many embodiments, the grating structure can include at least one grating formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a HPDLC recorded grating with another material having a refractive index higher than that of the extracted liquid crystal. In many embodiments, the grating structure can include at least one grating formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a HPDLC recorded grating with another material having a refractive index lower than that of the extracted liquid crystal. In many embodiments, the grating structure can include at least one reflection grating. In many embodiments, the grating structure can include at least one transmission grating. In many embodiments, the grating structure can be a switching grating. In many embodiments, the grating structure can further include the group of a beamsplitter layer, an anti-reflection coating, an optical bandpass filter, a polarization modification layer, and/or an alignment layer. In many embodiments, the grating structure can provide beam expansion and extraction of red, green and blue image modulated light over a field of view. In many other embodiments, the image modulated light is monochromatic. In many embodiments, the grating structure can be recorded into a material including a mixture including a liquid crystal and at least one monomer. The mixture may include at least one blue phase liquid crystal and at least one monomer. In many embodiments, the input coupler can be a grating or a prism. In many embodiments, the input coupler can include a plurality of gratings. In many embodiments, the input coupler can be a switching grating.

(156) In many embodiments, a photonic crystal formed by liquid crystal extraction may be used to form a multiplexed grating. FIG. 40 illustrates a multiplexed grating structure in accordance with

an embodiment of the invention. The multiplexed grating may be manufactured using a method. In a first step, a first mixture of liquid crystal is provided. A first grating **482** including alternating liquid crystal and polymer regions can be formed in the first mixture using a holographic exposure. The LC regions can be flushed to form a first set of polymer regions separated by air. A second mixture of liquid crystal can be provided in the air regions of the first grating. A second grating **484** including alternating liquid crystal and polymer regions can be formed in the second mixture using a holographic exposure. The LC regions can be flushed to form polymer regions separated by air. A third mixture of liquid crystal can be provided in the air regions formed by the first and second gratings. A third grating **486** comprising alternating liquid crystal and polymer regions can be formed in the second mixture using a holographic exposure. The LC regions can be flushed to form polymer regions separated by air. FIG. **40** shows the final grating structure formed by the above process. The first grating **482**, the second grating **484**, and the third grating **486** represent the polymer regions of the final grating. In many embodiments, the three superimposed gratings **482,484,486** may have the same refractive index. Air spaces **488** remaining after the above process has been completed. The structure of FIG. **40** multiplexes three gratings **482,484,486**. In some embodiments, the three gratings **482,484,486** may be formed from different monomers to provide a required spatial refractive index modulation variation. In some embodiments, the air regions **488** can be backfilled with an optical material for providing a desired refractive index contrast. It should be apparent from consideration of the embodiment of FIG. **40** that a similar process can be applied to multiplexing any number of grating structures subject to material and process limitations.

(157) One important advantage of the LC evacuated gratings is that they will not clear at elevated temperature so have advantage for automotive use, or any other higher temperature environment use. The LC evacuated grating principle can be applied to gratings of any scale. In some embodiments, the LC evacuated grating has spatially varying diffraction efficiency. The application of multiplexing, spatial varying thickness, k-vector directions, and diffraction efficiency in the present invention is based on the embodiments, drawings and teachings provided.

(158) In one embodiment the gratings are recorded in uniform modulation liquid crystal-polymer material system such as the ones disclosed in United State Patent Application Publication No.: US2007/0019152 by Caputo et al. and PCT Application No.: PCT/EP2005/006950 by Stumpe et al. both of which are incorporated herein by reference in their entireties. Uniform modulation gratings are characterized by high refractive index modulation (and hence high diffraction efficiency) and low scatter.

(159) The invention can be applied using grating arrays made up of grating elements of many different geometries that are limited only by geometrical constraints and the practical issues in implementing the arrays. In many embodiments, the grating array can include grating elements that are aperiodic (non-repeating). In such embodiments, the asymmetry in the geometry and the distribution of the grating elements can be used to produce uniformity in the output illumination from the waveguide. The optimal grating elements sizes and geometries can be determined using reverse vector raytracing from the eyebox through the output and input gratings (and fold gratings, if used) onto the grating array. A variety of asymmetric grating element patterns can be used in the invention. For example, FIG. **41A** conceptually illustrates a portion **490** of a grating element pattern including rectangular elements **490A-490F** of differing size and aspect ratio for use in an emissive display panel in accordance with an embodiment of the invention. In some embodiments, the grating element array can be based a non-repeating pattern based on a finite set of polygonal base elements. FIG. **41B** conceptually illustrates a portion **500** of a grating element pattern including Penrose tiles **500A-500J** in accordance with an embodiment of the invention. The tiles can be based on the principles disclosed in U.S. Pat. No. 4,133,152 by Penrose entitled "Set of tiles for covering a surface" which is hereby incorporated by reference in its entirety. Patterns occurring in nature, of which honeycombs are well known examples, can also be used in many embodiments.

(160) In many embodiments, the grating element can be configured in arrays of identical regular

polygons. For example, FIG. 41C conceptually illustrates a portion 510 of a grating element pattern including hexagons 510A-510C in accordance with an embodiment of the invention. FIG. 41D conceptually illustrates a portion 520 of a grating element pattern including squares 520A-520C in accordance with an embodiment of the invention. FIG. 41E conceptually illustrates a portion 530 of a grating element pattern comprising diamond-shaped elements 530A-530D in accordance with an embodiment of the invention. FIG. 41F conceptually illustrates a portion 540 of a grating element pattern including isosceles triangles 540A-540H in accordance with an embodiment of the invention.

(161) In many embodiments, the grating elements can have vertically or horizontally biased aspect ratios. FIG. 41G conceptually illustrates a portion 550 of a grating element pattern including hexagons 550A-550C of horizontally biased aspect ratio in one embodiment. FIG. 41H conceptually illustrates a portion 560 of a grating element pattern including rectangles 560A-560D of horizontally biased aspect ratio in accordance with an embodiment of the invention. FIG. 41I conceptually illustrates a portion 570 of a grating element pattern including diamond shaped elements 570A-570D of horizontally biased aspect ratio in accordance with an embodiment of the invention. FIG. 41J conceptually illustrates a portion 580 of a grating element pattern including triangles 580A-580H of horizontally biased aspect ratio in accordance with an embodiment of the invention.

(162) In many embodiments, using the grating element configurations discussed above, the grating elements can have differing spectral response. In many embodiments, using the grating element configurations discussed above, the grating elements can have differing angular response. In many embodiments, using the grating element configurations discussed above, the grating elements can have differing polarization response. In many embodiments, the grating element can have spectral, angular, and/or polarization response varying spatially across the grating element array. In many embodiments, grating elements of different sizes and geometries and different spectral, angular and polarization response can be arranged to provide a spatial emission variation for controlling uniformity in the final image.

(163) FIG. 42 conceptually illustrates a method for forming a waveguide color grating structure in accordance with an embodiment of the invention. As shown, the method 590 of forming a grating structure is provided. Referring to the flow diagram, the method 590 includes providing (591) a waveguide substrate. A master grating patterned into an array comprising red, green and blue diffracting grating elements can be provided (592). Coats of red, green and blue sensitized holographic recording material can be provided in the designated red, green and blue grating regions of the substrate correspond to the master grating array can be deposited (593). Red, green and blue laser sources can be provided (594). Red, green and blue exposure can be applied (595) to the holographic recording material using diffracted and zero order beams from the master. A grating structure comprising an array of red, green and blue differing gratings can be recorded (596).

(164) In many embodiments, the holographic recording material can be a mixture of at least one liquid crystal and at least one monomer. In many embodiments, the holographic recording material can further include at least one component selected from the group of a photoinitiator, a dye, a nano particle, a surfactant, a blue phase liquid crystal and a reactive mesogen.

(165) In many embodiments, the master is an amplitude grating with a hard overcoat. In many embodiments, the coated substrate is protected by a thin cover glass or a polymer layer. In many embodiments, the distance from the amplitude grating to the holographic recording plane is in the range 1-50 micron to avoid trichromatic pixel exposure overlap. In many embodiments, the grating elements thickness is approximately 2 microns. In many embodiments, the grating elements are printed with a resolution and accuracy of approximately 30 microns.

DOCTRINE OF EQUIVALENTS

(166) While the above description contains many specific embodiments of the invention, these

should not be construed as limitations on the scope of the invention, but rather as an example of one embodiment thereof. It is therefore to be understood that the present invention may be practiced in ways other than specifically described, without departing from the scope and spirit of the present invention. Thus, embodiments of the present invention should be considered in all respects as illustrative and not restrictive. Accordingly, the scope of the invention should be determined not by the embodiments illustrated, but by the appended claims and their equivalents.

Claims

1. A waveguide-based display device comprising: a waveguide; a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection (TIR) path within the waveguide; and a grating structure for providing beam expansion of the TIR light in at least one direction and extracting the TIR light from the waveguide, wherein the grating structure comprises a plurality of grating elements having at least two different grating prescriptions and at least one surface relief grating formed by a phase separation process; wherein the grating structure is configured to enable a plurality of ray path lengths within the grating structure that differ in length by a distance shorter than a coherence length of the source of image modulated light.
2. The waveguide-based display device of claim 1, wherein the plurality of grating elements is tiled in a repeating pattern or non-repeating pattern.
3. The waveguide-based display device of claim 1, wherein the grating structure further comprises a spatial refractive index modulation.
4. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises at least two different spectral responses, at least two different angular responses, at least two different birefringence characteristics, at least two different polarization responses, at least two different refractive index modulations, an average refractive index including at least two different refractive indices, at least two different grating thicknesses, and at least two different K-vectors.
5. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises diffracting grating elements arrayed with non-diffracting grating elements.
6. The waveguide-based display device of claim 1, wherein the plurality of grating elements is recorded in material containing at least two different materials.
7. The waveguide-based display device of claim 1, wherein the plurality of grating elements are recorded in material having dispersion curves selected from a group containing at least two different dispersion curves.
8. The waveguide-based display device of claim 1, wherein the plurality of grating elements is recorded in material having at least two different wavelength sensitivities, at least two different holographic exposure times, or at least two different holographic exposure energies.
9. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises at least one grating pitch greater than a wavelength of the TIR light.
10. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises a spatial variation of at least one selected from the group consisting of: spatial variation of grating thickness, average refractive index, refractive index modulation, and birefringence.
11. The waveguide-based display device of claim 1, wherein the grating structure comprises at least one selected from a group consisting of: a rolled K-vector grating, a dual interaction grating, a multiplexed grating, a lossy grating, a sub wavelength grating, and a chirped grating.
12. The waveguide-based display device of claim 1, wherein the at least one surface relief grating is formed by liquid crystal extraction from a grating recorded in holographic polymer dispersed liquid crystal (HPDLC).
13. The waveguide-based display device of claim 1, wherein the at least one surface relief grating

- is formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a holographic polymer dispersed liquid crystal (HPDLC) recorded grating with another material having a refractive index higher than or lower than that of the extracted liquid crystal.
14. The waveguide-based display device of claim 1, wherein the grating structure comprises at least one reflection grating or at least one transmission grating.
15. The waveguide-based display device of claim 1, wherein the grating structure provides beam expansion and extraction of red, green and blue image modulated light over a field of view.
16. The waveguide-based display device of claim 1, wherein the grating structure is recorded into a material selected from the group of a mixture comprising at least one liquid crystal and at least one monomer, a mixture comprising at least one blue phase liquid crystal and at least one monomer.
17. The waveguide-based display device of claim 1, wherein the input coupler is a grating or a prism.
18. The waveguide-based display device of claim 1, wherein the at least one surface relief grating comprises a grating modulation depth greater than a grating pitch.
19. The waveguide-based display device of claim 1, further comprising an alignment layer.
-